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Supplement
Pilot´s Operating Handbook
for the
(Reims) Cessna (F) 172
F,G,H,I,K,L,M
Equipped with TAE 125-02-114
Installation

Issue 2

MODEL No.	Cessna 172K
SERIAL No.	17267526
REGISTER No.	HA-SLY

This supplement must be attached to the EASA approved Pilot´s Operating Handbook when the TAE 125-02-114 installation has been installed in accordance with EASA STC 10014287.

The information contained in this supplement supersede or add to the information published in the EASA approved Pilot´s Operating Handbook only as set forth herein.
For limitations, procedures, performance and loading information not contained in this supplement, consult the EASA approved Pilot´s Operating Handbook.

This supplement Pilot's Operating Handbook is approved with EASA STC 10014287.

Doc.-No.: 20-0310-21122

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APPROVAL

The content of approved chapters is approved by EASA. All other content is approved by TAE under the authority of EASA DOA No. EASA.21J.010 in accordance with Part 21.

LOG OF REVISIONS

Revision	Section	Description	Approved	
			Date	Endorsed
2/0	all	new issue	May 21, 2010	EASA STC 10014287
2/1	1	New oil, editorial changes	April 14, 2011	Revision No. 1 to AFM supplement ref. 20-0310-21122 is approved under the authority of DOA ref. EASA.21J.010. Date: April 14, 2011 Office of Airworthiness
	2	New oil, editorial changes		
	3	Procedures updated		
	4	Procedures updated		
	5	Editorial changes		
	6	Editorial changes		
	7	New Section		
2/2	1	New gearbox oil, editorial changes	June 30, 2011	Revision No. 2 to AFM supplement ref. 20-0310-21122 is approved under the authority of DOA ref. EASA.21J.010. Date: June 30, 2011 Office of Airworthiness
	2	New gearbox oil		
	4	Procedures updated		

Revision	Section	Description	Approved	
			Date	Endorsed
2/3	1	New Fuel, new gearbox oil	March 16, 2012	Revision No. 3 to AFM supplement ref. 20-0310-21122 is approved under the authority of DOA ref. EASA.21J.010. Date: March 16, 2012 Office of Airworthiness
	2	New fuel, new gearbox oil		
	4	New fuel, Procedures updated		
	5	Procedures updated		
	6	New fuel		
2/4	1	New gearbox oil	March 11, 2013	Revision No. 4 to AFM supplement ref. 20-0310-21122 is approved under the authority of DOA ref. EASA.21J.010. Date: March 11, 2013 Office of Airworthiness
	2	New gearbox oil		
	5	Procedures updated		
2/5	---	EASA STC / AFM numbers corrected on the cover	May 27, 2013	Revision No. 5 to AFM supplement ref. 20-0310-21122 is approved under the authority of DOA ref. EASA.21J.010. Date: May 27, 2013 Office of Airworthiness

Revision	Section	Description	Approved	
			Date	Endorsed
2/6	1	Safety Recommendation New fuel New gearbox oil Note fuel additive	03.09.2014	Revision No. 6 to AFM supplement ref. 20-0310-22122 is approved under the authority of DOA ref. EASA.21J.010. Date: september 03, 2014 Office of Airworthiness
	2	Note added New fuel New gearbox oil Note fuel additive	03.09.2014	
	3	Description adapted wording	03.09.2014	
	4	Note added wording	03.09.2014	
	5	Wording	03.09.2014	
	6	Wording	03.09.2014	
2/7	4	Procedure added	26.01.2015	Revison No. 7 to AFM supplement ref. 20-0310-21122 is approved under the authority of DOA ref. EASA.21J.010. Date: January 26, 2015 Office of Airworthiness

Revision	Section	Description	Approved	
			Date	Endorsed
2/8	1	New propeller	April 08, 2015	EASA STC 10014287, Rev. 8
	5	splitted due to new propeller specs		
	5a	New section		
	5b	New section		
2/9	1	Update liquids	Jan. 22, 2018	Revision No. 9 to AFM supplement ref. 20-0310-21122 is approved under the authority of DOA ref. EASA.21.J.010. Date: Jan. 22, 2018 Office of Airworthiness
	2	Update Liquids		
	3	various minor corrections		
	4	Update FADEC Test above 5500ft		

Remark: The parts of the text which changed are marked with a vertical line on the margin of the page.

LIST OF EFFECTIVE SECTIONS

Sections	Issue/Revision	Date
1	2/8	Jan. 22, 2018
2	2/7	Jan. 22, 2018
3	2/7	Jan. 22, 2018
4	2/8	Jan. 22, 2018
5	2/7	April 08, 2015
5a	2/0	April 08, 2015
5b	2/0	April 08, 2015
6	2/6	Sept. 03, 2014
7	2/6	Sept. 03, 2014

GENERAL REMARK

The content of this POH supplement is developed on basis of the EASA-approved POH.

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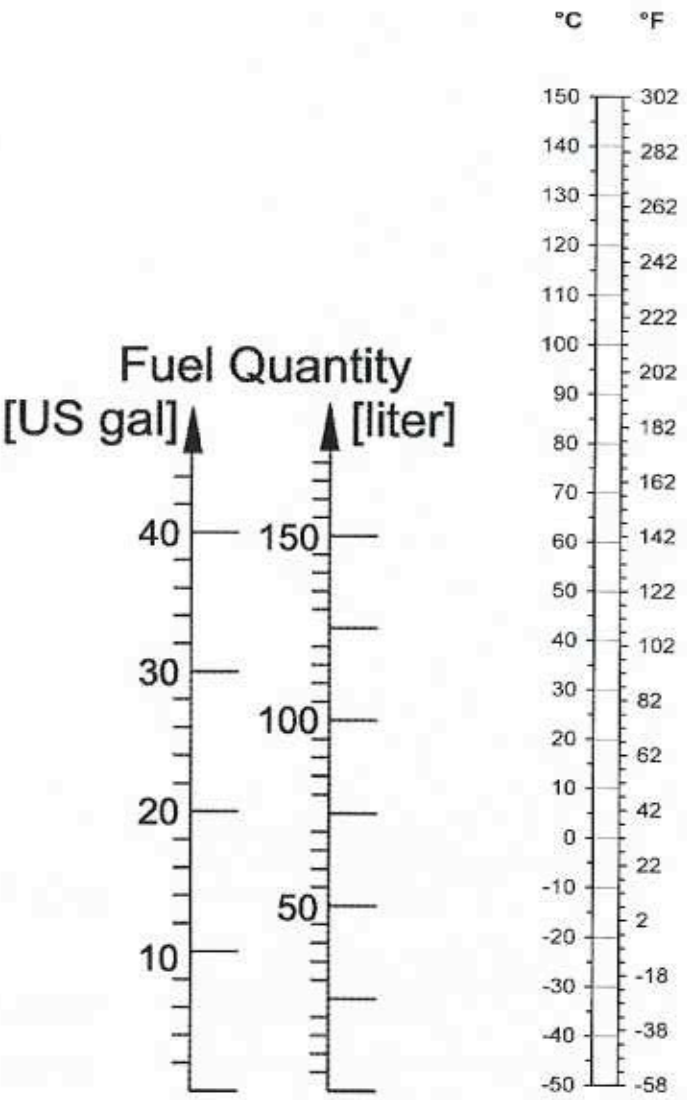
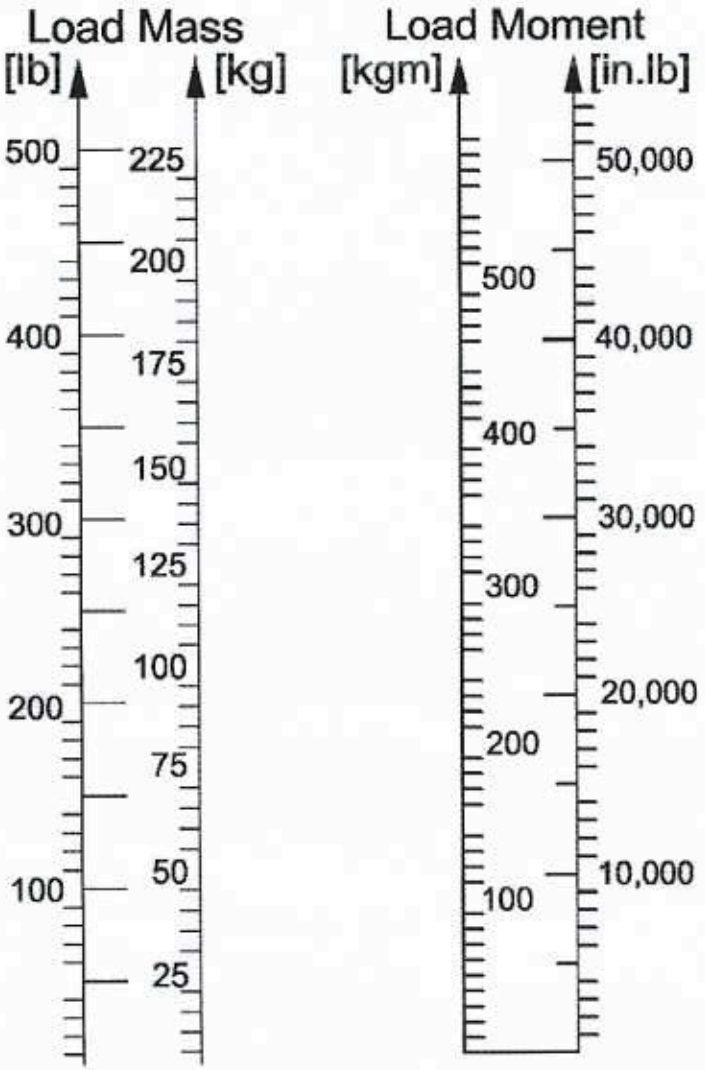
SECTION 6	APPENDIX (a non-approved chapter)
SECTION 7	SUPPLEMENTS

CONVERSION TABLES

VOLUME		
Unit [Abbr.]	Conversion factor SI to US / Imperial	Conversion factor US / Imperial to SI
Liter [l] US gallon [US gal] US quart [US qt] Imperial gallon [Imp gal] Cubic inch [in³]	[l] / 3.7854 = [US gal] [l] / 0.9464 = [US qt] [l] / 4.5459 = [Imp gal] [l] x 61.024 = [in³]	 [US gal] x 3.7854 = [l] [US qt] x 0.9464 = [l] [Imp gal] x 4.5459 = [l] [in³] / 61.024 = [l]
TORQUE		
Unit [Abbr.]	Conversion factor SI to US / Imperial	Conversion factor US / Imperial to SI
Kilopondmeter [kpm] Foot pound [ft.lb] Inch pound [in.lb]	[kpm] x 7.2331 = [ft.lb] [kpm] x 86.7962 = [in.lb]	 [ft.lb] / 7.2331 = [kpm] [in.lb] / 86.7962 = [kpm]
TEMPERATURE		
Unit [Abbr.]	Conversion factor SI to US / Imperial	Conversion factor US / Imperial to SI
Degree Celsius [°C] Degree Fahrenheit [°F]	[°C] x 1.8 + 32 = [°F]	 ([°F] - 32) / 1.8 = [°C]
SPEED		
Unit [Abbr.]	Conversion factor SI to US / Imperial	Conversion factor US / Imperial to SI
Kilometers per hour [km/h] Meters per second [m/s] Miles per hour [mph] Knots [kts] Feet per minute [fpm]	[km/h] / 1.852 = [kts] [km/h] / 1.609 = [mph] [m/s] x 196.85 = [fpm]	 [mph] x 1.609 = [km/h] [kts] x 1.852 = [km/h] [fpm] / 196.85 = [m/s]

PRESSURE		
Unit [Abbr.]	Conversion factor SI to US / Imperial	Conversion factor US / Imperial to SI
Bar [bar] Hectopascal [hpa] =Millibar [mbar] Pounds per square inch [psi] inches of mercury column [inHg]	[bar] x 14.5038 = [psi] [hpa] / 33.864= [inHg] [mbar] / 33.864 = [inHg]	 psi] / 14.5038 = [bar] [inHg] x 33.864 = [hPa] [inHg] x 33.864 = [mbar]
MASS		
Unit [Abbr.]	Conversion factor SI to US / Imperial	Conversion factor US / Imperial to SI
Kilogramm [kg] Pound [lb]	[kg] / 0.45359 = [lb]	 [lb] x 0.45359 = [kg]
LENGTH		
Unit [Abbr.]	Conversion factor SI to US / Imperial	Conversion factor US / Imperial to SI
Meter [m] Millimeter [mm] Kilometer [km] Inch [in] Foot [ft] Nautical mile [nm] Statute mile [sm]	[m] / = 0.3048 [ft] [mm] / = 25.4 [in] [km] / = 1.852 [nm] [km] / = 1.609 [sm]	 [in] x 25.4 = [mm] [ft] x 0.3048 = [m] [nm] x 1.852 = [km] [sm] x 1.609 = [km]
FORCE		
Unit [Abbr.]	Conversion factor SI to US / Imperial	Conversion factor US / Imperial to SI
Newton [N] Decanewton [daN] Pound [lb]	[N] / 4.448 = [lb] [daN] / 0.4448 = [lb]	 [lb] x 4.448 = [N] [lb] x 0.4448 = [daN]


°C
°F
150
140
130
120
110
100
90
80
70
60
50
40
30
20
10
0
-10
-20
-30
-40
-50
302
282
262
242
222
202
182
162
142
122
102
82
62
42
22
2
-18
-38
-58
Fuel Quantity
[US gal]
[liter]
40
30
20
10
150
100
50


Load Mass
[lb]
[kg]
500
400
300
200
100
225
200
175
150
125
100
75
50
25
Load Moment
[kgm]
[in.lb]
500
400
300
200
100
50,000
40,000
30,000
20,000
10,000

ABBREVIATIONS

FADEC	Full Authority Digital Engine Control
CED 125	Compact Engine Display Multifunctional instrument for indication of engine data of the TAE 125-02-114
AED 125	Auxiliary Engine Display Multifunctional instrument for indication of engine and airplane data

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SECTION 1
GENERAL

Safety Recommendations

The following symbols and warnings are used in this manual. They must be heeded strictly to prevent personal injury and material damage, to avoid impairment of the operational safety of the aircraft and to rule out any damage to the aircraft as a consequence of improper handling.

▲ **WARNING:**	Non-compliance with these safety rules could lead to injury or even death.
■ CAUTION:	Non-compliance with these special notes and safety measures could cause damage to the engine or to the other components.
◆ Note:	Information added for a better understanding of an instruction.

UPDATE AND REVISION OF THE MANUAL

▲ **WARNING:**	A safe operation is only assured with an up to date POH supplement. Information about actual POH supplement issues and revisions are published in the Service Bulletin TM TAE 000-0004.
◆ Note:	The Doc.-No of this POH supplement is published on the cover sheet of this supplement.

ENGINE

▲ **WARNING:** The engine requires an electrical power source for operation. If the main battery and alternator fail, the engine will only operate for a maximum of 30 minutes on FADEC backup battery power. Therefore, it is important to pay attention to indications of alternator failure.

Engine manufacturer:..... Technify Motors GmbH
Engine model:.....TAE 125-02-114

The TAE 125-02-114 is a liquid cooled in-line four-stroke 4-cylinder turbocharged engine with DOHC (double overhead camshaft), direct fuel injection and common-rail technology. It has a displacement of 1991 ccm (121.5 in^3). The engine is controlled by a FADEC system. The propeller is driven by a built-in gearbox
(i = 1.69) with mechanical vibration dampening and overload release. The engine has an electrical self starter and an alternator.

Due to this specific characteristic, all of the information from the flight manual recognized by EASA are no longer valid with reference to:

- carburetor and carburetor pre-heating
- ignition magnetos and spark plugs, and
- mixture control and priming system

PROPELLER

Manufacturer:.....MT Propeller Entwicklung GmbH
Model: MTV-6-A/187-129
..... MTV-6-A/190-69
Number of blades:.....3
Diameter: ..... 1.87 m (MTV-6-A/187-129
..... 1.90 m (MTV-6-A/190-69)
Type: .....constant speed

FUELS and LIQUIDS

▲ **WARNING:** The engine must not be started under any circumstances if any fluid level is too low.

■ CAUTION: Use of unapproved fuels may result in damage to the engine and fuel system components, resulting in possible engine failure.

■ CAUTION: Use approved oil with exact designation only!

■ CAUTION: Normally it is not necessary to fill the cooling liquid or gearbox oil between maintenance intervals. If the level is too low, please notify the service center immediately.

Fuel: .....JET A-1 (ASTM 1655)
.....JET A (ASTM 1655)
..... Jet Fuel No.3 (GB 6537-2006)
.....JP-8 (MIL-DTL-83133E)
.....JP-8+100 (MIL-DTL-83133E)
..... TS-1 (GOST 10227-86)
.....TS-1 (GSTU 320.00149943.011-99
Alternative: ..... Diesel (**DIN** EN 590)
..... SASOL GTL Diesel

◆ Note: The liquid fuel additive Biobor JF can be used in jet and diesel fuel systems to eliminate growth of fungi. For further details refer to the manufacturer specifications.

Engine oil: ..... AeroShell Oil Diesel Ultra
..... AeroShell Oil Diesel 10W-40
.....Shell Helix Ultra 5W-30
.....Shell Helix Ultra 5W-40

Gearbox oil:	Centurion Gearbox Oil N1
	Shell Spirax S6 ATF ZM
	Shell Spirax S6 GXME 75W-80, API GL-4
	Shell Spirax S4 G 75W-90, API GL-4
Coolant:	Water/Radiator Protection at a ratio of 50:50
Radiator Protection:	BASF Glysantin Protect Plus / G48
	Valvoline/Zerex Glysantin G48
	Mobil Antifreeze Extra (G48)
	Comma Xstream Green - Concentrate/G48

◆ Note: The freezing point of the coolant is -36°C.

INSTRUMENT PANEL

Components of the new installation can be seen as example in the following Figure.

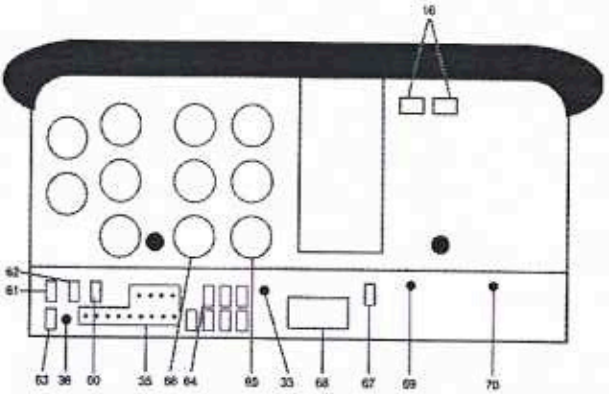

Figure 1-1 Example of Instrument panel

16. Fuel Quantity Gauges (Oil Temperature and Oil Pressure Gauge **N/A**)
33. "Alt. Air Door" Alternate Air Door (Carburetor Heat Button **N/A**)
35. Circuit Breakers
38. "Starter"- Push Button for Starter
60. "ALT" - Switch for Alternator
61. "BAT" - Switch for Battery
62. "MAIN" - Switch for Main Bus
63. "Engine Master" - Switch electrical supply FADEC
64. "Fuel Pump" - Switch for the Electric Fuel Pump
65. AED 125 SR (Voltmeter, Ammeter) with indication of Fuel Temperature, Voltage and a warning lamp "Water Level" (amber) for low coolant level
66. CED 125 (Tachometer **N/A**)
    The Compact Engine Display contains indication of Propeller Rotary Speed, Oil Pressure, Oil Temperature, Coolant Temperature, Gearbox Temperature and Load.
67. "Force B" - Switch for manually switching the FADEC

68. Lightpanel with:
"FADEC" Test Knob
"A FADEC B" Warning Lights for FADEC A and B (red)
"Alt" Alternator Warning Light (red)
"AED" Caution Light (amber) for AED 125
"CED" Caution Light (amber) for CED 125
"CED/AED" - Test/Confirm Knob for CED 125, AED 125 and Caution Lights (amber)
"Fuel L";"Fuel R" Caution Lights for low fuel level (amber)
"Glow" Glow Control Light (amber)

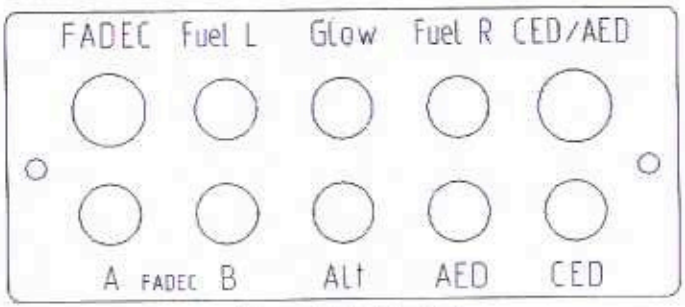

Figure 1-2 Lightpanel

69. "Shut-off Cabin Heat" control knob to close cabin heat in case emergency (Push for shut-off)
70. "Cabin Heat" Cabin heat control

FUEL SYSTEM (Left, Right, Both)

The fuel system of the engine includes the original standard or long-range tanks of the Cessna 172. Additional sensors for Fuel Temperature and "Low Level" Warning are installed.
The fuel flows out of the tanks to the fuel selector valve with the positions LEFT, RIGHT and BOTH, through a reservoir tank to the fuel shut-off valve and then via the electrically driven fuel pump to the fuel filter.
The electrically driven fuel pump supports the fuel flow to the filter module if required. Fuel can be shut off by the separate shutoff valve. Then, the engine-driven feed pump and the high-pressure pump supply the rail, from where the fuel is injected into the cylinders depending upon the position of the thrust lever and regulation by the FADEC. Surplus fuel flows to the fuel cooler and then through the fuel selector valve back into the pre-selected tank, if BOTH is selected the fuel returns to both tanks.
A temperature sensor in the filter module controls the heat exchange between the fuel feed and return. The fuel cooler reduces the fuel temperature in the return line.
The fuel cooler receives its cooling air through an inlet in the air duct to the heating radiator. This inlet is closed with a baffle which must be removed at high outside air temperatures (OAT higher than 20 °C (68 °F), see also Section 4).
Since the density of diesel and jet fuel (0.84 kg/l) is higher than AVGAS (0.715 kg/l), the usable fuel capacity was reduced by this factor through the fuel filler neck, to stay within the approved wing load.

Fuel Capacity				
Tanks		Total Capacity	Total Unusable Fuel	Total Usable Fuel
F,G,H	2 Standard-Tanks: each 62.85 l (16.60 US gal)	125.7 l (33.2 US gal)	11.4 l (3 US gal)	114.3 l (30.2 US gal)
I,K,L,M	2 Standard-Tanks each 67.55 l (17.85 US gal)	135.1 l (35.7 US gal)	15.1 l (4 US gal)	120.0 l (31.7 US gal)
F-M	2 Long-Range-Tanks: each 83.65 l (22.10 US gal)	167.3 l (44.2 US gal)	15.1 l (4 US gal)	152.2 l (40.2 US gal)

FUEL SYSTEM (Left, Right, Both)

■ CAUTION: In flight conditions with downward pointing wing, switch the fuel selector to the upper fuel tank or to the position BOTH

■ CAUTION: In turbulent air it is strongly recommended to use the BOTH position.

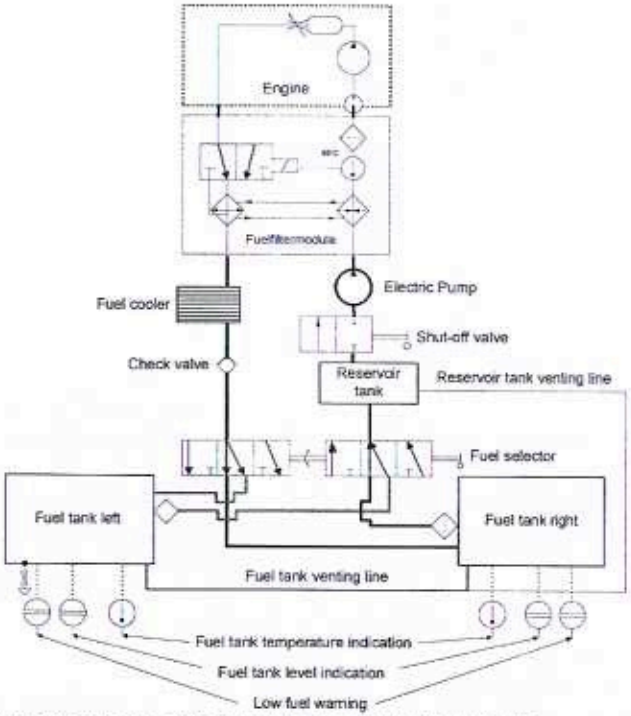

Figure 1-3 Scheme of the Fuel System (Left, Right, Both)

◆ Note: The handling of the fuel selector positions left, right and both are described in the original POH

ELECTRICAL SYSTEM

The electrical system differs from the previous installation and is equipped with the following operating and display elements:

1. Switch "Main Bus"
   This switch controls the Main Bus. The Main Bus is necessary to be able to run FADEC and engine with Battery/Alternator without disturbance in the event of onboard electrical system malfunctions. In normal operation Alternator, Main Bus ans Battery must be ON.
2. Switch "Alternator"
   Controls the alternator. Must be ON in normal operation.
3. Switch "Battery"
   Controls the Battery.
4. Push Button "Starter"
   Controls the magneto switch of the starter.
5. Ammeter
   The Ammeter shows the alternator current. In case of battery discharge if alternator inoperative the alternator warning light will illuminate.
6. Warning light "Alternator"
   Illuminates when the power output of the alternator is too low or the Switch "Alternator" is switched off. Normally, this warning light always illuminates when the "Engine Master" is switched on without revolution and extinguishes immediately after starting the engine.
7. Switch "Fuel Pump"
   This switch controls the electric fuel pump.
8. Switch "Engine Master"
   Controls the two redundant FADEC components and the Alternator Excitation Battery with two independent contacts. The Alternator Excitation Battery is used to ensure that the Alternator continues to function properly even if the main battery fails.

▲ **WARNING:** If the "Engine Master" is switched off, the power supply to the FADEC is interrupted and the engine will shut down.

9. Switch "Force B"
   If the FADEC does not automatically switch from A-FADEC to the B-FADEC in case of an emergency despite of obvious necessity, this switch allows to switch manually to the B-FADEC.

▲ **WARNING:** When operating on FADEC backup battery only, the "Force B" switch must not be activated. This will shut down the engine.

10. FADEC Backup Battery
    The electrical system includes a FADEC backup battery to ensure power supply to A-FADEC in case that supply from both battery and alternator is interrupted. The engine can be operated for a maximum of 30 minutes when powered by the FADEC backup battery only. Only A-FADEC is connected to the backup battery.

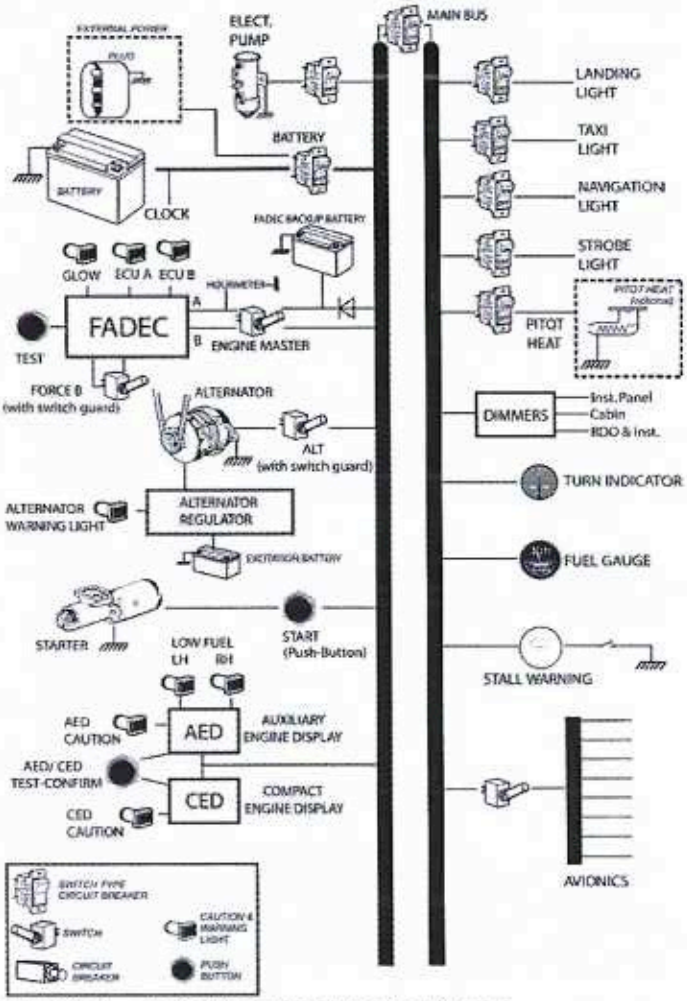

Figure 1-4 Basic Wiring of the Electrical System

FADEC RESET

In case of a FADEC warning, one or both FADEC warning lights are flashing. If then the FADEC test knob is pressed for at least 2 seconds,

a) the active warning lights will extinguish if it was a LOW category warning.
b) the active warning lights will be illuminated steady if it was a HIGH category warning.

■ CAUTION: If a FADEC-warning occurred, contact your service center.

When a high category warning occurs the pilot should land as soon as possible, since the affected FADEC ECU has diagnosed a severe fault. A low category fault has no significant impact on engine operation.
Refer also to the engine OM-02-02 for additional information.

COOLING

The installation is fitted with a fluid-cooling system whose three-way thermostat regulates the flow of coolant between the large and small cooling circuit.
The coolant exclusively flows through the small circuit up to a cooling water temperature of 84°C and then between 84 and 94°C both through the small and the large circuit.
If the cooling water temperature rises above 94°C, the complete volume of coolant flows through the large circuit and therefore through the radiator. This allows a maximum cooling water temperature of 105°C.
There is a sensor in the expansion reservoir which sends a signal to the warning light "Water level" on the instrument panel if the coolant level is low.
The cooling water temperature is measured in the cylinder head near the thermostat and passed on to the FADEC and CED 125.
The connection to the heat exchanger for cabin heating is always open; the warm air supply is regulated by the pilot over the heating valve. See Figure 1-5.

The supply of warm air into the cabin is controlled through the cabin heat control knob. In normal operation, the cabin heat control knob must be in the OPEN position.
In case of certain emergencies (refer to section 3), the control knob "Shut-off Cabin Heat" has to be closed according to the appropriate procedures.

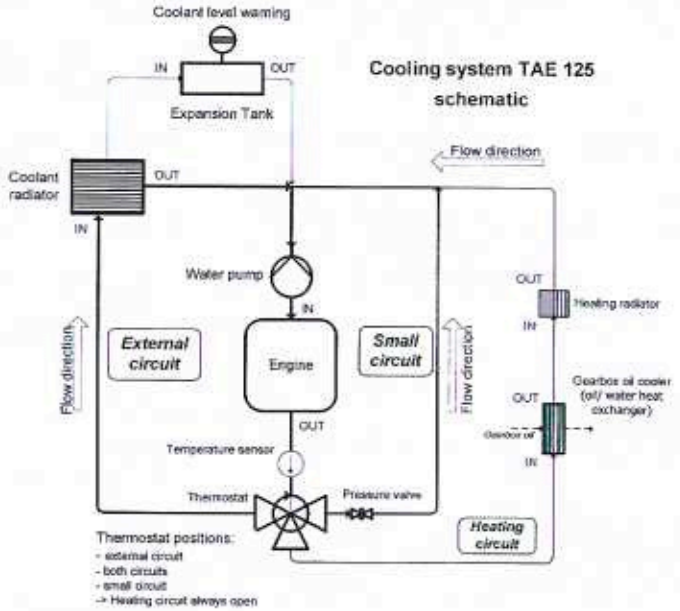

Figure 1-5 Cooling system

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SECTION 2
LIMITATIONS

▲ **WARNING:** It is not allowed to start up the engine using external power. If starting the engine is not possible using battery power, the condition of the battery must be verified before flight.

WEIGHT LIMITS

Normal Category:

Maximum Ramp Weight: ..... 1044 kg (2302 lbs)
Maximum Takeoff Weight: ..... 1043 kg (2300 lbs)
Maximum Landing Weight ..... 1043 kg (2300 lbs)

Utility Category:

Maximum Ramp Weight: ..... 908 kg (2002 lbs)
Maximum Takeoff Weight: ..... 907 kg (2000 lbs)
Maximum Landing Weight ..... 907 kg (2000 lbs)

MANEUVER LIMITS

■ CAUTION: Intentionally initiating negative G maneuvers is prohibited

Normal Category: No change

Utility Category: Intentionally initiating spins is prohibited

FLIGHT LOAD FACTORS

No change

■ CAUTION: Avoid extended negative g-loads duration. Extended negative g-loads can cause propeller control and engine problems.

◆ Note: The load factor limits for the engine must also be observed. Refer to the Operation & Maintenance Manual for the engine.

ENGINE OPERATING LIMITS

Engine manufacturer:..... Technify Motors GmbH
Engine model: TAE125-02-114
Take-off and Max. continuous power:..... 114 kW (155 HP)
Take-off and Max. continuous RPM:..... 2300 min^{-1}
Max. recommended cruise.....85%

◆ Note: In the absence of any other explicit statements, all of the information on RPM in this supplement to the Pilot's Operating Handbook are propeller RPM.

◆ Note: This change of the original aircraft is certified up to an altitude of 18,000 ft.

Engine operating limits for take-off and continuous operation:

▲ **WARNING:** It is not allowed to start the engine outside of these temperature limits.

◆ Note: The operating limit temperature is a temperature limit below which the engine may be started, but not operated at the Take-off RPM. The warm-up RPM to be selected can be found in Section 4 of this supplement.

Oil temperature:
Minimum engine starting temperature: .....-32 °C
Minimum operating limit temperature:.....50 °C
Maximum operating limit temperature:.....140 °C

Coolant temperature:
Minimum engine starting temperature: .....-32 °C
Minimum operating limit temperature: .....60 °C
Maximum operating limit temperature:.....105 °C

Gearbox temperature:

Mininum operating limit temperature:.....-30 °C
Maximum operating limit temperature:.....120 °C

Min. fuel temperature limits in the fuel tank:

Fuel	Minimum fuel temperature in the fuel tank before Take-off	Minimum fuel temperature in the fuel tank during the flight
JET A-1, JET A, Fuel No.3 JP-8, JP-8+100 TS-1	-30°C	-35°C
Diesel, Sasol GTL Diesel	greater than 0°C	-5°C

Table 2-3a Minimum fuel temperature limits in the fuel tank

▲ **WARNING:** The fuel temperature of the fuel tank not used should be monitored if its later use is intended.

▲ **WARNING:** The following applies to Diesel and JET fuel mixtures in the tank:
As soon as the proportion of Diesel in the tank is more than 10% Diesel, the fuel temperature limits for Diesel operation must be monitored. If there is uncertainty about which fuel is in the tank, the assumption should be made that it is Diesel.

Oil Pressure

Minimum oil pressure: ..... 1.2 bar
Minimum oil pressure (at Take-off power) ..... 2.3 bar
Minimum oil pressure (in flight) ..... 2.3 bar
Maximum oil pressure ..... 6.0 bar
Maximum oil pressure (cold start < 20 sec.): ..... 6.5 bar
Maximum oil consumption: ..... 0.1 l/h (0.1 quart/h)

ENGINE INSTRUMENT MARKINGS

The engine data of the TAE 125-02-114 installation to be monitored are integrated in the compact engine instrument CED-125. The ranges of the individual engine monitoring parameters are shown in the following table.

◆ Note: "Load" describes the available percentage of maximum engine power.

Instrument AED/CED		Red range	Amber range	Green range	Amber range	Red range
Tachometer	[RPM]	--------	--------	0-2300	--------	> 2300
Oil pressure	[bar]	0 - 1.1	1.2 - 2.2	2.3 - 5.1	5.2 - 6.5	> 6.5
	[psi]	0 - 16	17.4 - 32	33.4 - 74	75.4 - 87.0	> 87.0
Coolant temperature	[°C]	< -32	-32...+59	60 - 100	101 - 105	> 105
Oil temperature	[°C]	< -32	-32...+49	50 - 129	130 - 140	> 140
Gearbox temperature	[°C]	--------	--------	< 115	115 - 120	> 120
Load	[%]	--------	--------	0 - 100	--------	--------
Fuel Temperature (left and right)	[°C]	< -30	-30...-1	0 - 69	70 - 75	> 75
Alternator Current	[A]	--------	--------	0 - 84	85 - 90	>90
Electrical System Voltage	[V]	0 - 10	11 - 12.5	12.6 - 14	15	>15

Table 2-3b Markings of the engine instruments

Figure 2-1 AED/CED

◆ Note: The AED/CED caution lamp is switched on if an engine reading is in the amber or red range.
The AED/CED caution lamp remains on even when the parameter returns to the green/normal operating range and must be confirmed by pressing the Confirm/Test knob.
After being confirmed, the AED/CED caution lamp will switch on again whenever another parameter enters amber/red range.
Pressing the Confirm/Test knob longer than one second will initiate the power-up test sequence.

PERMISSIBLE FUEL GRADES

■ CAUTION: Using non-approved fuels and additives can lead to dangerous engine malfunctions.

Fuel:
- JET A-1 (ASTM 1655)
- JET A (ASTM 1655)
- Jet Fuel No.3 (GB 6537-2006)
- JP-8 (MIL-DTL-83133E)
- JP-8+100 (MIL-DTL-83133E)
- TS-1 (GOST 10227-86)
- TS-1 (GSTU 320.00149943.011-99

Alternative:
- Diesel (**DIN** EN 590)
- SASOL GTL Diesel

◆ Note: The liquid fuel additive Biobor JF can be used in jet and diesel fuel systems to eliminate growth of fungi. For further details refer to the manufacturer specifications.

MAXIMUM FUEL QUANTITIES

Due to the higher specific density of Kerosene and Diesel in comparison to Aviation Gasoline (AVGAS) the permissible tank capacity has been reduced.

Fuel Capacitiy				
	Tanks	Total Capacity	Total Unusable Fuel	Total Usable Fuel
F,G,H	2 Standard-Tanks: each 62.85 l (16.60 US gal)	125.7 l (33.2 US gal)	11.4 l (3 US gal)	114.3 l (30.2 US gal)
I,K,L,M	2 Standard-Tanks each 68.55 l (17.85 US gal)	135.1 l (35.7 US gal)	15.1 l (4 US gal)	120.0 l (31.7 US gal)
F-M	2 Long-Range-Tanks: each 83.65 l (22.10 US gal)	167.3 l (44.2 US gal)	15.1 l (4 US gal)	152.2 l (40.2 US gal)

■ CAUTION: To prevent air from penetrating into the fuel system avoid running one tank dry. As soon as the "Low Level" caution light illuminates, switch to the tank with sufficient fuel or land as soon as possible.

■ CAUTION: With ¼ tank or less, prolonged uncoordinated flight is prohibited when operating on either left or right tank.

■ CAUTION: In turbulent air it is strongly recommended to use the BOTH position.

◆ Note: The tanks are equipped with a low fuel sensor. If the fuel level is below 10 l (2.6 US gal) usable fuel per tank, the "Fuel L" or "Fuel R" Caution light illuminates respectively.

PERMISSIBLE OIL TYPES

Engine oil: ..... AeroShell Oil Diesel Ultra
..... AeroShell Oil Diesel 10W-40
..... Shell Helix Ultra 5W-30
..... Shell Helix Ultra 5W-40

Gearbox oil: ..... Centurion Gearbox Oil N1
..... Shell Spirax S6 ATF ZM
..... Shell Spirax S6 GXME 75W-80, API GL-4
..... Shell Spirax S4 G 75W-90, API GL-4

■ CAUTION: Use approved oil with exact designation only!

PERMISSIBLE COOLING LIQUID

Coolant: ..... Water/Radiator Protection at a ratio of 50:50

Radiator Protection: ..... BASF Glysantin Protect Plus / G48
..... Valvoline/Zerex Glysantin G48
..... Mobil Antifreeze Extra (G48)
..... Comma Xstream Green - Concentrate/G4 48

PLACARDS

Near the fuel tank caps:

Cessna 172 F,G,H (With standard tanks):

JET FUEL ONLY
JET A-1/ DIESEL
CAP. 57.2 LITER (15.1 U.S. GAL.)
USABLE TO BOTTOM OF FILLER INDICATOR TAB

Cessna 172 I,K,L,M (With standard tanks):

JET FUEL ONLY
JET A-1/ DIESEL
CAP. 60.0 LITER (15.85 U.S. GAL.)
USABLE TO BOTTOM OF FILLER INDICATOR TAB

Cessna 172 F-M (With long-range tanks):

JET FUEL ONLY
JET A-1/ DIESEL
CAP. 76.1 LITER (20.1 U.S. GAL.)
USABLE TO BOTTOM OF FILLER INDICATOR TAB

At the fuel selector valve:

With standard tanks F,G,H:

Left and Right position: 57.2 Ltr./ 15.1 gal
Both position: 114.4 Ltr./ 30.2 gal

With standard tanks I,K,L,M:

Left and Right position: 60 Ltr./ 15.85 gal
Both position: 120 Ltr./ 31.7 gal

With long-range tanks F-M:

Left and Right position: 76.1 Ltr./ 20.1 gal
Both position: 152.2 Ltr./ 40.2 gal

On the oil funnel or at the flap of the engine cowling:

"Oil, see POH supplement"

If installed, at the flap of the engine cowling to the External Power Receptacle:

"ATTENTION 12 V DC OBSERVE CORRECT POLARITY"

All further placards contained in this section of the EASA-approved POH remain valid.

SECTION 3
EMERGENCY PROCEDURES

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GENERAL

In addition to the original AFM/POH, the following applies:

▲ **WARNING:** Due to failures indicated by the FADEC warning lights there might be a loss propeller valve current which leads in a low pitch setting of the propeller. This might result in overspeed.
Airspeeds below 100 KIAS/ 115 mph are suitable to avoid overspeed in failure case.
If the propeller speed control fails, climbs can be performed at 65 KIAS/ 75 mph and a powersetting of 100%.

EMERGENCY PROCEDURES CHECK LIST

ENGINE MALFUNCTION

DURING TAKE-OFF (WITH SUFFICENT RUNWAY AHEAD)

(1) Thrust Lever - IDLE
(2) Brakes - APPLY
(3) Wing flaps (if extended) - RETRACT to increase the braking effect on the runway
(4) Engine Master - OFF
(5) Alternator, Main Bus and Battery switch - OFF

IMMEDIATELY AFTER TAKE-OFF

If there is an engine malfunction after take-off, at first lower the nose to keep the airspeed and attain gliding attitude. In most cases, landing should be executed straight ahead with only small corrections in direction to avoid obstacles.

▲ **WARNING:** Altitude and airspeed are seldom sufficient for a return to the airfield with a 180° turn while gliding.

(1) Airspeed..... 65 KIAS/ 75 mph (wing flaps retracted)
.....60 KIAS/ 69 mph (wing flaps extended)
(2) Fuel Shut-off Valve - CLOSED
(3) Engine Master - OFF
(4) Wing flaps - as required (Full down recommended)
(5) Alternator, Main Bus and Battery switch - OFF

DURING FLIGHT

◆ Note: Running a tank dry activates both FADEC warning lights flashing.

In case that one fuel tank was flown empty, at the first signs of insufficient fuel feed proceed as follows:

(1) Fuel Shut-off Valve - OPEN (push full in)
(2) Immediately switch the Fuel Selector to BOTH position
(3) Electric Fuel Pump - ON
(4) Check the engine (engine parameters, airspeed/altitude change, whether the engine responds to changes in the Thrust Lever position).
(5) If the engine acts normally, continue the flight and land as soon as possible.

▲ **WARNING:** The high-pressure pump must be checked by an authorized service center before the next flight.

RESTART AFTER ENGINE FAILURE

Whilst gliding to a suitable landing strip, try to determine the reason for the engine malfunction. If time permits and a restart of the engine is possible, proceed as follows:

(1) Airspeed between 65 and 85 KIAS/ 75 and 98 mph
(2) Glide below 13,000 ft
(3) Fuel Shut-off Valve - OPEN (push full in)
(4) Fuel Selector switch to BOTH position
(5) Electric Fuel Pump - ON
(6) Thrust Lever - IDLE
(7) Engine Master OFF and then ON
(if the propeller does not turn, then additionally Starter ON)

◆ Note: The propeller will normally continue to turn as long as the airspeed is above 65 KIAS/ 75 mph. Should the propeller stop at an airspeed of more than 65 KIAS/ 127 mph or more, the reason for this should be found out before attempting a restart.
If it is obvious that the engine or propeller is blocked, do not use the Starter.

◆ Note: If the Engine Master is in position OFF, the Load Display shows no value even if the propeller is turning.

(8) Check the engine power: Thrust lever 100%, engine parameters, check altitude and airspeed.

FADEC WARNING

◆ Note: The FADEC consists of two components that are independent of each other: FADEC A and FADEC B. In case of malfunctions in the active FADEC, it automatically switches to the other.

a) One FADEC Light is flashing

1. Press FADEC test knob/switch at least 2 seconds
2. FADEC warning extinguished (LOW category warning):
   a) Continue flight normally
   b) Inform service center after landing
3. Steady FADEC Light (HIGH category warning)
   a) Observe the other FADEC light
   b) Land as soon as possible
   c) Select an airspeed to avoid engine overspeed
   d) Inform service center after landing

b) Both FADEC Lights are flashing

◆ Note: CED load display should be considered unreliable with both FADEC lights illuminated. Use other indications to assess engine condition.

1. Press FADEC test knob at least 2 seconds
2. FADEC Lights extinguished (LOW category warning):
   a) Continue flight normally
   b) Inform service center after landing
3. Steady FADEC Lights/warnings (HIGH category warning):
   a) Check the available engine power
   b) Expect engine failure
   c) Flight can be continued, however the pilot should
      i) Select an appropriate airspeed to avoid engine overspeed.
      ii) Land as soon as possible
      iii) Be prepared for an emergency landing
   d) Inform service center after landing
4. In case a fuel tank was flown empty, proceed at the first signs of insufficient fuel feed as follows:
   a) Immediately switch the Fuel Selector to BOTH
   b) Electric Fuel Pump - ON
   c) Select an airspeed to avoid engine overspeed.
   d) Check the engine (engine parameters, airspeed/altitude change, whether the engine responds to changes in the Thrust Lever position).
   e) If the engine acts normally, continue the flight and land as soon as possible.

▲ **WARNING:** The high-pressure pump must be checked by an authorized service center before the next flight.

ABNORMAL ENGINE BEHAVIOR

If the engine acts abnormal during flight and the system does not automatically switch to the B-FADEC, it is possible switch to the B-FADEC manually.

▲ **WARNING:** It is only possible to switch from the automatic position to B-FADEC (A-FADEC is active in normal operation, B-FADEC is active in case of malfunction). This only becomes necessary when no automatic switching occurred in case of abnormal engine behavior.

(1) Select an appropriate airspeed to avoid engine overspeed.

▲ **WARNING:** When operating on FADEC backup battery only, the "Force B" switch MUST not be activated. This will shut down the engine.

(2) "FORCE-B" switch to B-FADEC

(3) Flight may be continued, but the pilot should:

i) Select an airspeed to avoid engine overspeed

ii) Land as soon as possible

iii) Be prepared for an emergency landing

FIRES

ENGINE FIRE WHEN STARTING ENGINE ON GROUND

(1) Engine Master - OFF
(2) Fuel Shut-off Valve - CLOSED
(3) Electric Fuel Pump - OFF
(4) Battery Switch - OFF
(5) Extinguish the flames with a fire extinguisher, wool blankets or sand.
(6) Inform service center after landing for examination of fire damages.

ENGINE FIRE DURING TAKE-OFF (ON GROUND)

(1) Engine Master - OFF
(2) Fuel Selector - CLOSED
(3) Electric Fuel Pump - OFF
(4) Battery switch - OFF
(5) Extinguish the flames with a fire extinguisher, wool blankets or sand.
(6) Inform service center after landing for examination of fire damages.

ENGINE FIRE IN FLIGHT

(1) Engine Master - OFF
(2) Fuel Shut-off Valve - CLOSED
(3) Select an airspeed to avoid engine overspeed
(4) Electric Fuel Pump - OFF
(5) Main Bus Switch - OFF
(6) Cabin heat and ventilation OFF resp. CLOSE (except the fresh air nozzles on the ceiling)
(7) Perform emergency landing (as described in the procedure "Emergency Landing With Engine Out")

ELECTRICAL FIRE IN FLIGHT

The first sign of an electrical fire is an unmistakable sharp, acrid smell. As the fire grows, electrical load might be higher than normal or circuit breakers start to trip. In this event proceed as follows:

(1) Main Bus switch - OFF
(2) Avionics Master - OFF
(3) Fresh air nozzles, Cabin Heat and Ventilation - OFF (closed)
(4) Fire Extinguisher - Activate (if available)
(5) All electrical consumers - Switch OFF, leave Alternator, battery and Engine Master ON

▲ **WARNING:** After the fire extinguisher has been used, make sure that the fire is extinguished before exterior air is used to remove smoke from the cabin.

(6) If there is evidence of continued electrical fire, consider turning off battery and alternator.

▲ **WARNING:** If both alternator and main battery are turned OFF, continued engine operation is dependent on the remaining capacity of the FADEC backup battery. The engine has been demonstrated to continue operating for a maximum of 30 minutes when powered by the FADEC backup battery only.

(7) Fresh Air Nozzles, Cabin Heat and Ventilation - ON (open)
(8) Check Circuit Breakers, do not reset if open

If the fire has been extinguished:

(9) Main Bus - ON
(10) Avionics Master - ON

▲ **WARNING:** Turn on electrical equipment required to continue flight depending on the situation and land as soon as possible. Switch circuit breakers switch ON one at a time, with delay after each.

ENGINE SHUT DOWN IN FLIGHT

If it is necessary to shut down the engine in flight (for instance, abnormal engine behavior does not allow continued flight or there is a fuel leak, etc.), proceed as follows:

(1) Select an airspeed to avoid engine overspeed (best glide recommended)
(2) Engine Master - OFF
(3) Fuel Shut-off Valve - CLOSED
(4) Electric Fuel Pump - OFF
(5) If the propeller also has to be stopped (for instance, due to excessive vibrations)
    i) Reduce airspeed below 55 KIAS
    ii) When the propeller is stopped, continue to glide at 65 KIAS

EMERGENCY LANDING

EMERGENCY LANDING WITH ENGINE OUT

If all attempts to restart the engine fail and an emergency landing is immanent, select suitable site and proceed as follows:

(1) Airspeed
    i) 65 KIAS (flaps retracted)
    ii) 60 KIAS (flaps extended)
(2) Fuel Shut-off Valve - CLOSED,
(3) Engine Master - OFF
(4) Wing Flaps - as required (Full down recommended)
(5) Alternator, Main Bus and Battery switch - OFF
(6) Cabin Doors - unlock before touch-down
(7) Touch-down - slightly nose up attitude
(8) Brake firmly

◆ Note: Gliding Distance. Refer to Figure 5-5 "Maximum Glide" Section 5 of this supplement.

FLIGHT IN ICING CONDITIONS

▲ **WARNING:** It is prohibited to fly in known icing conditions.

In case of inadvertent icing encounter proceed as follows:

(1) Pitot Heat switch - ON (if installed)
(2) Turn back or change the altitude to obtain an outside air temperature that is less conducive to icing.
(3) Pull the cabin heat control full out and open defroster outlets to obtain maximum windshield defroster airflow. Adjust cabin air control to get maximum defroster heat and airflow.
(4) Advance the Thrust Lever to increase the propeller speed and keep ice accumulation on the propeller blades as low as possible.
(5) Watch for signs of air filter icing and pull the "Alternate Air Door" control if necessary. An unexplaned loss in engine power could be caused by ice blocking the air intake filter. Opening the "Alternate Air Door" allows preheated air from the engine compartment to be aspirated.
(6) Plan a landing at the nearest airfield. With an extremely rapid ice build up, select a suitable "off airfield" landing site.
(7) With an ice accumulation of 0.5 cm or more on the wing leading edges, a significantly higher stall speed should be expected.
(8) Leave wing flaps retracted. With a severe ice build up on the horizontal tail, the change in wing wake airflow direction caused by wing flap extension could result in a loss of elevator effectiveness.
(9) Perform a landing approach using a forward slip, if necessary, for improved visibility.
(10) Approach at 65 to 75 KIAS/ 75 to 85 mph depending upon the amount of the accumulation.
(11) Perform a landing in level attitude.

RECOVERY FROM SPIRAL DIVE

If a spiral is encountered in the clouds, proceed as follows:

(1) Retard Thrust Lever to idle position
(2) Stop the turn by using coordinated aileron and rudder control to align the symbolic airplane in the turn coordinator with the horizontal reference line.
(3) Cautiously apply elevator back pressure to slowly reduce the airspeed to 70 KIAS/ 80 mph.
(4) Adjust the elevator trim control to maintain an 70 KIAS/ 80 mphglide.
(5) Keep hands off the control wheel, using rudder control to hold a straight heading.
(6) Readjust the rudder trim (if installed) to relieve the rudder of asymmetric forces.
(7) Clear the engine occasionally, but avoid using enough power to disturb the trimmed glide.
(8) Upon breaking out of clouds, resume normal cruising flight and continue the flight.

ELECTRICAL POWER SUPPLY SYSTEM MALFUNCTIONS

▲ **WARNING:** If the power supply from both alternator and main battery is interrupted, continued engine operation is dependent on the remaining capacity of the FADEC backup battery. The engine has been demonstrated to continue operating for a maximum of 30 minutes when powered by the FADEC backup battery only. In this case, all electrical equipment will not operate:

- land as soon as possible
- do not switch the FORCE-B switch, this will shut down the engine

■ CAUTION:	The TAE 125-02-114 requires an electrical power source for its operation. If the alternator fails, continued engine operation time is dependent upon the remaining capacity of the main battery, the FADEC backup battery and equipment powered. The engine has been demonstrated to continue operating for approximately 120 minutes based upon the following assumptions:

■ CAUTION:	This table only gives a reference point. The pilot should turn off all nonessential items and supply power only to equipment which is absolutely necessary for continued flight depending upon the situation. Deviating from this recommendation, the remaining engine operating time may change.

Equipment		Time switched on	
		in [min]	in [%]
NAV/COM 1 receiving	ON	120	100
NAV/COM 1 transmitting	ON	12	10
NAV/COM 2 receiving	OFF	0	0
NAV/COM 2 transmitting	OFF	0	0
GPS	ON	60	50
Transponder	ON	120	100
Fuel Pump	OFF	0	0
AED-125	ON	120	100
Battery	ON	120	100
CED-125	ON	120	100
Landing Light	ON	12	10
Flood Light	ON	1.2	1
Pitot Heat	ON	24	20
Wing Flaps	ON	1.2	1
Interior Lighting	OFF	0	0
Nav Lights	OFF	0	0
Beacon	OFF	0	0
Strobes	OFF	0	0
ADF	OFF	0	0
Intercom	OFF	0	0
Engine Control	ON	120	100

Table 3-1a

Emergency Procedures Check List

ALTERNATOR WARNING DURING NORMAL ENGINE OPERATION

(1) Ammeter - CHECK
(2) Alternator switch CHECK - ON
(3) Battery Switch CHECK - ON

■ CAUTION: If the FADEC was supplied by battery only until this point, the RPM can momentarily drop, when the alternator will be switched on. In any case: leave the alternator switched ON!

(4) Electrical load - REDUCE IMMEDIATELY as follows:
   i) NAV / COM 2 – OFF
   ii) Fuel Pump – OFF
   iii) Landing Light – OFF (use as required for landing)
   iv) Taxi Light – OFF
   v) Strobe Light – OFF
   vi) Nav Lights – OFF
   vii) Beacon – OFF
   viii) Interior Lights – OFF
   ix) Intercom – OFF
   x) Pitot Heat – OFF (use as required)
   xi) Autopilot – OFF
   xii) Non-essential equipment – OFF
(5) The pilot should:
   i) Land as soon as possible.
   ii) Be prepared for an emergency landing.
   iii) Expect an engine failure.

AMMETER SHOWS BATTERY DISCHARGE DURING NORMAL ENGINE OPERATION FOR MORE THAN 5 MINUTES

◆ Note: When the AED 125 Ammeter indication decreases to the far left and the voltage indication is decreasing simultaneously, the battery is being discharged.

(1) Alternator switch CHECK - ON
(2) Battery Switch CHECK - ON

■ CAUTION: If the FADEC was supplied by battery only until this point, the RPM can momentarily drop, when the alternator will be switched on. In any case: leave the alternator switched ON!

(3) Electrical load - REDUCE IMMEDIATELY as follows:
- i) NAV / COM 2 – OFF
- ii) Fuel Pump – OFF
- iii) Landing Light – OFF (use as required for landing)
- iv) Taxi Light – OFF
- v) Strobe Light – OFF
- vi) Nav Lights – OFF
- vii) Beacon – OFF
- viii) Interior Lights – OFF
- ix) Intercom – OFF
- x) Pitot Heat – OFF (use as required)
- xi) Autopilot – OFF
- xii) Non-essential equipment – OFF

(4) The pilot should:
- i) Land as soon as possible
- ii) Be prepared for an emergency landing
- iii) Expect an engine failure

TOTAL ELECTRICAL FAILURE
(all equipment inoperative, except engine)

▲ **WARNING:** If the power supply from both alternator and main battery is interrupted simultaneously, continued engine operation is dependent on the remaining capacity of the FADEC backup battery. The engine has been demonstrated to continue operating for a maximum of 30 minutes when powered by the FADEC backup battery only. In this case, all other electrical equipment will not operate.

▲ **WARNING:** If the aircraft was operated on battery power only until this point (alternator warning light illuminated), the remaining engine operating time may be less than 30 minutes.

▲ **WARNING:** Do not activate the FORCE-B switch, this will shut down the engine.

(1) Alternator switch CHECK - ON
(2) Battery Switch CHECK – ON
(3) Land as soon as possible
    i) Be prepared for an emergency landing
    ii) Expect an engine failure

ROUGH ENGINE OPERATION OR LOSS OF POWER

DECREASE IN POWER

(1) Push Thrust Lever full forward (Take-off position)
(2) Fuel Selector to BOTH Position.
(3) Electric Fuel Pump - ON
(4) Reduce airspeed to 65-85 KIAS/ 75-98 mph (best glide recommended), [max. 100 KIAS)
(5) Check engine parameters (FADEC lights, oil pressure and temperature, fuel quantity)

If normal engine power is not achieved, the pilot should:

i) Land as soon as possible
ii) Be prepared for an emergency landing
iii) Expect an engine failure

▲ **WARNING:** The high pressure pump must be checked before the next flight.

ICE FORMATION IN THE CARBURETOR

- N/A, since this is a Diesel engine -

SOILED SPARK PLUGS

- N/A, since this is a Diesel engine -

IGNITION MAGNET MALFUNCTIONS

- N/A, since this is a Diesel engine -

OIL PRESSURE TOO LOW (< 2.3 BAR IN CRUISE (AMBER RANGE) OR < 1.2 BAR AT IDLE (RED RANGE)):

(1) Reduce power as quickly as possible
(2) Check oil temperature: If the oil temperature is high or near operating limits,
    i) Land as soon as possible
    ii) Be prepared for an emergency landing
    iii) Expect an engine failure

◆ Note: During warm-weather operation or long climbs at low airspeed engine temperatures could rise into the amber range and trigger the "Caution" light. This warning allows the pilot to avoid overheating of the engine as follows:

(3) Increase the climbing airspeed, reduce angle of climb
(4) Reduce power, if the engine temperatures approach the red range

OIL TEMPERATURE TOO HIGH (RED RANGE):

(1) Increase airspeed and reduce power as quickly as possible
(2) Check oil pressure: if the oil pressure is lower than normal (< 2.3 bar in cruise or < 1.2 bar at idle),
    i) Land as soon as possible
    ii) Be prepared for an emergency landing
    iii) Expect an engine failure
(3) If the oil pressure is in the normal range:
    i) Land as soon as possible

COOLANT TEMPERATURE TOO HIGH (RED RANGE):

(1) Increase airspeed and reduce power as quickly as possible
(2) Cabin Heat - COLD (push Lever "Cabin Heat")
(3) If coolant temperature reduces rapidly to normal range, continue to fly normally and monitor coolant temperature, Cabin Heat.
(4) If coolant temperature does not decrease,
    i) Land as soon as possible
    ii) Be prepared for an emergency landing
    iii) Expect an engine failure

"WATER LEVEL" LIGHT ILLUMINATES/Caution COOL LVL

(1) Increase airspeed and reduce power as quickly as possible
(2) Coolant temperature "CT" check and observe
(3) Oil temperature "OT" check and observe
(4) If coolant temperature and/or oil temperature are rising into amber or red range,
    i) Land as soon as possible
    ii) Be prepared for an emergency landing
    iii) Expect an engine failure

GEARBOX TEMPERATURE TOO HIGH (RED RANGE):

(1) Reduce power to 55% - 75% as quickly as possible
(2) Land as soon as possible

FUEL TEMPERATURE TOO HIGH:

(1) Switch to fuel tank with lower fuel temperature, if this contains sufficient fuel
(2) Reduce engine power, if possible
(3) If fuel temperature remains in Red Range, land as soon as possible

FUEL TEMPERATURE TOO LOW (AMBER RANGE for Diesel Operation, RED RANGE for Kerosine Operation):

(1) Switch to fuel tank with higher fuel temperature, if this contains sufficient fuel
(2) Change to altitude with higher outside air temperature
(3) If use of the non-active tank is intended, switch fuel selector to BOTH

◆ Note: Low fuel temperature may be caused when flying in cold weather with fuel cooler in operation (baffle removed).

PROPELLER RPM TOO HIGH:

With propeller RPM between 2,400 and 2,500 for more than 10 seconds or over 2,500:

(1) Reduce power
(2) Reduce airspeed below 100 KIAS or as appropriate to prevent propeller overspeed
(3) Set power as required to maintain altitude and land as soon as possible.

◆ Note: If the propeller speed control fails, climbs be performed at 65 KIAS/ 75 mph and a power setting of 100%.
In case of overspeed the FADEC will reduce the engine power at higher airspeeds to avoid propeller speeds above 2500 rpm.

FLUCTUATIONS IN PROPELLER RPM:

If the propeller RPM fluctuates by more than + / - 100 RPM with a constant Thrust Lever position:

(1) Change the power setting and attempt to find a power setting where the propeller RPM no longer fluctuates.
(2) If this does not work, set the maximum power at an airspeed < 100 KIAS until the propeller speed stabilizes.
(3) If the problem is resolved, continue the flight
(4) If the problem continues, select a power setting where the propeller RPM fluctuations are minimum. Fly at an airspeed below 100 KIAS and land as soon as possible.

SECTION 4
NORMAL PROCEDURES

PREFLIGHT INSPECTION

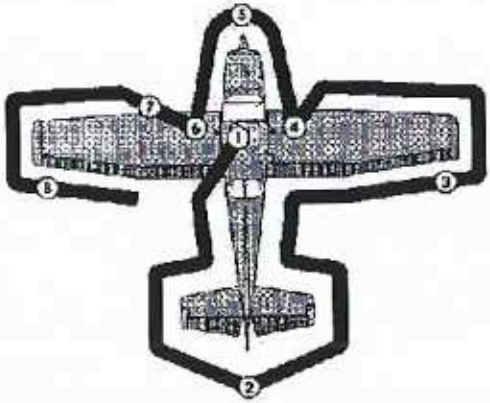

Figure 4-1a Preflight Inspection

◆ Note: Visually check airplane for general condition during walk around inspection. In cold weather, remove even small accumulations of frost, ice or snow from wing, tail and control surfaces. Also, make sure that control surfaces contain no internal accumulations of ice or debris. Prior to flight, check that pitot heater (if installed) is warm to touch within 30 seconds with battery and pitot heat switches on. If a night flight is planned, check operation of all lights, and make sure a flashlight is available.

(3) RIGHT WING Trailing Edge

(1) Aileron - CHECK freedom of movement and security

(2) Flap - CHECK for security and condition

(4) RIGHT WING

(1) Wing Tie-Down - DISCONNECT

(2) Main Wheel Tire - CHECK for proper inflation and general condition (weather checks, tread depth and wear, etc.).

▲ **WARNING:** If, after repeated sampling, evidence of contamination still exists, the airplane should not be flown. Tanks should be drained and system purged by qualified maintenance personnel. All evidence of contamination must be removed before further flight.

(3) Fuel Tank Sump Quick Drain Valves - DRAIN at least a cupful of fuel (using sampler cup) from each sump location to check for water, sediment and the right type of fuel (Diesel or JET-A1) before each flight and after each refueling. If water is observed, take further samples until clear and then gently rock wings and lower tail to the ground to move any additional contaminants to the sampling points. Take repeated samples from all fuel drain points until all contamination has been removed. If contaminants are still present, refer to above WARNING and do not fly airplane.

(4) Fuel Quantity - CHECK VISUALLY for desired level not above marking in fuel filler

(5) Fuel Filler Cap - SECURE

(1) CABIN

(1) Pilot´s Operating Handbook - AVAILABLE IN THE AIRPLANE
(2) Airplane Weight and Balance - CHECKED
(3) Parking Brake - SET
(4) Control Wheel Lock - REMOVE
(5) "Engine Master" - OFF
(6) Avionics Power Switch - OFF
(7) "Shut-off Cabin Heat" - OFF (Push Full Forward

▲ **WARNING:** When turning on the Battery switch, using an external power source, or pulling the propeller through by hand, treat the propeller as if the Engine Master was on.

(8) Battery and Main Bus switches - ON
(9) Fuel Quantity Indicators and Fuel Temperature CHECK
(10) Light "Water Level" - CHECK OFF
(11) Battery and Main Bus switches - OFF
(12) Entry in log-book concerning type of fuel filled - CHECK
(13) Static Pressure Alternate Source Valve - CHECK
(14) Fuel Selector Valve - BOTH
(15) Fuel Shut-off Valve - ON (Push Full In)
(16) Baggage Door - CHECK, lock with key if the child's seat is supposed to be occupied

(2) EMPENNAGE

(1) Rudder Gust Lock (if attached) - REMOVE
(2) Tail Tie-Down - DISCONNECT
(3) Control Surfaces - CHECK freedom of movement and security

(8) Nose Wheel Strut and Tire - CHECK for proper inflation of strut and general condition (weather checks, tread depth and wear, etc.) of tire

(9) Left Static Source Opening - CHECK for blockage

(10) Fuel cooler baffle - CHECK

- REMOVE, if OAT on ground in higher than 20°C (68°F)
- INSTALL, if OAT on ground in lower than 20°C (68°F)

(6) LEFT WING

(1) Fuel Quantity - CHECK VISUALLY for desired level not above marking in fuel filler

(2) Fuel Filler Cap - SECURE

(3) Fuel Tank Sump Quick Drain Valves - DRAIN at least a cupful of fuel (using sampler cup) from each sump location to check for water, sediment and the right type of fuel (Diesel or JET-A1) before each flight and after each refueling. If water is observed, take further samples until clear and then gently rock wings and lower tail to the ground to move any additional contaminants to the sampling points. Take repeated samples from all fuel drain points until all contamination has been removed. If contaminants are still present, refer to previous WARNING (see right wing) and do not fly airplane.

(4) Main Wheel Tire - CHECK for proper inflation and general condition (weather checks, tread depth and wear, etc.)

(7) LEFT WING Leading Edge

(1) Pitot Tube Cover (if mounted) - REMOVE and CHECK for pitot blockage

(2) Fuel Tank Vent Opening - CHECK for blockage

(3) Stall Warning Opening - CHECK for blockage

(4) Wing Tie-Down - DISCONNECT

(5) NOSE

(1) Reservoir-tank Quick Drain Valve - DRAIN at least a cupful of fuel (using sampler cup) from valve to check for water, sediment and proper fuel grade (Diesel or JET-A1) before each flight and after each refueling. If water is observed, take further samples until clear and then gently rock wings and lower tail to the ground to move any additional contaminants to the sampling point. Take repeated samples until all contamination has been removed.

◆ Note: The reservoir tank drain is located in the fuselage on the co-pilot side of the aircraft.

(2) Before first flight of the day and after each refueling - DRAIN the Fuel Strainer Quick Drain Valve with the sampler cup to remove water and sediment from the screen. Ensure that the screen drain is properly closed again. If water is discovered, there might be even more water in the fuel system. Therefore, take further samples from Fuel Strainer and the Tank Sumps.

◆ Note: The fuel strainer drain is located on the left-hand side of the firewall (flight direction).

(3) Engine Oil Dipstick/Filler Cap:
   a) Oil level - CHECK
   b) Dipstick/filler cap - SECURE

   Do not operate below the minimum dipstick indication.

(4) Engine Air and Cooling Inlets - CLEAR of obstructions.

(5) Landing Light - CHECK for condition and cleanliness

(6) Propeller and Spinner - CHECK for nicks and security.

(7) Gearbox Oil Level - CHECK the oil has to cover at least half of the inspection glass

PROCEDURES UP TO 5500ft AIRFIELD ELEVATION

STARTING ENGINE

▲ **WARNING:** Do not use ground power unit for engine starts. It is not allowed to start up the engine using external power. If starting the engine is not possible using battery power, the condition of the battery must be verified before flight.

(1) Electric Fuel Pump - ON
(2) Navigation Lights and Flashing Beacon - ON (as required).
(3) Thrust Lever - IDLE
(4) Area Aircraft / Propeller - CLEAR
(5) "Engine Master" - ON , wait until the Glow Control light extinguishes
(6) Starter - ON, keep starter engaged until min. 500rpm
Release when engine starts, leave Thrust Lever in idle

■ CAUTION: Do not overheat the starter motor. Do not operate the starter motor for more than 10 seconds. After operating the starter motor, let it cool off for 20 seconds. After 6 attempts to start the engine, let the starter cool off for half an hour.

(7) Oil Pressure - CHECK

■ CAUTION: If after 3 seconds the minimum oil pressure of 1 bar is not indicated:
shut down the engine immediately!

(8) CED - Test Knob - PRESS (to delete Caution light)
(9) Ammeter - CHECK for positive charging current
(10) Voltmeter - CHECK for green range

(8) LEFT WING Trailing Edge

(1) Aileron - CHECK freedom of movement and security

(2) Flap - Check for security and conditions

BEFORE STARTING ENGINE

(1) Preflight Inspection - COMPLETE

(2) Seats and Seat Belts - ADJUST and LOCK

(3) Brakes - TEST and SET

(4) Avionics Power Switch, Autopilot (if installed) and Electrical Equipment - OFF

■ CAUTION: The Avionics Power Switch must be off during engine start to prevent possible damage to avionics.

(5) Circuit Breakers - CHECK IN

(6) Switch Alternator - CHECK ON

(7) Battery and Main Bus Switches - ON

■ CAUTION: The electronic engine control needs an electrical power source for its operation. For normal operation Battery, Alternator and Main Bus have to be switched on. Separate switching is only allowed for tests and in the event of emergencies.

(8) Fuel Quantity and Temperature - CHECK

(9) Fuel Selector Valve - SET to BOTH position. The fuel temperature limitations must be observed.

(10) Fuel Shut-off Valve - OPEN (Push Full In)

(11) Alternate Air Door - CLOSED

(12) Thrust Lever - CHECK for freedom of movement

(13) Load Display - CHECK 0% at Propeller RPM 0

(8) FADEC and propeller adjustment function check:
   a) Thrust Lever - IDLE (both FADEC lights should be OFF)
   b) FADEC Test Button - PRESS and HOLD button for entire test
   c) Both FADEC lights - ON, RPM increases.

▲ **WARNING:** If the FADEC lights do not come on at this point, it means that the test procedure has failed and take off should not be attempted.

   d) The FADEC automatically switches to B-component (only FADEC B light is ON)
   e) The propeller control is excited, RPM decreases
   f) The FADEC automatically switches to channel A (only FADEC A light is ON), RPM increases
   g) The propeller control is excited, RPM decreases
   h) FADEC A light goes OFF, idle RPM is reached, the test is completed.
   i) FADEC Test Button - RELEASE

(9) Force B Switch - switch to FADEC B

(10) Engine - check running without a change

(11) Force B Switch - switch back to Automatic

▲ **WARNING:** If there are prolonged engine misfires or the engine shuts down during the test, take off may not be attempted.

▲ **WARNING:** The whole test procedure has to be performed without any failure. In case the engine shuts down or the FADEC lights are flashing, take off is prohibited. This applies even if the engine seems to run without failure after the test.

◆ Note: If the test button is released before the self test is over, the FADEC immediately switches over to normal operation.

◆ Note: While switching from one FADEC to another, it is normal to hear and feel a momentary surge in the engine.

(11) FADEC Backup Battery test
   a) Alternator - OFF, engine must operate normally
   b) Battery - OFF, for min. 10 seconds;
      engine must operate normally, the red FADEC lamps must not be illuminated
   c) Battery - ON
   d) Alternator - ON

▲ **WARNING:** It must be ensured that both battery and alternator are ON!
If the guarded alternator switch is installed, the switch guard must be closed.

(12) Master Switch - ON
(13) Radios- ON
(14) Ammeter – Check positive charge, alternator warning light must be OFF
(15) Voltmeter – Check in green range
(16) Electric Fuel Pump - OFF
(17) Flaps - RETRACT

WARM UP

(1) Let the engine warm up about 2 minutes at 890 RPM.
(2) Increase RPM to 1,400 until Oil Temperature 50°C (122°F), Coolant Temperature 60°C (140°F).

BEFORE TAKE-OFF

(1) Parking Brake - SET
(2) Cabin Doors and Windows - CLOSED and LOCKED
(3) Flight Controls - FREE and CORRECT
(4) Flight Instruments - CHECK and SET
(5) Fuel quantity - CHECK
(6) Fuel Selector Valve - SET to BOTH
(7) Elevator Trim and Rudder Trim (if installed) - SET for Takeoff

PROCEDURES OVER 5500ft AIRFIELD ELEVATION

◆ Note: Due to the increase of the idle speed with increasing pressure altitudes, the FADEC test is only possible to a limited extent from an airfield elevation of approximately 5500ft.
Over 5500ft, the FADEC test is only possible if the load selector lever remains in the idle position after engine start until the FADEC test is starting.
If the load selector lever is moved from the idle position, a FADEC test is no longer possible at pressure altitudes above 5500ft. For this purpose, the engine has to be stopped and re-started to perform the FADEC test.

STARTING ENGINE

▲ **WARNING:** Do not use ground power unit for engine starts. It is not allowed to start up the engine using external power. If starting the engine is not possible using battery power, the condition of the battery must be verified before flight.

(1) Electric Fuel Pump - ON
(2) Navigation Lights and Flashing Beacon - ON (as required)
(3) Thrust Lever - IDLE
(4) Area Aircraft / Propeller - CLEAR
(5) "Engine Master" - ON, wait until the Glow Control light extinguishes
(6) Starter - ON, keep starter engaged until min. 500rpm
Release when engine starts, leave Thrust Lever in idle

■ CAUTION: Do not overheat the starter motor. Do not operate the starter motor for more than 10 seconds. After operating the starter motor, let it cool off for 20 seconds. After 6 attempts to start the engine, let the starter cool off for half an hour.

(12) Thrust Lever - FULL FORWARD, load display min. 94%, RPM 2240 - 2300
(13) Thrust Lever - IDLE
(14) Engine Instruments and Ammeter - CHECK
(15) Suction gage - CHECK
(16) Wing Flaps - SET for Take-off (0° or 10°).
(17) Electric Fuel Pump - ON
(18) Strobe Lights - AS DESIRED
(19) Radios and Avionics - ON and SET
(20) Autopilot (if installed) - OFF
(21) Air Conditioning (if installed) - OFF
(22) Thrust Lever Friction Control - ADJUST
(23) Brakes - RELEASE

(8) Ammeter - CHECK for positive charging current
(9) Voltmeter - CHECK for green range
(10) FADEC and propeller adjustment function check:
  a) Thrust Lever - IDLE (both FADEC lights should be OFF).
  b) FADEC Test Button - PRESS and HOLD button for entire test.
  c) Both FADEC lights - ON, RPM increases

▲ **WARNING:** If the FADEC lights do not come on at this point, it means that the test procedure has failed and take off should not be attempted.

  d) The FADEC automatically switches to B-component (only FADEC B light is ON)
  e) The propeller control is excited, RPM decreases
  f) The FADEC automatically switches to channel A (only FADEC A light is ON), RPM increases
  g) The propeller control is excited, RPM decreases
  h) FADEC A light goes OFF, idle RPM is reached, the test is completed.
  i) FADEC Test Button - RELEASE
(11) Force B Switch - switch to FADEC B
(12) Engine - check running without a change
(13) Force B Switch - switch back to Automatic

▲ **WARNING:** If there are prolonged engine misfires or the engine shuts down during the test, take off may not be attempted.

▲ **WARNING:** The whole test procedure has to be performed without any failure. In case the engine shuts down or the FADEC lights are flashing, take off is prohibited. This applies even if the engine seems to run without failure after the test.

(7) Oil Pressure - CHECK

■ CAUTION: If after 3 seconds the minimum oil pressure of 1 bar is not indicated: shut down the engine immediately!

(8) CED-Test Knob - PRESS (to delete Caution light)
(9) Ammeter - CHECK for positive charging current
(10) Voltmeter - CHECK for green range
(11) FADEC Backup Battery test
   a) Alternator - OFF, engine must operate normally
   b) Battery - OFF, for min. 10 seconds; engine must operate normally, the red FADEC lamps must not be illuminated
   c) Battery - ON
   d) Alternator - ON

▲ **WARNING:** It must be ensured that both battery and alternator are ON! If the guarded alternator switch is installed, the switch guard must be closed.

(12) Ammeter – Check positive charge, alternator warning light must be OFF
(13) Voltmeter – Check in green range
(14) Flaps – RETRACT

WARM UP AND FADEC-TEST

(1) Let the engine warm up about 2 minutes at 890 RPM.
(2) Increase RPM to 1,400 until Oil Temperature 50°C, Coolant Temperature 60°C.
(3) Thrust Lever - IDLE
(4) "Engine Master" - OFF
(5) Area Aircraft / Propeller - CLEAR
(6) "Engine Master" - ON, wait until the Glow Control light extinguishes
(7) Starter - ON, keep starter engaged until min. 500rpm Release when engine starts, leave Thrust Lever in idle

TAKE-OFF

NORMAL TAKEOFF

(1) Wing Flaps - 0° or 10°
(2) Thrust Lever - FULL FORWARD
(3) Elevator Control - LIFT NOSE WHEEL at 55 KIAS/ 63 mph.
(4) Climb Speed - 65 to 80 KIAS/ 75 to 92 mph

SHORT FIELD TAKEOFF

(1) Wing Flaps - 10°
(2) Brakes - APPLY
(3) Thrust Lever - FULL FORWARD
(4) Brakes - RELEASE
(5) Airplane Attitude - SLIGHTLY TAIL LOW
(6) Elevator Control - LIFT NOSE WHEEL at 45 KIAS/ 52 mph
(7) Climb Speed - 59 KIAS/ 68 mph (until all obstacles are cleared)

AFTER TAKEOFF

(1) Altitude about 300 ft, Airspeed more than 65 KIAS/ 75 mph - Wing Flaps - RETRACT
(2) Electric Fuel Pump - OFF

◆ Note: If the test button is released before the self test is over, the FADEC immediately switches over to normal operation.

◆ Note: While switching from one FADEC to another, it is normal to hear and feel a momentary surge in the engine.

(14) Avionic Switch – ON
(15) Radios – ON
(16) Electric Fuel Pump – OFF

BEFORE TAKE-OFF

(1) Parking Brake – SET
(2) Cabin Doors and Windows – CLOSED and LOCKED
(3) Flight Controls – FREE and CORRECT
(4) Flight Instruments – CHECK and SET
(5) Fuel quantity – CHECK
(6) Fuel Selector Valve – SET to BOTH position
(7) Elevator Trim and Rudder Trim (if installed) - SET for Takeoff
(8) Thrust Lever - FULL FORWARD, load display min. 94%, RPM 2240 - 2300
(9) Thrust Lever - IDLE
(10) Engine Instruments and Ammeter - CHECK
(11) Suction gage - CHECK
(12) Wing Flaps - SET for Take-off (0° or 10°).
(13) Electric Fuel Pump - ON
(14) Strobe Lights - AS DESIRED
(15) Radios and Avionics - ON and SET
(16) Autopilot (if installed) - OFF
(17) Air Conditioning (if installed) - OFF
(18) Thrust Lever Friction Control - ADJUST
(19) Brakes - RELEASE

CRUISE

(1) Power - maximum load 100% (maximum continuous power), 75% or less is recommended.
For economic cruise set load 70% or less.

(2) Elevator trim and Rudder trim (if installed) - ADJUST

(3) Compliance with Limits for Oil Pressure, Oil Temperature, Coolant Temperature and Gearbox Temperature (CED 125 and Caution light) - MONITOR closely

(4) Fuel Quantity and Temperature (Display and LOW LEVEL caution lights) - MONITOR.

Whenever possible, the airplance should be flown with the fuel selector in the BOTH position to empty and heat both fuel tanks evenly. However, operation in the LEFT or RIGHT position may be desirable to correct a fuel quantity imbalance or during periods of intentional uncoordinated flight maneuvres. During prolonged operation with the fuel selector in either the LEFT or RIGHT position the fuel balance and temperatures should be closely monitored.

■ CAUTION: Do not use any fuel tank below the minimum permissible fuel temperature!

■ CAUTION: In turbulent air it is strongly recommended to use the BOTH position.

■ CAUTION: With ¼ tank or less prolonged or uncoordinated flight is prohibited when operating on either the left or right tank.

(5) FADEC and Alternator Warning lights - MONITOR

CLIMB

(1) Airspeed - 70 to 85 KIAS.

◆ Note: If a maximum performance climb is necessary, use speeds shown in the "Maximum Rate Of Climb" chart in Section 5. In case that Oil Temperature and/or Coolant Temperature are approaching the upper limit, continue at a lower climb angle for better cooling if possible.

◆ Note: It is recommended to set the fuel selector valve to the BOTH position.The fuel temperatures have to be monitored.

(2) Thrust Lever - FULL FORWARD

BALKED LANDING

(1) Thrust Lever - FULL FORWARD
(2) Wing Flaps - RETRACT TO 20° (immediately after Thrust Lever FULL FORWARD)
(3) Climb Speed - 58 KIAS/ 68 mph
(4) Wing Flaps - 10° (until all obstacles are cleared)
(5) Wing Flaps - RETRACT after reaching a safe altitude and 65 KIAS/ 75 mph

AFTER LANDING

(1) Wing Flaps - RETRACT
(2) Electric Fuel Pump - OFF

SECURING AIRPLANE

(1) Parking Brake - SET
(2) Thrust Lever - IDLE
(3) Avionics Power Switch, Electrical Equipment, Autopilot (if installed) - OFF
(4) Main Bus switch - OFF
(5) "Engine Master" - OFF
(6) Switch Battery - OFF
(7) Control Lock - INSTALL
(8) Fuel Selector Valve - LEFT or RIGHT (to prevent crossfeeding between tanks)

DESCENT

(1) Fuel Selector Valve - SELECT BOTH position
(2) Power - AS DESIRED

BEFORE LANDING

(1) Pilot and Passenger Seat Backs - MOST UPRIGHT POSITION
(2) Seats and Seat Belts - SECURED and LOCKED
(3) Fuel Selector Valve - SELECT BOTH position
(4) Electric Fuel Pump - ON
(5) Landing / Taxi Lights - ON
(6) Autopilot (if installed) - OFF
(7) Air Conditioning (if installed) - OFF

LANDING

NORMAL LANDING

(1) Airspeed - 69 to 80 KIAS/ 80 to 92 mph (wing flaps UP)
(2) Wing Flaps - AS DESIRED (0°-10° below 110 KIAS/ 126 mph; 10°-40° below 85 KIAS/ 98 mph)
(3) Airspeed in Final Approach:
    - wing flaps 20°: 63 KIAS/72 mph
    - wing flaps 40°: 60 KIAS/69 mph
(4) Touchdown - MAIN WHEELS FIRST
(5) Landing Roll - LOWER NOSE WHEEL GENTLY
(6) Brakes - MINIMUM REQUIRED

SHORT FIELD LANDING

(1) Airspeed - 69 to 80 KIAS/ 80 to 92 mph (Flaps UP)
(2) Wing Flaps - 40°
(3) Airspeed in the Final Approach - 60 KIAS/ 69 mph (until flare)
(4) Power - REDUCE to idle after clearing obstacles.
(5) Touchdown - MAIN WHEELS FIRST
(6) Brakes - APPLY HEAVILY
(7) Wing Flaps - RETRACT

TAXIING

When taxiing, it is important that speed and use of brakes be held to a minimum and that all controls be utilized to maintain directional control and balance.
The Alternate Air Door should be always for ground operation to ensure that no unfiltered air is sucked in.
Taxiing over loose gravel or cinders should be done at low engine speed to avoid abrasion and stone damage to the propeller tips.

BEFORE TAKE-OFF

WARM UP

To warm up the engine, operate the engine for about 2 minutes at 890 RPM.
Let the engine run at propeller RPM of 1,400 until it reaches an engine oil temperature of 50°C (green range) and a coolant temperature of 60°C (green range to ensure normal operation).

MAGNETO CHECK

N/A since this is a Diesel engine.

ALTERNATOR CHECK

Prior to flights where verification of proper alternator and alternator control unit operation is essential (such as night and instrument flights), a positive verification can be made by loading the electrical system momentarily (3 to 5 seconds) with the landing light or by operating the wing flaps during the engine runup (20% load). The ammeter will remain within a needle width of zero if the alternator and alternator control unit are operating properly.

AMPLIFIED PROCEDURES

STARTING ENGINE

The TAE 125-02-114 is a direct Diesel injection engine with common-rail technology and a turbocharger. It is controlled automatically by the FADEC, which makes a proper performance of the FADEC test important for safe flight operation.
All information relating to the engine are compiled in the CED 125 multifunction instrument.
Potentiometers within the Thrust Lever transmit the load value selected by the pilot to the FADEC.
If the engine master is switched to ON, the preheating relay is actuated by the FADEC and the glow plugs are supplied with power. The glow duration depends on the engine temperature.
If the engine master is switched to OFF, the injection valves are not supplied with power and remain closed.
The switch/push button "Starter" controls the Starter.

EXTERNAL POWER

External power may be used to charge the battery or for maintenance purposes. To charge the battery with external power the battery switch must be ON.

When using an External Power Source, the Battery Switch must be in the OFF position before connecting the External Power Source to the airplane receptacle.

It is not allowed to start up the engine using external power. If starting the engine is not possible using battery power, the condition of the battery must be verified before flight.

CLIMB

Normal climbs are performed with flaps up and full load and at speeds 5 to 10 knots/ 7 to 12 mph higher than best rate-of-climb speeds for the best combination of engine cooling, climb speed and visibility. The speed for best climb is about 69 KIAS/ 79 mph. If an obstruction dictates the use of a steep climb angle, climb at 62 KIAS and flaps up.

◆ Note: Climbs at low speeds should be of short duration to improve engine cooling.

CRUISE

As guidance for calculation of the optimum altitude and power setting for a given flight use the tables in chapter 5.

LANDING

NORMAL LANDING

Remarks in Pilot´s Operating Handbook concerning carburetor pre-heating are **N/A**

BALKED LANDING

In a balked landing (go around) climb, reduce the flap setting to 20° immediately after full power is applied. If obstacles must be cleared during the go-around climb, reduce wing flap setting to 10° and maintain a safe airspeed until the obstacles are cleared. After clearing any obstacles, the flaps may be retracted as the airplane accelerates to the normal flaps up climb speed.

CARBURETOR ICING

N/A since this is a Diesel engine.

FLIGHT IN HEAVY RAIN

N/A since no special procedures are necessary for heavy rain.

BATTERY CHECK

If there is doubt regarding the battery conditions or functionality the battery has to be checked after warm-up as follows:

Switch off the alternator while the engine is running (battery remains "ON")"

Perform a 10 sec. engine run. The voltmeter must remain in the green range. If not, the battery has to be charged or, if necessary, exchanged.

After this test the alternator has to be switched on again.

TAKE-OFF

POWER CHECK

It is important to check full load engine operation early in the takeoff roll. Any signs of rough engine operation or sluggish engine acceleration is good cause for discontinuing the take-off. If this occurs, you are justified in making a thorough full load static runup before another take-off is attempted.
After full load is applied, adjust the Thrust Lever Friction Control to prevent the Thrust Lever from creeping back from a maximum power position. Similar friction lock adjustments should be made as required in other flight conditions to maintain a fixed Thrust Lever setting.

WING FLAP SETTINGS

Flap deflections greater than 10° are not approved for normal and short field takeoffs. Using 10° wing flaps reduces the ground roll and total distance over a 15 m obstacle by approximately 10%.

◆ Note: It is advisable to refuel before each flight and to enter the type of fuel filled and the additives used in the log-book of the airplane.

If snow or slush covers the take-off surface, allowance must be made for take-off distances which will be increasingly extended as snow or slush depth increases. The depth and consistency of this cover can, in fact, prevent take-off in many instances.

Cold weather starting procedures are the same as the normal starting procedures. Use caution to prevent inadvertent forward movement of the airplane during starting when parked on snow or ice.

HOT WEATHER OPERATION

Engine temperatures may rise into the amber range and activate the "Caution" light when operating in hot weather or longer climbouts at low speed. This indication gives the pilot the opportunity to keep the engine from possibly overheating by doing the following:

i) decrease rate of climb
ii) increase airspeed
iii) reduce power, if the engine temperatures approach the red range

Should the seldom case occur that the fuel temperature is rising into the amber or red range, switch to the other tank or to the BOTH position

COLD WEATHER OPERATION

Special attention should be paid to operation of the aircraft and the fuel system in winter or before any flight at low temperatures. Correct preflight draining of the fuel system is particularly important and will prevent the accumulation of water.
The following limitations for cold weather operation are established due to temperature. "Operating limits".
(Refer also to Section 2 "Limitations")

Fuel	Minimum permissible fuel temperature in the fuel tank before Take-off	Minimum permissible fuel temperature in the fuel tank during the flight
JET A-1, JET-A, Fuel No.3 JP-8 JP8+100 TS-1	-30°C	-35°C
Diesel Sasol GTL Diesel	greater than 0°C	-5°C

Figure 4-1a Minimum fuel temperature limits in the fuel tank

▲ **WARNING:** The fuel temperature of the fuel tank not in use should be observed if it is intended for later use.

▲ **WARNING:** The following applies to Diesel and Jet fuel mixtures in the tank:
As soon as the proportion of Diesel in the tank is more than 10% Diesel, the fuel temperature limits have to be observed for Diesel operation. If there is uncertainty about the type of fuel in the tank, the assumption should be made that it is Diesel.

SECTION 5
PERFORMANCE

SAMPLE PROBLEM

The following sample flight problem utilizes information from the various tables and diagrams of this section to determine the predicted performance data for a typical flight.
Assume the following information has already been determined:

AIRPLANE CONFIGURATION

Takeoff Weight..... 1043 kg
Usable Fuel ..... 120.0 l (31.7 US gal)

TAKEOFF CONDITIONS

Field Pressure Altitude..... 1000 ft
Temperature ..... 28°C (ISA +15°C)
Wind Component along Runway ... 12 Knot Headwind
Field Length ..... 1067 m (3500 ft)

CRUISE CONDITIONS

Total Distance ..... 841 km (400 NM)
Pressure Altitude..... 6000 ft
Temperature ..... 23°C (ISA + 20°C)
Expected Wind Enroute ..... 10 Knot Headwind

LANDING CONDITIONS

Field Pressure Altitude..... 2000 ft
Temperature ..... 25°C
Field Length ..... 914 m (3000 ft)

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CRUISE

The cruising altitude should be selected based on a consideration of trip length, winds aloft and the airplanes performance. A typical cruising altitude and the expected wind enroute have been given for this sample problem. However, the power setting selection for cruise must be determined based on several considerations. These include the cruise performance characteristics presented in Figures 5-4. Considerable fuel savings and longer range result when lower power settings are used.

Figure 5-4b shows a range of 527 NM at zero wind, a power setting of 70% and altitude of 6,000 ft.

With an expected headwind of 10 Knot at 6,000 ft altitude the range has to be corrected as follows:

Range at zero wind (standard tanks) ..... 527 NM

Reduction due to Headwind.....(4.2 h x 10 Knots) = - 42NM

Corrected Range..... 485 NM

This shows that the flight can be performed at a power setting of approximately 70% with full tanks without an intermediate fuel stop.

Figure 5-4a is based on ISA conditions. For a temperature of 20°C above ISA temperature, according to Note 3, true airspeed and maximum range are increased by 2 %.

The following values most nearly correspond to the planned altitude and expected temperature conditions. Engine Power setting chosen is 70%.

The resultants are:

Engine Power:.....70%

True Airspeed: ..... 120 kt + 2% = 122 kt

Fuel Consumption in cruise:22.1 l/h (5.8 US gal/h)

GROUND ROLL AND TAKE-OFF

The ground roll and take-off distance chart, Figure 5-1a ff (Ground Roll and Take-off Distance), should be consulted, keeping in mind that distances shown are based on the short field technique. Conservative distances can be established by reading the chart at the next higher value of weight, temperature and altitude. For example, in this particular sample problem, the takeoff distance information presented for a weight of 1043 kg, pressure altitude of 1000 ft and a temperature of 30°C should be used and results in the following:

Ground Roll263 m (863 ft)

Total Distance to clear a 15 m obstacle451 m (1478 ft)

These distances are well within the available takeoff field length. However, a correction for the effect of wind may be made based on Note 2 of the takeoff chart. The correction for a 12 Knot Headwind is:

$$\frac{12\ \text{Kt}}{9\ \text{Kt}} \times 10\ \% = 13\ \%\ \text{(Decrease)}$$

This results in the following distances, corrected for wind:

Ground Roll, zero wind ..... 263 m (863 ft)
Decrease at 12 Knot Headwind (263m x 13%)= - 34 m (112 ft)
Corrected Ground Roll..... 229 m (751 ft)

Total Distance to clear a 15 m obstacle,
zero wind..... 451 m (1478 ft)
Decrease at 12 Knot Headwind (451 m x 13%)= - 58 m (192 ft)
Corrected Total Distance to clear a ..... 393 m (1286 ft)
15 m obstacle

The resultant cruise distance is:

Total Distance	400.0 NM
Climbout Distance	- 9.1 NM
Cruise Distance	390.9 NM

With an expected 10 Knot headwind, the ground speed for cruise is predicted to be:

122 Knot
- 10 Knot
112 Knot

Therefore, the time required for the cruise portion of the trip is:

$$\frac{390.9 \text{ NM}}{112 \text{ Kt}} = 3.5 \text{ hrs}$$

The fuel required for cruise is:

3.5 h x 22.1 l/h = 77.4 l (20.5 US gal)

The total estimated fuel required is as follows:

Engine Start, Taxi and Takeoff	4.0 l (1.1 US gal)
Climb	+ 4.0 l (1.1 US gal)
Cruise	+ 77.4 l (20.5 US gal)
Total fuel required	85.4 l (22.7 US gal)

This gives with full tanks a reserve of:

120.0 l (31.7 US gal)
- 85.4 l (22.7 US gal)
34.6 l (9.0 US gal)

Once the flight is underway, ground speed checks will provide a more accurate basis for estimating the time enroute and the corresponding fuel required.

LANDING DISTANCE

Refer to Pilot's Operating Handbook

FUEL REQUIRED

The total fuel requirement for the flight may be estimated using the performance information in Figures 5-3 and 5-4. For this sample problem, Figure 5-3b shows that a climb from 1000 ft to 6,000 ft requires 3.3 l (0.9 US gal) of fuel. The corresponding distance during the climb is 7.6 NM. These values are for a standard temperature and are sufficiently accurate for most flight planning purposes.
However, a further correction for the effect of temperature may be made as noted in Note 2 of the climb chart in Figure 5-3b. An effect of 10°C above the standard temperature is to increase time and distance by 10%.

In this case, assuming a temperature 20°C above standard, the correction would be:

$$\frac{20\ °C}{10\ °C} \times 10\ \% = 20\ \%\ \text{(Increase)}$$

With this factor included, the fuel estimate would be calculated as follows:

Fuel to climb, standard temperature:
3.3 l (0.9 US gal)

Increase due to non-standard temperature:
3.3 l (0.9 US gal) x 20.0% = 0.7 l (0.2 US gal)

Corrected fuel to climb:
4.0 l (1.1 US gal)

Using a similar procedure for the distance to climb results in 9.1 NM.

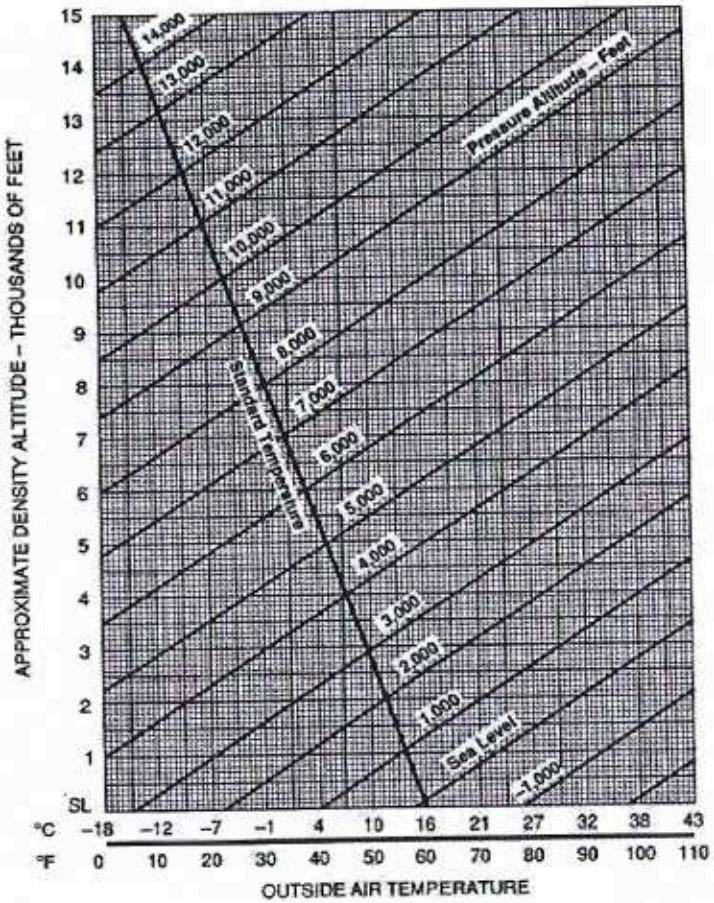

Figure 5-2 Density Altitude Chart

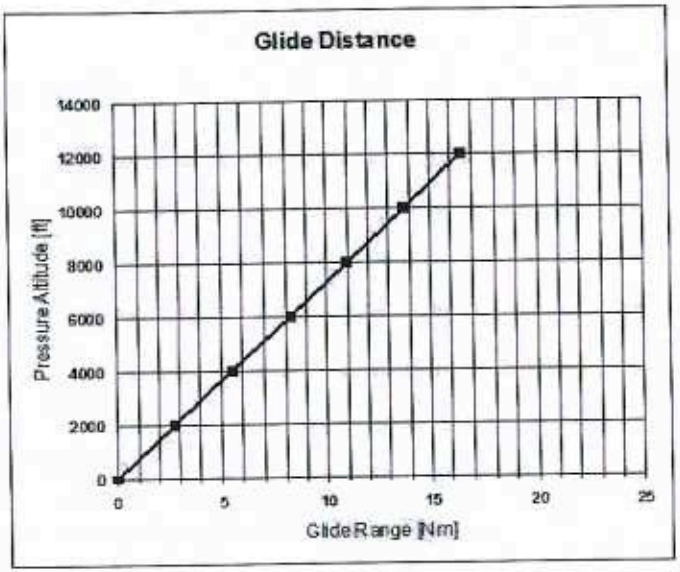

Figure 5-1 Maximum Glide at 73 KIAS/ 84 mph

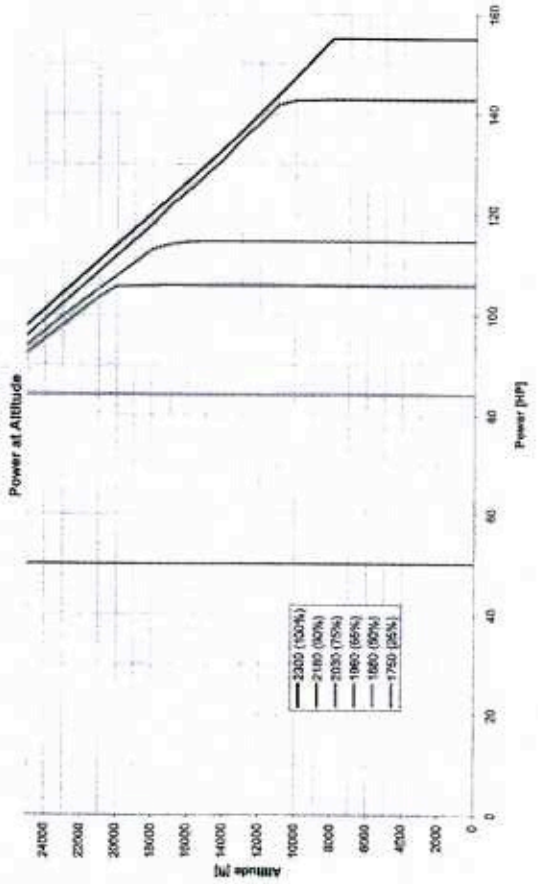

Figure 5-3 Engine Power Over Altitude

SECTION 5a
PERFORMANCE

◆ Note:	This chapter applies to aircraft with propellers **MTV-6-A/187-129**. The correct propeller designation can be found on the blades.

◆ Note:	The chapter not relevant to the respective propeller can be omitted.

GROUND ROLL AND TAKE-OFF DISTANCE at 907 kg (2000 lbs)

SHORT FIELD TAKE-OFFS

Conditions:

Take-off weight 907 kg (2000 lbs)
Flaps 10°
Full Power Prior to Brake Release
Paved, level, dry runway
Zero Wind

Lift Off: ..... 44 KIAS/ 51 mph

Speed at 15 m / 50 ft: ..... 50 KIAS/ 58 mph

Notes:

1. Short field technique
2. Decrease distances 10% for each 9 Knot headwind. For operation with tailwinds up to 10 Knot increase distances by 10% for each 2 Knot.
3. For operation on dry, grass runway, increase distances by 15% of the "ground roll" figure.
4. Consider additional distances (min. 20%) for wet grass runway, softened ground or snow.

Press. alt	Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C]							
[ft]	---	-20°C	0°C	10°C	20°C	30°C	40°C	50°C
0	Gnd Roll	120	138	148	158	171	190	213
	50 ft (15 m) obstacle	205	237	253	270	293	326	369
1000	Gnd Roll	128	148	158	169	183	203	229
	50 ft (15 m) obstacle	219	253	271	289	313	349	395
2000	Gnd Roll	137	159	170	181	196	218	245
	50 ft (15 m) obstacle	235	272	291	310	336	374	424
3000	Gnd Roll	147	170	182	194	210	233	263
	50 ft (15 m) obstacle	252	291	312	332	360	401	454
4000	Gnd Roll	158	183	195	208	226	250	282
	50 ft (15 m) obstacle	271	313	334	357	387	431	487
5000	Gnd Roll	170	196	210	224	242	269	303
	50 ft (15 m) obstacle	290	336	359	383	415	462	523
6000	Gnd Roll	182	211	225	240	260	289	325
	50 ft (15 m) obstacle	312	360	385	411	446	496	562
7000	Gnd Roll	200	231	247	263	285	317	356
	50 ft (15 m) obstacle	342	395	423	451	489	544	616
8000	Gnd Roll	219	253	271	289	313	347	391
	50 ft (15 m) obstacle	375	434	464	495	536	597	676
9000	Gnd Roll	243	281	301	321	347	385	434
	50 ft (15 m) obstacle	417	482	515	549	595	663	751
10000	Gnd Roll	270	312	334	356	386	428	482
	50 ft (15 m) obstacle	463	535	573	611	662	738	835

Figure 5-1a Take-Off Distance [m] at take-off weight 907 kg (2000 lbs)

Press. alt	Ground Roll and Take-Off Distance [ft] Outside Air Temperature [°C]							
[ft]	---	-20°C	0°C	10°C	20°C	30°C	40°C	50°C
0	Gnd Roll	392	453	485	517	560	622	700
	50 ft (15 m) obstacle	672	776	830	885	960	1069	1210
1000	Gnd Roll	420	486	520	554	600	666	750
	50 ft (15 m) obstacle	720	831	889	948	1028	1145	1296
2000	Gnd Roll	451	521	557	594	643	714	804
	50 ft (15 m) obstacle	771	891	953	1016	1102	1228	1389
3000	Gnd Roll	483	558	597	637	690	766	862
	50 ft (15 m) obstacle	827	956	1022	1090	1182	1316	1490
4000	Gnd Roll	518	599	641	683	740	821	925
	50 ft (15 m) obstacle	887	1025	1097	1169	1268	1412	1598
5000	Gnd Roll	556	643	688	734	794	882	993
	50 ft (15 m) obstacle	952	1100	1177	1255	1361	1516	1716
6000	Gnd Roll	598	691	739	788	853	947	1066
	50 ft (15 m) obstacle	1023	1182	1264	1348	1461	1628	1843
7000	Gnd Roll	655	757	810	864	935	1038	1169
	50 ft (15 m) obstacle	1122	1296	1386	1478	1602	1785	2020
8000	Gnd Roll	719	831	889	948	1026	1139	1283
	50 ft (15 m) obstacle	1231	1422	1521	1622	1759	1959	2217
9000	Gnd Roll	798	922	986	1052	1139	1264	1424
	50 ft (15 m) obstacle	1367	1580	1689	1802	1953	2176	2463
10000	Gnd Roll	887	1024	1096	1169	1265	1405	1582
	50 ft (15 m) obstacle	1520	1756	1879	2003	2172	2420	2739

Figure 5-1b Take-Off Distance [ft] at take-off weight 907 kg (2000 lbs)

GROUND ROLL AND TAKE-OFF DISTANCE at 1043 kg (2300 lbs)

SHORT FIELD TAKE-OFFS

Conditions:

Take-off weight 1043 kg (2300 lbs)
Flaps 10°
Full Power Prior to Brake Release
Paved, level, dry runway
Zero Wind

Lift Off: ..... 48 KIAS/ 55 mph

Speed at 15 m / 50 ft: ..... 54 KIAS/ 62 mph

Notes:

1. Short field technique
2. Decrease distances 10% for each 9 Knot headwind. For operation with tailwinds up to 10 Knot increase distances by 10% for each 2 Knot.
3. For operation on dry, grass runway, increase distances by 15% of the "ground roll" figure.
4. Consider additional distances (min. 20%) for wet grass runway, softened ground or snow.

Press. alt	Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C]							
[ft]	—	-20°C	0°C	10°C	20°C	30°C	40°C	50°C
0	Gnd Roll	172	199	213	227	246	273	307
	50 ft (15 m) obstacle	294	340	364	388	421	469	530
1000	Gnd Roll	184	213	228	243	263	292	329
	50 ft (15 m) obstacle	315	364	390	416	451	502	568
2000	Gnd Roll	198	228	244	260	282	313	352
	50 ft (15 m) obstacle	338	391	418	446	483	538	609
3000	Gnd Roll	212	245	262	279	302	336	378
	50 ft (15 m) obstacle	363	419	448	478	518	577	653
4000	Gnd Roll	227	263	281	300	324	360	406
	50 ft (15 m) obstacle	389	449	481	513	556	619	701
5000	Gnd Roll	244	282	302	322	348	387	435
	50 ft (15 m) obstacle	418	482	516	550	597	665	752
6000	Gnd Roll	262	303	324	345	374	415	468
	50 ft (15 m) obstacle	448	518	554	591	641	714	808
7000	Gnd Roll	287	332	355	379	410	455	513
	50 ft (15 m) obstacle	492	568	608	648	703	783	886
8000	Gnd Roll	315	364	390	416	450	500	563
	50 ft (15 m) obstacle	540	624	667	711	771	859	972
9000	Gnd Roll	350	404	432	461	499	554	624
	50 ft (15 m) obstacle	599	692	741	790	856	954	1080
10000	Gnd Roll	389	449	480	512	555	616	694
	50 ft (15 m) obstacle	666	770	824	878	952	1061	1201

Figure 5-1c Take-Off Distance [m] at take-off weight 1043 kg (2300 lbs)

Press. alt [ft]	Ground Roll and Take-Off Distance [ft] Outside Air Temperature [°C]							
	—	-20°C	0°C	10°C	20°C	30°C	40°C	50°C
0	Gnd Roll	564	652	697	744	805	894	1007
	50 ft (15 m) obstacle	966	1116	1194	1273	1380	1537	1740
1000	Gnd Roll	605	698	747	797	863	958	1079
	50 ft (15 m) obstacle	1035	1195	1279	1364	1478	1647	1864
2000	Gnd Roll	648	749	801	854	925	1027	1156
	50 ft (15 m) obstacle	1109	1281	1371	1462	1585	1765	1998
3000	Gnd Roll	695	803	859	916	992	1101	1240
	50 ft (15 m) obstacle	1189	1374	1470	1567	1699	1893	2142
4000	Gnd Roll	745	861	921	983	1064	1181	1330
	50 ft (15 m) obstacle	1276	1474	1577	1682	1823	2031	2299
5000	Gnd Roll	800	925	989	1055	1142	1268	1428
	50 ft (15 m) obstacle	1370	1582	1693	1805	1957	2180	2467
6000	Gnd Roll	859	993	1062	1133	1227	1362	1533
	50 ft (15 m) obstacle	1471	1700	1818	1939	2102	2341	2650
7000	Gnd Roll	942	1089	1165	1242	1345	1493	1681
	50 ft (15 m) obstacle	1613	1863	1993	2126	2304	2567	2905
8000	Gnd Roll	1034	1195	1278	1363	1476	1638	1845
	50 ft (15 m) obstacle	1770	2045	2187	2333	2529	2817	3189
9000	Gnd Roll	1148	1326	1418	1513	1638	1818	2048
	50 ft (15 m) obstacle	1966	2271	2429	2591	2809	3129	3541
10000	Gnd Roll	1275	1473	1576	1681	1820	2020	2275
	50 ft (15 m) obstacle	2186	2526	2701	2881	3123	3480	3938

Figure 5-1d Take-Off Distance [ft] at take-off weight 1043 kg (2300 lbs)

MAXIMUM RATE-OF-CLIMB at 907 kg (2000 lbs)

Conditions:

Take-off weight 907 kg (2000 lbs)
Climb speed v_y = 69 KIAS/ 79 mph
Flaps Up
Full Power

Notes:

1. For operation in air colder than this table provides, use coldest data shown.
2. For operation in air warmer than this table provides, use extreme caution.

PRESS ALT [FT]	Climb speed [KIAS]	Rate of Climb [ft/min] Outside Air Temperature [°C]				
		-20°C	0°C	+20°C	+40°C	+50°C
0	69	1089	1076	1064	899	751
1000	69	1083	1070	1057	892	744
2000	69	1077	1063	1050	885	737
3000	69	1070	1057	1043	878	729
4000	69	1064	1050	1036	870	722
5000	69	1057	1043	1029	863	714
6000	69	1050	1035	1021	855	706
7000	69	1043	1028	1013	847	698
8000	69	1035	1020	1005	839	690
9000	69	999	984	969	805	659
10000	69	963	947	932	771	627
11000	69	926	910	895	736	595
12000	69	889	873	857	702	563
13000	69	852	836	820	667	531
14000	69	815	798	782	631	499
15000	69	778	760	744	596	466
16000	69	740	722	705	560	433
17000	69	702	684	666	524	400
18000	69	664	645	627	488	366

Figure 5-2a Maximum Rate of Climb ISA at take-off weight 907 kg (2000 lbs)

MAXIMUM RATE-OF-CLIMB at 1043 kg (2300 lbs)

Conditions:

Take-off weight 1043 kg (2300 lbs)
Climb speed v_y = 69 KIAS/ 79 mph
Flaps Up
Full Power

Notes:

1. For operation in air colder than this table provides, use coldest data shown.
2. For operation in air warmer than this table provides, use extreme caution.

PRESS ALT [FT]	Climb speed [KIAS]	Rate of Climb [ft/min] Outside Air Temperature [°C]				
		-20°C	0°C	+20°C	+40°C	+50°C
0	69	879	865	851	706	576
1000	69	872	858	844	698	568
2000	69	865	851	837	691	561
3000	69	858	844	829	683	553
4000	69	851	836	822	675	545
5000	69	844	829	814	667	536
6000	69	836	821	806	658	528
7000	69	829	813	797	650	519
8000	69	821	805	789	641	510
9000	69	788	771	755	610	481
10000	69	755	738	721	578	452
11000	69	721	704	687	547	423
12000	69	688	670	653	515	393
13000	69	654	636	619	482	363
14000	69	620	601	584	450	333
15000	69	585	567	549	417	302
16000	69	551	531	513	384	271
17000	69	516	496	477	350	240
18000	69	480	461	441	317	209

Figure 5-2b Maximum Rate of Climb ISA at take-off weight 1043 kg (2300 lbs)

TIME, FUEL AND DISTANCE TO CLIMB at 907 kg (2000 lbs)

Conditions:

Take-off weight 907 kg (2000 lbs)
Climb speed v_y = 69 KIAS/ 79 mph
Flaps Up
Full Power
Standard Temperature (ISA)

Notes :

1. Add 4 l (1.1 US gal) of fuel for engine start, taxi and takeoff allowance.
2. Increase time and distance by 10% for 10°C above standard temperature.
3. Distances shown are based on zero wind.
4. Time, distance and fuel required are only valid from the point where the airplane climbs at v_y = 69 KIAS.

Press Alt [ft]	OAT [°C]	Vy [KIAS]	ROC [FPM]	Time [MIN]	Distance [NM]	Fuel used [l]	Fuel used [US Gal]
0	15	69	1067	0.0	0.0	0.0	0.0
1000	13	69	1061	0.9	1.1	0.5	0.1
2000	11	69	1056	1.9	2.2	1.1	0.3
3000	9	69	1050	2.8	3.4	1.6	0.4
4000	7	69	1045	3.8	4.6	2.1	0.6
5000	5	69	1039	4.7	5.8	2.7	0.7
6000	3	69	1033	5.7	7.1	3.2	0.8
7000	1	69	1027	6.7	8.5	3.7	1.0
8000	-1	69	1021	7.7	9.9	4.3	1.1
9000	-3	69	986	8.7	11.3	4.7	1.2
10000	-5	69	951	9.7	12.9	5.2	1.4
11000	-7	69	916	10.8	14.5	5.6	1.5
12000	-9	69	880	11.9	16.3	6.0	1.6
13000	-11	69	845	13.0	18.2	6.4	1.7
14000	-13	69	809	14.2	20.2	6.8	1.8
15000	-15	69	773	15.5	22.4	7.2	1.9
16000	-17	69	737	16.8	24.7	7.6	2.0
17000	-19	69	701	18.2	27.2	8.0	2.1
18000	-21	69	664	19.7	29.9	8.4	2.2

Figure 5-3a Time, Fuel and Distance to Climb at 907 kg (2000 lbs)

TIME, FUEL AND DISTANCE TO CLIMB at 1043 kg (2300 lbs)

Conditions:

Take-off weight 1043 kg (2300 lbs)
Climb speed v_y = 69 KIAS/ 79 mph
Flaps Up
Full Power
Standard Temperature (ISA)

Notes :

1. Add 4 l (1.1 US gal) of fuel for engine start, taxi and takeoff allowance.
2. Increase time and distance by 10% for 10°C above standard temperature.
3. Distances shown are based on zero wind.
4. Time, distance and fuel required are only valid from the point where the airplane climbs at v_y = 69 KIAS.

Press. Alt. [ft]	OAT [°C]	Vy [KIAS]	ROC [FPM]	Time [MIN]	Distance [NM]	Fuel used [l]	Fuel used [US Gal]
0	15	69	855	0.0	0.0	0.0	0.0
1000	13	69	849	1.2	1.4	0.7	0.2
2000	11	69	843	2.4	2.8	1.3	0.3
3000	9	69	837	3.5	4.2	2.0	0.5
4000	7	69	831	4.7	5.8	2.6	0.7
5000	5	69	825	6.0	7.3	3.3	0.9
6000	3	69	818	7.2	9.0	4.0	1.1
7000	1	69	812	8.4	10.7	4.7	1.2
8000	-1	69	805	9.6	12.4	5.4	1.4
9000	-3	69	774	10.9	14.3	5.9	1.6
10000	-5	69	742	12.2	16.3	6.5	1.7
11000	-7	69	710	13.6	18.4	7.0	1.9
12000	-9	69	678	15.0	20.7	7.6	2.0
13000	-11	69	645	16.6	23.1	8.1	2.1
14000	-13	69	613	18.1	25.8	8.7	2.3
15000	-15	69	580	19.8	28.6	9.2	2.4
16000	-17	69	547	21.6	31.7	9.8	2.6
17000	-19	69	514	23.5	35.1	10.3	2.7
18000	-21	69	481	25.5	38.7	10.8	2.9

Figure 5-3b Time, Fuel and Distance to Climb at 1043 kg (2300 lbs)

CRUISE PERFORMANCE, RANGE AND ENDURANCE WITH STANDARD TANKS, Cessna 172 F,G,H at 1043 kg (2300 lbs)

Conditions:
Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

Notes:

1. Endurance information is based on standard tanks with 114.3 l (30.2 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb 45 min. reserve.
3. Increase true airspeed (KTAS/mph)) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
SL	100	131	151	33.6	8.9	332	2.5
SL	90	126	145	29.6	7.8	375	3.0
SL	80	120	138	25.8	6.8	423	3.5
SL	70	114	131	22.1	5.8	483	4.2
SL	60	106	122	18.6	4.9	549	5.2
SL	50	97	112	15.3	4.0	627	6.5
2000	100	133	154	33.6	8.9	334	2.5
2000	90	128	147	29.6	7.8	378	2.9
2000	80	122	141	25.8	6.8	427	3.4
2000	70	116	133	22.1	5.8	488	4.1
2000	60	108	124	18.6	4.9	555	5.1
2000	50	99	114	15.3	4.0	634	6.3
4000	100	136	156	33.6	8.9	339	2.4
4000	90	131	150	29.6	7.8	384	2.8
4000	80	124	143	25.8	6.8	430	3.3
4000	70	118	135	22.1	5.8	492	4.0
4000	60	110	126	18.6	4.9	560	5.0
4000	50	100	115	15.3	4.0	634	6.2
6000	100	139	159	33.6	8.9	344	2.3
6000	90	133	153	29.6	7.8	386	2.7
6000	80	127	146	25.8	6.8	437	3.3
6000	70	120	138	22.1	5.8	496	3.9
6000	60	111	128	18.6	4.9	560	4.8
6000	50	102	117	15.3	4.0	641	6.1
8000	100	141	162	33.6	8.9	346	2.2
8000	90	135	156	29.6	7.8	389	2.6
8000	80	129	148	25.8	6.8	440	3.2
8000	70	122	140	22.1	5.8	499	3.8
8000	60	113	130	18.6	4.9	564	4.7
8000	50	103	119	15.3	4.0	641	5.9
10000	90	138	159	29.6	7.8	394	2.5
10000	80	131	151	25.8	6.8	442	3.1

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
10000	70	124	142	22.1	5.8	503	3.7
10000	60	115	132	18.6	4.9	569	4.6
10000	50	105	120	15.3	4.0	647	5.8
12000	90	140	162	29.6	7.8	397	2.4
12000	80	134	154	25.8	6.8	448	3.0
12000	70	126	145	22.1	5.8	506	3.6
12000	60	117	134	18.6	4.9	573	4.5
12000	50	106	122	15.3	4.0	646	5.7
14000	90	143	165	29.6	7.8	402	2.3
14000	80	136	157	25.8	6.8	451	2.8
14000	70	128	147	22.1	5.8	509	3.5
14000	60	119	137	18.6	4.9	577	4.4
14000	50	108	124	15.3	4.0	652	5.5
16000	80	138	159	25.8	6.8	454	2.7
16000	70	130	150	22.1	5.8	513	3.4
16000	60	121	139	18.6	4.9	581	4.2
16000	50	109	126	15.3	4.0	651	5.4
18000	80	141	162	25.8	6.8	459	2.6
18000	70	132	152	22.1	5.8	515	3.2
18000	60	123	141	18.6	4.9	585	4.1
18000	50	111	127	15.3	4.0	656	5.2

Figure 5-4a Cruise Performance, Range and Endurance with standard tanks F,G,H; 1043 kg (2300 lbs)

CRUISE PERFORMANCE, RANGE AND ENDURANCE WITH STANDARD TANKS, Cessna 172 I,K,L,M at 1043 kg (2300 lbs)

Conditions:

Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

Notes:

1. Endurance information is based on standard tanks with 120.0 l (31.7 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb 45 min. reserve.
3. Increase true airspeed (KTAS/mph)) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
SL	100	131	151	33.6	8.9	354	2.7
SL	90	126	145	29.6	7.8	399	3.2
SL	80	120	138	25.8	6.8	450	3.7
SL	70	114	131	22.1	5.8	513	4.5
SL	60	106	122	18.6	4.9	582	5.5
SL	50	97	112	15.3	4.0	663	6.8
2000	100	133	154	33.6	8.9	357	2.6
2000	90	128	147	29.6	7.8	403	3.1
2000	80	122	141	25.8	6.8	454	3.7
2000	70	116	133	22.1	5.8	518	4.4
2000	60	108	124	18.6	4.9	588	5.4
2000	50	99	114	15.3	4.0	671	6.7
4000	100	136	156	33.6	8.9	362	2.5
4000	90	131	150	29.6	7.8	409	3.0
4000	80	124	143	25.8	6.8	457	3.6
4000	70	118	135	22.1	5.8	522	4.3
4000	60	110	126	18.6	4.9	593	5.3
4000	50	100	115	15.3	4.0	671	6.6
6000	100	139	159	33.6	8.9	368	2.5
6000	90	133	153	29.6	7.8	412	2.9
6000	80	127	146	25.8	6.8	465	3.5
6000	70	120	138	22.1	5.8	527	4.2
6000	60	111	128	18.6	4.9	594	5.2
6000	50	102	117	15.3	4.0	679	6.5
8000	100	141	162	33.6	8.9	370	2.4
8000	90	135	156	29.6	7.8	415	2.8
8000	80	129	148	25.8	6.8	468	3.4
8000	70	122	140	22.1	5.8	531	4.1
8000	60	113	130	18.6	4.9	599	5.0
8000	50	103	119	15.3	4.0	679	6.3
10000	90	138	159	29.6	7.8	421	2.7
10000	80	131	151	25.8	6.8	471	3.3

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
10000	70	124	142	22.1	5.8	535	4.0
10000	60	115	132	18.6	4.9	604	4.9
10000	50	105	120	15.3	4.0	686	6.2
12000	90	140	162	29.6	7.8	423	2.6
12000	80	134	154	25.8	6.8	478	3.2
12000	70	126	145	22.1	5.8	539	3.9
12000	60	117	134	18.6	4.9	609	4.8
12000	50	106	122	15.3	4.0	686	6.0
14000	90	143	165	29.6	7.8	429	2.5
14000	80	136	157	25.8	6.8	481	3.1
14000	70	128	147	22.1	5.8	542	3.8
14000	60	119	137	18.6	4.9	614	4.7
14000	50	108	124	15.3	4.0	692	5.9
16000	80	138	159	25.8	6.8	484	3.0
16000	70	130	150	22.1	5.8	546	3.6
16000	60	121	139	18.6	4.9	618	4.5
16000	50	109	126	15.3	4.0	692	5.7
18000	80	141	162	25.8	6.8	490	2.8
18000	70	132	152	22.1	5.8	550	3.5
18000	60	123	141	18.6	4.9	622	4.4
18000	50	111	127	15.3	4.0	697	5.6

Figure 5-4b Cruise Performance, Range and Endurance
with standard tanks I,K,L,M; 1043 kg (2300 lbs)

CRUISE PERFORMANCE, RANGE AND ENDURANCE WITH LONG-RANGE TANKS Cessna 172 F-M at 1043 kg (2300 lbs)

Conditions:

Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

Notes:

1. Endurance information is based on long-range tanks with 152.2 l (40.2 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb 45 min. reserve.
3. Increase true airspeed (KTAS/mph)) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
SL	100	131	151	33.6	8.9	480	3.7
SL	90	126	145	29.6	7.8	536	4.3
SL	80	120	138	25.8	6.8	599	5.0
SL	70	114	131	22.1	5.8	679	6.0
SL	60	106	122	18.6	4.9	765	7.2
SL	50	97	112	15.3	4.0	867	8.9
2000	100	133	154	33.6	8.9	484	3.6
2000	90	128	147	29.6	7.8	542	4.2
2000	80	122	141	25.8	6.8	606	4.9
2000	70	116	133	22.1	5.8	687	5.9
2000	60	108	124	18.6	4.9	775	7.1
2000	50	99	114	15.3	4.0	879	8.8
4000	100	136	156	33.6	8.9	493	3.5
4000	90	131	150	29.6	7.8	552	4.1
4000	80	124	143	25.8	6.8	612	4.8
4000	70	118	135	22.1	5.8	694	5.8
4000	60	110	126	18.6	4.9	784	7.0
4000	50	100	115	15.3	4.0	882	8.7
6000	100	139	159	33.6	8.9	501	3.4
6000	90	133	153	29.6	7.8	557	4.0
6000	80	127	146	25.8	6.8	623	4.7
6000	70	120	138	22.1	5.8	702	5.7
6000	60	111	128	18.6	4.9	786	6.9
6000	50	102	117	15.3	4.0	893	8.6
8000	100	141	162	33.6	8.9	505	3.3
8000	90	135	156	29.6	7.8	562	3.9
8000	80	129	148	25.8	6.8	629	4.6
8000	70	122	140	22.1	5.8	709	5.6
8000	60	113	130	18.6	4.9	795	6.8
8000	50	103	119	15.3	4.0	896	8.4
10000	90	138	159	29.6	7.8	571	3.8
10000	80	131	151	25.8	6.8	635	4.5

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
10000	70	124	142	22.1	5.8	715	5.4
10000	60	115	132	18.6	4.9	803	6.6
10000	50	105	120	15.3	4.0	907	8.3
12000	90	140	162	29.6	7.8	576	3.7
12000	80	134	154	25.8	6.8	645	4.4
12000	70	126	145	22.1	5.8	722	5.3
12000	60	117	134	18.6	4.9	811	6.5
12000	50	106	122	15.3	4.0	909	8.1
14000	90	143	165	29.6	7.8	585	3.6
14000	80	136	157	25.8	6.8	651	4.3
14000	70	128	147	22.1	5.8	729	5.2
14000	60	119	137	18.6	4.9	820	6.4
14000	50	108	124	15.3	4.0	919	8.0
16000	80	138	159	25.8	6.8	657	4.2
16000	70	130	150	22.1	5.8	736	5.1
16000	60	121	139	18.6	4.9	828	6.3
16000	50	109	126	15.3	4.0	921	7.8
18000	80	141	162	25.8	6.8	666	4.1
18000	70	132	152	22.1	5.8	742	5.0
18000	60	123	141	18.6	4.9	835	6.1
18000	50	111	127	15.3	4.0	931	7.7

Figure 5-4c Cruise Performance, Range and Endurance with long range tanks, 1043 kg (2300 lbs)

SECTION 5b
PERFORMANCE

◆ Note:	This chapter applies to aircraft with propellers **MTV-6-A/190-69**. The correct propeller designation can be found on the blades.

◆ Note:	The chapter not relevant to the respective propeller can be omitted.

GROUND ROLL AND TAKE-OFF DISTANCE at 907 kg (2000 lbs)

SHORT FIELD TAKE-OFFS

Conditions:

Take-off weight 907 kg (2000 lbs)
Flaps 10°
Full Power Prior to Brake Release
Paved, level, dry runway
Zero Wind

Lift Off: ..... 44 KIAS/ 51 mph

Speed at 15 m / 50 ft: ..... 50 KIAS/ 58 mph

Notes:

1. Short field technique
2. Decrease distances 10% for each 9 Knot headwind. For operation with tailwinds up to 10 Knot increase distances by 10% for each 2 Knot.
3. For operation on dry, grass runway, increase distances by 15% of the "ground roll" figure.
4. Consider additional distances (min. 20%) for wet grass runway, softened ground or snow.

Press. alt	Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C]							
[ft]	---	-20°C	0°C	10°C	20°C	30°C	40°C	50°C
0	Gnd Roll	107	123	132	142	151	166	187
	50 ft (15 m) obstacle	167	193	206	222	236	261	295
1000	Gnd Roll	114	132	141	152	162	178	200
	50 ft (15 m) obstacle	179	206	221	237	253	280	316
2000	Gnd Roll	122	141	151	163	174	191	215
	50 ft (15 m) obstacle	191	221	237	254	272	300	339
3000	Gnd Roll	131	152	162	174	186	205	230
	50 ft (15 m) obstacle	205	237	254	273	291	321	363
4000	Gnd Roll	141	163	174	187	200	220	247
	50 ft (15 m) obstacle	220	255	273	293	312	345	389
5000	Gnd Roll	151	175	187	201	214	236	265
	50 ft (15 m) obstacle	236	273	293	314	335	370	418
6000	Gnd Roll	162	188	201	216	230	253	285
	50 ft (15 m) obstacle	254	293	314	337	360	397	449
7000	Gnd Roll	178	206	220	236	252	278	312
	50 ft (15 m) obstacle	278	322	344	370	395	436	492
8000	Gnd Roll	195	225	241	259	276	304	342
	50 ft (15 m) obstacle	305	352	377	405	433	477	539
9000	Gnd Roll	216	250	267	287	306	337	379
	50 ft (15 m) obstacle	338	391	418	449	480	529	598
10000	Gnd Roll	239	276	296	318	339	373	420
	50 ft (15 m) obstacle	375	433	464	498	532	587	663

Figure 5-1a Take-Off Distance [m] at take-off weight 907 kg (2000 lbs)

Press. alt		Ground Roll and Take-Off Distance [ft] Outside Air Temperature [°C]						
[ft]	---	-20°C	0°C	10°C	20°C	30°C	40°C	50°C
0	Gnd Roll	350	404	433	464	496	546	614
	50 ft (15 m) obstacle	547	632	677	727	776	856	967
1000	Gnd Roll	375	433	463	497	531	584	658
	50 ft (15 m) obstacle	586	677	725	778	831	917	1036
2000	Gnd Roll	401	464	497	533	569	626	705
	50 ft (15 m) obstacle	628	726	777	834	891	983	1110
3000	Gnd Roll	431	497	533	572	610	672	756
	50 ft (15 m) obstacle	673	778	833	895	955	1054	1191
4000	Gnd Roll	462	534	571	613	655	721	811
	50 ft (15 m) obstacle	722	835	894	960	1025	1131	1278
5000	Gnd Roll	496	573	613	658	703	774	870
	50 ft (15 m) obstacle	776	896	959	1030	1100	1214	1371
6000	Gnd Roll	532	615	659	707	755	831	935
	50 ft (15 m) obstacle	833	962	1030	1107	1181	1303	1473
7000	Gnd Roll	584	674	722	775	827	911	1025
	50 ft (15 m) obstacle	913	1055	1129	1213	1295	1429	1614
8000	Gnd Roll	640	739	791	850	907	998	1123
	50 ft (15 m) obstacle	1001	1156	1238	1329	1419	1566	1769
9000	Gnd Roll	708	819	876	941	1004	1105	1244
	50 ft (15 m) obstacle	1109	1281	1372	1473	1573	1735	1961
10000	Gnd Roll	785	907	971	1042	1113	1225	1378
	50 ft (15 m) obstacle	1230	1421	1521	1634	1744	1924	2174

Figure 5-1b Take-Off Distance [ft] at take-off weight 907 kg (2000 lbs)

GROUND ROLL AND TAKE-OFF DISTANCE at 1043 kg (2300 lbs)

SHORT FIELD TAKE-OFFS

Conditions:

Take-off weight 1043 kg (2300 lbs)
Flaps 10°
Full Power Prior to Brake Release
Paved, level, dry runway
Zero Wind

Lift Off: ..... 48 KIAS/ 55 mph

Speed at 15 m / 50 ft: ..... 54 KIAS/ 62 mph

Notes:

1. Short field technique
2. Decrease distances 10% for each 9 Knot headwind. For operation with tailwinds up to 10 Knot increase distances by 10% for each 2 Knot.
3. For operation on dry, grass runway, increase distances by 15% of the "ground roll" figure.
4. Consider additional distances (min. 20%) for wet grass runway, softened ground or snow.

Press. alt	Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C]							
[ft]	—	-20°C	0°C	10°C	20°C	30°C	40°C	50°C
0	Gnd Roll	153	177	190	204	217	239	269
	50 ft (15 m) obstacle	240	277	297	319	340	375	424
1000	Gnd Roll	164	190	203	218	233	256	289
	50 ft (15 m) obstacle	257	297	318	342	365	402	455
2000	Gnd Roll	176	204	218	234	250	275	309
	50 ft (15 m) obstacle	276	318	341	366	391	431	487
3000	Gnd Roll	189	218	234	251	268	295	332
	50 ft (15 m) obstacle	295	341	366	393	419	462	522
4000	Gnd Roll	203	234	251	269	287	316	356
	50 ft (15 m) obstacle	317	366	392	421	450	496	561
5000	Gnd Roll	218	251	269	289	308	339	382
	50 ft (15 m) obstacle	340	393	421	452	483	532	602
6000	Gnd Roll	234	270	289	310	331	365	410
	50 ft (15 m) obstacle	365	422	452	486	518	572	646
7000	Gnd Roll	256	296	317	340	363	400	450
	50 ft (15 m) obstacle	401	463	495	532	568	627	708
8000	Gnd Roll	281	324	347	373	398	438	493
	50 ft (15 m) obstacle	439	507	543	583	623	687	776
9000	Gnd Roll	311	359	385	413	441	485	546
	50 ft (15 m) obstacle	487	562	602	646	690	761	860
10000	Gnd Roll	344	398	426	457	488	537	605
	50 ft (15 m) obstacle	539	623	667	717	765	844	954

Figure 5-1c Take-Off Distance [m] at take-off weight 1043 kg (2300 lbs)

Press. alt	Ground Roll and Take-Off Distance [ft] Outside Air Temperature [°C]							
[ft]	---	-20°C	0°C	10°C	20°C	30°C	40°C	50°C
0	Gnd Roll	503	581	622	668	713	785	883
	50 ft (15 m) obstacle	787	909	974	1046	1116	1232	1392
1000	Gnd Roll	539	623	667	716	764	841	946
	50 ft (15 m) obstacle	843	974	1043	1120	1196	1319	1491
2000	Gnd Roll	578	668	715	767	819	901	1014
	50 ft (15 m) obstacle	904	1044	1118	1201	1282	1414	1598
3000	Gnd Roll	620	716	766	823	878	967	1088
	50 ft (15 m) obstacle	969	1120	1199	1288	1374	1517	1714
4000	Gnd Roll	665	768	822	883	942	1037	1167
	50 ft (15 m) obstacle	1040	1201	1286	1381	1475	1627	1839
5000	Gnd Roll	714	824	883	948	1011	1113	1253
	50 ft (15 m) obstacle	1116	1290	1381	1483	1583	1747	1973
6000	Gnd Roll	766	885	948	1018	1086	1196	1345
	50 ft (15 m) obstacle	1199	1385	1483	1592	1700	1876	2119
7000	Gnd Roll	840	970	1039	1115	1191	1310	1474
	50 ft (15 m) obstacle	1314	1518	1625	1745	1863	2056	2323
8000	Gnd Roll	921	1064	1139	1223	1305	1436	1616
	50 ft (15 m) obstacle	1440	1664	1782	1913	2042	2253	2546
9000	Gnd Roll	1020	1178	1261	1354	1445	1591	1790
	50 ft (15 m) obstacle	1596	1844	1974	2120	2263	2497	2822
10000	Gnd Roll	1130	1305	1397	1500	1601	1762	1983
	50 ft (15 m) obstacle	1769	2044	2189	2351	2510	2769	3129

Figure 5-1d Take-Off Distance [ft] at take-off weight 1043 kg (2300 lbs)

MAXIMUM RATE-OF-CLIMB at 907 kg (2000 lbs)

Conditions:

Take-off weight 907 kg (2000 lbs)
Climb speed v_y = 69 KIAS/ 79 mph
Flaps Up
Full Power

Notes:

1. For operation in air colder than this table provides, use coldest data shown.
2. For operation in air warmer than this table provides, use extreme caution.

PRESS ALT [FT]	Climb speed [KIAS]	Rate of Climb [ft/min] Outside Air Temperature [°C]				
		-20°C	0°C	+20°C	+40°C	+50°C
0	69	1157	1144	1132	1003	846
1000	69	1151	1138	1125	996	839
2000	69	1145	1131	1118	989	832
3000	69	1138	1125	1111	982	825
4000	69	1132	1118	1104	975	817
5000	69	1125	1111	1097	967	810
6000	69	1118	1104	1090	959	802
7000	69	1111	1096	1082	951	794
8000	69	1104	1089	1074	943	785
9000	69	1066	1051	1036	907	752
10000	69	1029	1013	998	871	719
11000	69	991	975	960	835	686
12000	69	953	937	921	798	652
13000	69	915	898	882	762	618
14000	69	876	859	843	725	584
15000	69	838	820	804	687	549
16000	69	799	781	764	650	515
17000	69	759	742	724	612	480
18000	69	720	702	684	574	444

Figure 5-2a Maximum Rate of Climb ISA at take-off weight 907 kg (2000 lbs)

MAXIMUM RATE-OF-CLIMB at 1043 kg (2300 lbs)

Conditions:

Take-off weight 1043 kg (2300 lbs)
Climb speed v_y = 69 KIAS/ 79 mph
Flaps Up
Full Power

Notes:

1. For operation in air colder than this table provides, use coldest data shown.
2. For operation in air warmer than this table provides, use extreme caution.

PRESS ALT [FT]	Climb speed [KIAS]	Rate of Climb [ft/min] Outside Air Temperature [°C]				
		-20°C	0°C	+20°C	+40°C	+50°C
0	69	938	924	911	796	659
1000	69	932	917	904	789	651
2000	69	925	910	897	782	644
3000	69	918	903	889	774	636
4000	69	911	896	881	766	628
5000	69	904	888	874	758	620
6000	69	896	881	866	750	611
7000	69	888	873	857	741	603
8000	69	881	864	849	732	594
9000	69	847	830	814	700	563
10000	69	812	796	779	666	533
11000	69	778	761	744	633	502
12000	69	743	726	709	599	471
13000	69	708	691	673	566	439
14000	69	673	655	638	531	408
15000	69	638	619	601	497	376
16000	69	602	583	565	462	343
17000	69	566	547	528	427	311
18000	69	530	510	491	392	278

Figure 5-2b Maximum Rate of Climb ISA at take-off weight 1043 kg (2300 lbs)

TIME, FUEL AND DISTANCE TO CLIMB at 907 kg (2000 lbs)

Conditions:

Take-off weight 907 kg (2000 lbs)
Climb speed v_y = 69 KIAS/ 79 mph
Flaps Up
Full Power
Standard Temperature (ISA)

Notes :

1. Add 4 l (1.1 US gal) of fuel for engine start, taxi and takeoff allowance.
2. Increase time and distance by 10% for 10°C above standard temperature.
3. Distances shown are based on zero wind.
4. Time, distance and fuel required are only valid from the point where the airplane climbs at v_y = 69 KIAS.

Press. Alt.	OAT	Vy	ROC	Time	Distance	Fuel used	
[ft]	[°C]	[KIAS]	[FPM]	[MIN]	[NM]	[l]	[US Gal]
0	15	69	1135	0,0	0,0	0,0	0,0
1000	13	69	1130	0,9	1,0	0,5	0,1
2000	11	69	1124	1,8	2,1	1,0	0,3
3000	9	69	1119	2,7	3,2	1,5	0,4
4000	7	69	1113	3,6	4,4	2,0	0,5
5000	5	69	1107	4,5	5,5	2,5	0,7
6000	3	69	1101	5,4	6,8	3,0	0,8
7000	1	69	1095	6,3	8,0	3,5	0,9
8000	-1	69	1089	7,2	9,4	4,0	1,1
9000	-3	69	1053	8,1	10,8	4,4	1,2
10000	-5	69	1017	9,1	12,2	4,8	1,3
11000	-7	69	980	10,1	13,8	5,2	1,4
12000	-9	69	944	11,1	15,5	5,6	1,5
13000	-11	69	907	12,2	17,3	6,0	1,6
14000	-13	69	870	13,3	19,2	6,4	1,7
15000	-15	69	833	14,5	21,2	6,8	1,8
16000	-17	69	796	15,7	23,4	7,1	1,9
17000	-19	69	758	17,0	25,7	7,5	2,0
18000	-21	69	721	18,4	28,3	7,8	2,1

Figure 5-3a Time, Fuel and Distance to Climb at 907 kg (2000 lbs)

TIME, FUEL AND DISTANCE TO CLIMB at 1043 kg (2300 lbs)

Conditions:

Take-off weight 1043 kg (2300 lbs)
Climb speed v_y = 69 KIAS/ 79 mph
Flaps Up
Full Power
Standard Temperature (ISA)

Notes :

1. Add 4 l (1.1 US gal) of fuel for engine start, taxi and takeoff allowance.
2. Increase time and distance by 10% for 10°C above standard temperature.
3. Distances shown are based on zero wind.
4. Time, distance and fuel required are only valid from the point where the airplane climbs at v_y = 69 KIAS.

Press. Alt [ft]	OAT [°C]	Vy [KIAS]	ROC [FPM]	Time [MIN]	Distance [NM]	Fuel used [l]	Fuel used [US Gal]
0	15	69	914	0,0	0,0	0,0	0,0
1000	13	69	908	1,1	1,3	0,6	0,2
2000	11	69	903	2,2	2,6	1,2	0,3
3000	9	69	897	3,3	4,0	1,9	0,5
4000	7	69	891	4,4	5,4	2,5	0,7
5000	5	69	885	5,6	6,9	3,1	0,8
6000	3	69	878	6,7	8,5	3,7	1,0
7000	1	69	872	7,8	10,1	4,4	1,2
8000	-1	69	865	9,0	11,8	5,0	1,3
9000	-3	69	832	10,2	13,5	5,6	1,5
10000	-5	69	800	11,4	15,4	6,1	1,6
11000	-7	69	767	12,7	17,4	6,6	1,7
12000	-9	69	733	14,0	19,5	7,1	1,9
13000	-11	69	700	15,4	21,8	7,6	2,0
14000	-13	69	667	16,9	24,3	8,1	2,1
15000	-15	69	633	18,4	27,0	8,6	2,3
16000	-17	69	599	20,0	29,8	9,1	2,4
17000	-19	69	565	21,7	32,9	9,6	2,5
18000	-21	69	530	23,6	36,3	10,1	2,7

Figure 5-3b Time, Fuel and Distance to Climb at 1043 kg (2300 lbs)

CRUISE PERFORMANCE, RANGE AND ENDURANCE WITH STANDARD TANKS, Cessna 172 F,G,H at 1043 kg (2300 lbs)

Conditions:

Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

Notes:

1. Endurance information is based on standard tanks with 114.3 l (30.2 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb 45 min. reserve.
3. Increase true airspeed (KTAS/mph)) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
SL	100	125	144	33,6	8,9	317	2,5
SL	90	120	139	29,6	7,8	358	3,0
SL	80	115	133	25,8	6,8	407	3,5
SL	70	110	126	22,1	5,8	465	4,2
SL	60	103	119	18,6	4,9	535	5,2
SL	50	95	110	15,3	4,0	616	6,5
2000	100	127	147	33,6	8,9	320	2,4
2000	90	123	141	29,6	7,8	362	2,9
2000	80	118	135	25,8	6,8	411	3,4
2000	70	112	129	22,1	5,8	469	4,1
2000	60	105	121	18,6	4,9	539	5,1
2000	50	97	112	15,3	4,0	620	6,3
4000	100	130	149	33,6	8,9	324	2,4
4000	90	125	144	29,6	7,8	366	2,8
4000	80	120	138	25,8	6,8	414	3,3
4000	70	114	131	22,1	5,8	473	4,0
4000	60	107	123	18,6	4,9	543	4,9
4000	50	98	113	15,3	4,0	622	6,2
6000	100	132	152	33,6	8,9	327	2,3
6000	90	127	147	29,6	7,8	370	2,7
6000	80	122	140	25,8	6,8	418	3,2
6000	70	116	133	22,1	5,8	477	3,9
6000	60	109	125	18,6	4,9	546	4,8
6000	50	100	115	15,3	4,0	625	6,0
8000	100	135	155	33,6	8,9	331	2,2
8000	90	130	150	29,6	7,8	373	2,6
8000	80	124	143	25,8	6,8	422	3,1
8000	70	118	136	22,1	5,8	481	3,8
8000	60	111	127	18,6	4,9	550	4,7
8000	50	101	117	15,3	4,0	627	5,9
10000	90	133	153	29,6	7,8	377	2,5
10000	80	127	146	25,8	6,8	426	3,0

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
10000	70	120	138	22,1	5,8	484	3,6
10000	60	113	130	18,6	4,9	553	4,5
10000	50	103	118	15,3	4,0	628	5,7
12000	90	135	156	29,6	7,8	380	2,3
12000	80	129	149	25,8	6,8	429	2,8
12000	70	123	141	22,1	5,8	488	3,5
12000	60	115	132	18,6	4,9	555	4,4
12000	50	104	120	15,3	4,0	628	5,5
14000	90	138	159	29,6	7,8	384	2,2
14000	80	132	152	25,8	6,8	433	2,7
14000	70	125	144	22,1	5,8	491	3,4
14000	60	117	134	18,6	4,9	558	4,2
14000	50	105	121	15,3	4,0	628	5,3
16000	80	134	155	25,8	6,8	436	2,6
16000	70	127	146	22,1	5,8	494	3,2
16000	60	118	136	18,6	4,9	560	4,0
16000	50	107	123	15,3	4,0	626	5,1
18000	80	137	158	25,8	6,8	442	2,4
18000	70	130	149	22,1	5,8	500	3,0
18000	60	120	139	18,6	4,9	564	3,8
18000	50	107	124	15,3	4,0	626	4,9

Figure 5-4a Cruise Performance, Range and Endurance with standard tanks F,G,H; 1043 kg (2300 lbs)

CRUISE PERFORMANCE, RANGE AND ENDURANCE WITH STANDARD TANKS, Cessna 172 I,K,L,M at 1043 kg (2300 lbs)

Conditions:
Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

Notes:

1. Endurance information is based on standard tanks with 120.0 l (31.7 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb 45 min. reserve.
3. Increase true airspeed (KTAS/mph)) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
SL	100	125	144	33,6	8,9	338	2,7
SL	90	120	139	29,6	7,8	382	3,2
SL	80	115	133	25,8	6,8	432	3,7
SL	70	110	126	22,1	5,8	494	4,5
SL	60	103	119	18,6	4,9	567	5,5
SL	50	95	110	15,3	4,0	652	6,8
2000	100	127	147	33,6	8,9	342	2,6
2000	90	123	141	29,6	7,8	386	3,1
2000	80	118	135	25,8	6,8	436	3,6
2000	70	112	129	22,1	5,8	498	4,4
2000	60	105	121	18,6	4,9	571	5,4
2000	50	97	112	15,3	4,0	656	6,7
4000	100	130	149	33,6	8,9	346	2,5
4000	90	125	144	29,6	7,8	390	3,0
4000	80	120	138	25,8	6,8	441	3,5
4000	70	114	131	22,1	5,8	503	4,3
4000	60	107	123	18,6	4,9	575	5,2
4000	50	98	113	15,3	4,0	659	6,5
6000	100	132	152	33,6	8,9	350	2,4
6000	90	127	147	29,6	7,8	394	2,9
6000	80	122	140	25,8	6,8	445	3,4
6000	70	116	133	22,1	5,8	507	4,2
6000	60	109	125	18,6	4,9	580	5,1
6000	50	100	115	15,3	4,0	662	6,4
8000	100	135	155	33,6	8,9	354	2,3
8000	90	130	150	29,6	7,8	398	2,8
8000	80	124	143	25,8	6,8	449	3,3
8000	70	118	136	22,1	5,8	511	4,0
8000	60	111	127	18,6	4,9	584	5,0
8000	50	101	117	15,3	4,0	665	6,2
10000	90	133	153	29,6	7,8	402	2,7
10000	80	127	146	25,8	6,8	454	3,2

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
10000	70	120	138	22,1	5,8	515	3,9
10000	60	113	130	18,6	4,9	587	4,8
10000	50	103	118	15,3	4,0	666	6,1
12000	90	135	156	29,6	7,8	406	2,5
12000	80	129	149	25,8	6,8	458	3,1
12000	70	123	141	22,1	5,8	519	3,8
12000	60	115	132	18,6	4,9	591	4,7
12000	50	104	120	15,3	4,0	667	5,9
14000	90	138	159	29,6	7,8	410	2,4
14000	80	132	152	25,8	6,8	462	2,9
14000	70	125	144	22,1	5,8	523	3,6
14000	60	117	134	18,6	4,9	594	4,5
14000	50	105	121	15,3	4,0	667	5,7
16000	80	134	155	25,8	6,8	466	2,8
16000	70	127	146	22,1	5,8	527	3,4
16000	60	118	136	18,6	4,9	596	4,3
16000	50	107	123	15,3	4,0	666	5,5
18000	80	137	158	25,8	6,8	472	2,6
18000	70	130	149	22,1	5,8	533	3,3
18000	60	120	139	18,6	4,9	601	4,1
18000	50	107	124	15,3	4,0	666	5,3

Figure 5-4b Cruise Performance, Range and Endurance
with standard tanks I,K,L,M; 1043 kg (2300 lbs)

CRUISE PERFORMANCE, RANGE AND ENDURANCE WITH LONG-RANGE TANKS Cessna 172 F-M at 1043 kg (2300 lbs)

Conditions:
Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

Notes:

1. Endurance information is based on long-range tanks with 152.2 l (40.2 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb 45 min. reserve.
3. Increase true airspeed (KTAS/mph)) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

Press Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
SL	100	125	144	33,6	8,9	458	3,7
SL	90	120	139	29,6	7,8	513	4,3
SL	80	115	133	25,8	6,8	576	5,0
SL	70	110	126	22,1	5,8	654	6,0
SL	60	103	119	18,6	4,9	745	7,2
SL	50	95	110	15,3	4,0	853	8,9
2000	100	127	147	33,6	8,9	464	3,6
2000	90	123	141	29,6	7,8	519	4,2
2000	80	118	135	25,8	6,8	583	4,9
2000	70	112	129	22,1	5,8	661	5,8
2000	60	105	121	18,6	4,9	753	7,1
2000	50	97	112	15,3	4,0	860	8,8
4000	100	130	149	33,6	8,9	470	3,5
4000	90	125	144	29,6	7,8	526	4,1
4000	80	120	138	25,8	6,8	590	4,8
4000	70	114	131	22,1	5,8	669	5,7
4000	60	107	123	18,6	4,9	761	7,0
4000	50	98	113	15,3	4,0	866	8,6
6000	100	132	152	33,6	8,9	477	3,4
6000	90	127	147	29,6	7,8	533	4,0
6000	80	122	140	25,8	6,8	598	4,7
6000	70	116	133	22,1	5,8	676	5,6
6000	60	109	125	18,6	4,9	768	6,8
6000	50	100	115	15,3	4,0	872	8,5
8000	100	135	155	33,6	8,9	483	3,3
8000	90	130	150	29,6	7,8	539	3,9
8000	80	124	143	25,8	6,8	605	4,6
8000	70	118	136	22,1	5,8	683	5,5
8000	60	111	127	18,6	4,9	775	6,7
8000	50	101	117	15,3	4,0	878	8,3
10000	90	133	153	29,6	7,8	546	3,7
10000	80	127	146	25,8	6,8	612	4,4

Press. Alt.	Load	Speed		Fuel Flow		Distance	Endu-rance Time
[ft]	[%]	[KTAS]	[mph]	[l/h]	[US Gal/h]	[NM]	[Hrs]
10000	70	120	138	22,1	5,8	691	5,4
10000	60	113	130	18,6	4,9	782	6,5
10000	50	103	118	15,3	4,0	883	8,2
12000	90	135	156	29,6	7,8	553	3,6
12000	80	129	149	25,8	6,8	619	4,3
12000	70	123	141	22,1	5,8	698	5,2
12000	60	115	132	18,6	4,9	789	6,4
12000	50	104	120	15,3	4,0	896	8,0
14000	90	138	159	29,6	7,8	560	3,5
14000	80	132	152	25,8	6,8	626	4,2
14000	70	125	144	22,1	5,8	705	5,1
14000	60	117	134	18,6	4,9	795	6,2
14000	50	105	121	15,3	4,0	869	7,8
16000	80	134	155	25,8	6,8	634	4,0
16000	70	127	146	22,1	5,8	712	4,9
16000	60	118	136	18,6	4,9	801	6,1
16000	50	107	123	15,3	4,0	890	7,6
18000	80	137	158	25,8	6,8	644	3,9
18000	70	130	149	22,1	5,8	722	4,8
18000	60	120	139	18,6	4,9	810	5,9
18000	50	107	124	15,3	4,0	892	7,4

Figure 5-4c Cruise Performance, Range and Endurance with long range tanks, 1043 kg (2300 lbs)

SECTION 6
APPENDIX

▲ **WARNING:**	Do not start the engine in any case when filling levels are below the corresponding minimum marking.
■ CAUTION:	Normally, a refill of coolant or gearbox oil between service intervals is not necessary. In case of low coolant or gearbox oil levels, inform the maintenance company immediately.

ENGINE OIL

The TAE 125-02-114 engine variants are filled with 4.5 - 6 l engine oil (refer to section 1 of this supplement for specification).
A dip stick is used to check the oil level. It is accessible by a flap on the upper right-hand side of the engine cowling.
Notice that on warm engines 5 minutes after engine shut-off there are 80% of the entire engine oil in the oil pan and therefore visible on the oil dipstick. On warm engines oil should be added if the oil dip stick shows oil levels below 50%. After 30 minutes the real oil level is visible on the dip stick.
The drain screw is located on the lower left-hand outside of the oil pan, the oil filter is on the upper left-hand side of the housing.
The oil system has to be checked for sealing after the first 5 operating hours (visual inspection).
Checks and changes of oil and oil filter have to be performed regularly according to the engine Operation and Maintenance Manual, see OM-02-02. The Supplement of the Aircraft Maintenance Manual has to be considered as well, see AMM-20-02.

GEARBOX OIL

To ensure the necessary propeller speed, the engine is equipped with a reduction gearbox filled with gearbox oil. (refer to section 1 of this supplement for specification).
The level can be checked through a viewing glass on the lower leading edge of the gearbox. To do so open the flap on the left front side of the engine cowling.
The drain screw is located at the lowest point of the gearbox. A filter is installed upstream of the pump, as well as microfilter in the Constant Speed Unit. Check the gearbox for sealing after the first 5 hours of operation (visual inspection). Regular checks as well as oil and filter changes have to be performed in according with the engine Operation and Maintenance Manual, see OM-02-02. The Supplement of the Aircraft Maintenance Manual has to be considered as well, see AMM-20-02.

▲ **WARNING:**	It is not allowed to start the engine with low gearbox oil level.
■ CAUTION:	Between scheduled maintenance topping up gearbox oil should not be necessary. If low gearbox oil level is detected, inform your service centre immediately.

FUEL

The TAE 125-02-114 engine can be operated with kerosene (JET A-1, Jet A, Fuel No.3, JP-8, TS-1) or Diesel fuel.
Due to the higher specific density of turbine engine fuel or Diesel in comparison to aviation gasoline (AVGAS) the permissible capacity for standard tanks was reduced as mentioned in Section 1.
Appropriate placards are attached near the fuel filler connections. For temperature limitations refer to Section 2 "Limitations" and Section 4 "Normal Operation".

It is recommended to refuel before each flight and to enter the type of fuel into the log-book.

COOLANT

To cool the engine a liquid cooling system was installed with a water/approved radiator protection mixture at a ratio of 1:1.
A heat exchanger for cabin heating is part of the cooling system.
Check the cooling system for sealing after the first 5 hours of operation (visual inspection).
The coolant has to be changed in accordance with the engine Operations and Maintenance Manual, see OM-02-02. The Supplement of the Aircraft Maintenance Manual has to be considered as well, see AMM-20-02.

▲ **WARNING:** It is not allowed to start the engine with low coolant level.

■ CAUTION: Between scheduled maintenance topping-up coolant should not be necessary. If low coolant level is detected, inform your service centre immediately.

■ CAUTION: The water has to satisfy the following requirements:

1. visual appearance: colorless, clear and no deposits allowed
2. pH-value: 6.5 to 8.5
3. maximum water hardness: 2.7 mmol/l
4. maximum hydrogen carbonate concentration: 100 mg/l
5. maximum chloride concentration: 100 mg/l
6. maximum sulfate concentration: 100 mg/l

◆ Note: The freezing point of the coolant is -36°C.

◆ Note: The waterworks also provide information. In general, tap water may be diluted with distilled water. Pure distilled water may not be used to mix with approved radiator protection.

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SECTION 7
SUPPLEMENTS

TABLE OF CONTENTS

No supplements

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CESSNA

MORE PEOPLE BUY AND FLY CESSNA AIRPLANES THAN ANY OTHER MAKE

1970

WORLD'S LARGEST PRODUCER OF GENERAL AVIATION AIRCRAFT SINCE 1956

MODEL 172 AND SKYHAWK

OWNER'S MANUAL

PERFORMANCE - SPECIFICATIONS

	Model 172*	Skyhawk*
GROSS WEIGHT	2300 lbs	2300 lbs
SPEED:		
Top Speed at Sea Level	139 mph	140 mph
Cruise, 75% Power at 9000 ft	131 mph	132 mph
RANGE:		
Cruise, 75% Power at 9000 ft 38 Gal., No Reserve	615 mi 4.7 hrs	620 mi 4.7 hrs
	131 mph	132 mph
Cruise, 75% Power at 9000 ft 48 Gal., No Reserve	775 mi 5.9 hrs	780 mi 5.9 hrs
	131 mph	132 mph
Optimum Range at 10,000 ft 38 Gal., No Reserve	640 mi 5.5 hrs	655 mi 5.5 hrs
	117 mph	118 mph
Optimum Range at 10,000 ft 48 Gal., No Reserve	820 mi 7.0 hrs	830 mi 7.0 hrs
	117 mph	118 mph
RATE OF CLIMB AT SEA LEVEL	645 fpm	645 fpm
SERVICE CEILING	13,100 ft	13,100 ft
TAKE-OFF:		
Ground Run	865 ft	865 ft
Total Distance Over 50-Foot Obstacle	1525 ft	1525 ft
LANDING:		
Ground Roll	520 ft	520 ft
Total Distance Over 50-Foot Obstacle	1250 ft	1250 ft
STALL SPEED:		
Flaps Up, Power Off	57 mph	57 mph
Flaps Down, Power Off	49 mph	49 mph
EMPTY WEIGHT (Approximate)	1245 lbs	1315 lbs
USEFUL LOAD	1055 lbs	985 lbs
BAGGAGE	120 lbs	120 lbs
WING LOADING: Pounds/Sq Foot	13.2	13.2
POWER LOADING: Pounds/HP	15.3	15.3
FUEL CAPACITY: Total		
Standard Tanks	42 gal.	42 gal.
Optional Long Range Tanks	52 gal.	52 gal.
OIL CAPACITY: Total	8 qts	8 qts
PROPELLER: Fixed Pitch (Diameter)	76 inches	76 inches
ENGINE:		
Lycoming Engine 150 rated HP at 2700 RPM	O-320-E2D	O-320-E2D

*This manual covers operation of the Model 172/Skyhawk which is certificated as Model 172K under FAA Type Certificate No. 3A12.

D742-13-RAND-1500-10/73

CONGRATULATIONS . . . . . .

Welcome to the ranks of Cessna owners! Your Cessna has been designed and constructed to give you the most in performance, economy, and comfort. It is our desire that you will find flying it, either for business or pleasure, a pleasant and profitable experience.

This Owner's Manual has been prepared as a guide to help you get the most pleasure and utility from your Model 172/Skyhawk. It contains information about your Cessna's equipment, operating procedures, and performance; and suggestions for its servicing and care. We urge you to read it from cover to cover, and to refer to it frequently.

Our interest in your flying pleasure has not ceased with your purchase of a Cessna. World-wide, the Cessna Dealer Organization backed by the Cessna Service Department stands ready to serve you. The following services are offered by most Cessna Dealers:

FACTORY TRAINED PERSONNEL to provide you with courteous expert service.

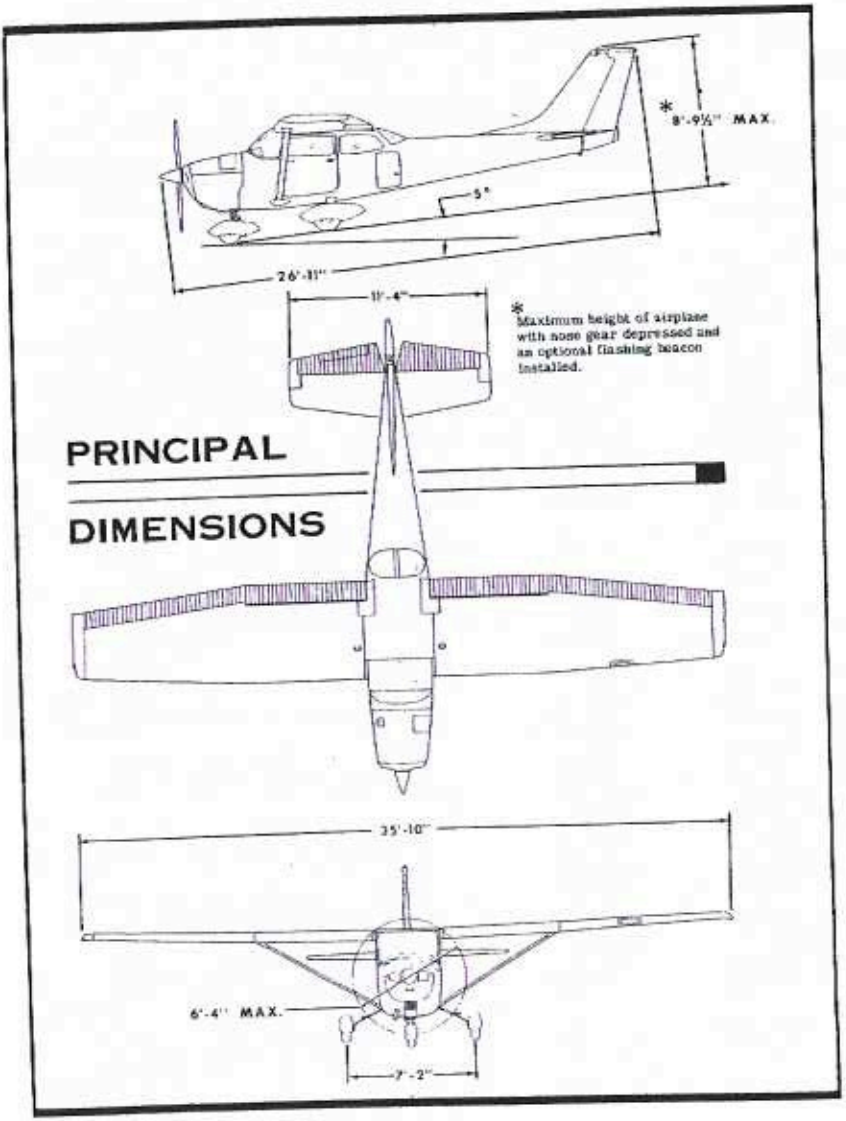
FACTORY APPROVED SERVICE EQUIPMENT to provide you with the most efficient and accurate workmanship possible.

A STOCK OF GENUINE CESSNA SERVICE PARTS on hand when you need them.

THE LATEST AUTHORITATIVE INFORMATION FOR SERVICING CESSNA AIRPLANES, since Cessna Dealers have all of the Service Manuals and Parts Catalogs, kept current by Service Letters and Service News Letters, published by Cessna Aircraft Company.

We urge all Cessna owners to use the Cessna Dealer Organization to the fullest.

A current Cessna Dealer Directory accompanies your new airplane. The Directory is revised frequently, and a current copy can be obtained from your Cessna Dealer. Make your Directory one of your cross-country flight planning aids; a warm welcome awaits you at every Cessna Dealer.


* 8'-9½" MAX.
5°
26'-11"
11'-4"
* Maximum height of airplane with nose gear depressed and an optional flashing beacon installed.
PRINCIPAL
DIMENSIONS
35'-10"
6'-4" MAX.
7'-2"

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This manual describes the operation and performance of both the Cessna Model 172 and Skyhawk. Equipment described as "Optional" denotes that the subject equipment is optional on the Model 172. Much of this equipment is standard on the Skyhawk model.

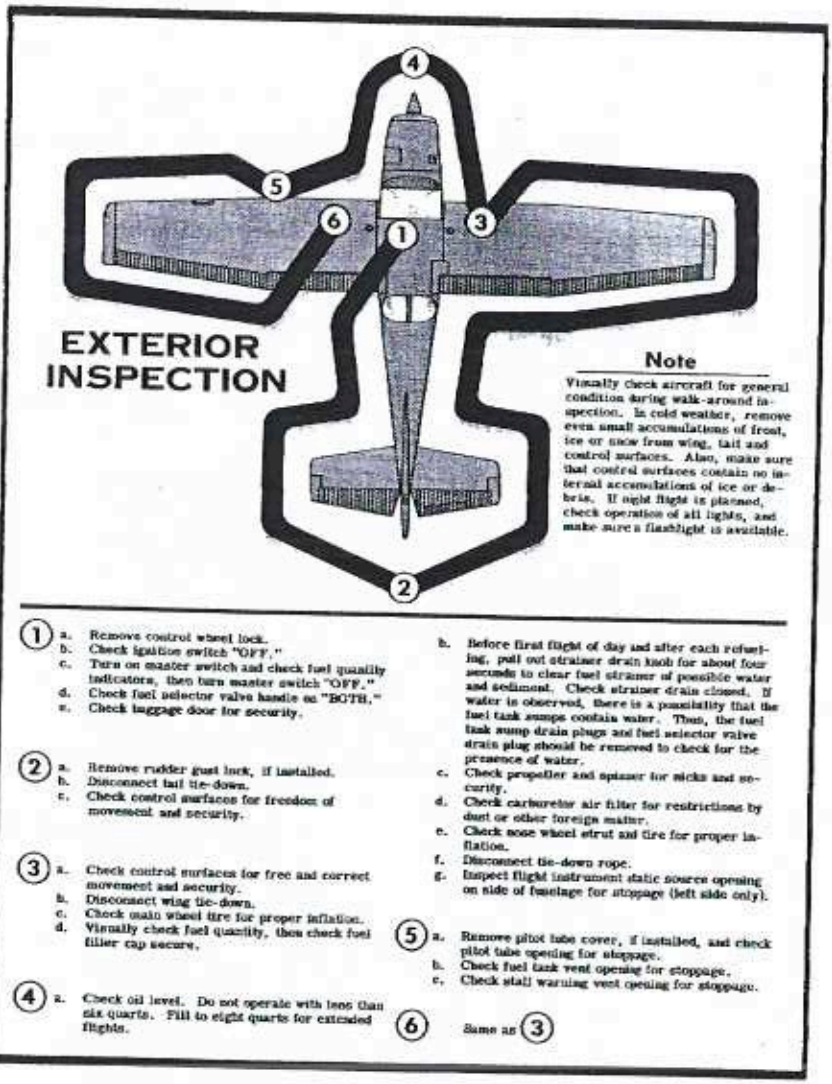

EXTERIOR INSPECTION
Note
Visually check aircraft for general condition during walk-around inspection. In cold weather, remove even small accumulations of frost, ice or snow from wing, tail and control surfaces. Also, make sure that control surfaces contain no internal accumulations of ice or debris. If night flight is planned, check operation of all lights, and make sure a flashlight is available.
1
a. Remove control wheel lock.
b. Check ignition switch "OFF."
c. Turn on master switch and check fuel quantity indicators, then turn master switch "OFF."
d. Check fuel selector valve handle on "BOTH."
e. Check luggage door for security.
2
a. Remove rudder gust lock, if installed.
b. Disconnect tail tie-down.
c. Check control surfaces for freedom of movement and security.
3
a. Check control surfaces for free and correct movement and security.
b. Disconnect wing tie-down.
c. Check main wheel tire for proper inflation.
d. Visually check fuel quantity, then check fuel filler cap secure.
4
a. Check oil level. Do not operate with less than six quarts. Fill to eight quarts for extended flights.
b. Before first flight of day and after each refueling, pull out strainer drain knob for about four seconds to clear fuel strainer of possible water and sediment. Check strainer drain closed. If water is observed, there is a possibility that the fuel tank sumps contain water. Thus, the fuel tank sump drain plugs and fuel selector valve drain plug should be removed to check for the presence of water.
c. Check propeller and spinner for nicks and security.
d. Check carburetor air filter for restrictions by dust or other foreign matter.
e. Check nose wheel strut and tire for proper inflation.
f. Disconnect tie-down rope.
g. Inspect flight instrument static source opening on side of fuselage for stoppage (left side only).
5
a. Remove pitot tube cover, if installed, and check pitot tube opening for stoppage.
b. Check fuel tank vent opening for stoppage.
c. Check stall warning vent opening for stoppage.
6
Same as 3

Figure 1-1.

Section I

OPERATING CHECK LIST

One of the first steps in obtaining the utmost performance, service, and flying enjoyment from your Cessna is to familiarize yourself with your airplane's equipment, systems, and controls. This can best be done by reviewing this equipment while sitting in the airplane. Those items whose function and operation are not obvious are covered in Section II.

Section I lists, in Pilot's Check List form, the steps necessary to operate your airplane efficiently and safely. It is not a check list in its true form as it is considerably longer, but it does cover briefly all of the points that you should know for a typical flight.

The flight and operational characteristics of your airplane are normal in all respects. There are no "unconventional" characteristics or operations that need to be mastered. All controls respond in the normal way within the entire range of operation. All airspeeds mentioned in Sections I, II and III are indicated airspeeds. Corresponding calibrated airspeed may be obtained from the Airspeed Correction Table in Section VI.

BEFORE ENTERING THE AIRPLANE.

(1) Make an exterior inspection in accordance with figure 1-1.

BEFORE STARTING THE ENGINE.

(1) Seats and Seat Belts -- Adjust and lock.
(2) Fuel Selector Valve -- "BOTH."
(3) Brakes -- Test and set.
(4) Radios and Electrical Equipment -- "OFF."

STARTING THE ENGINE.

(1) Mixture -- Rich.
(2) Carburetor Heat -- Cold.
(3) Primer -- 2 - 6 strokes (as required; none if engine is warm). Close and lock primer.
(4) Throttle -- Open 1/8".
(5) Master Switch -- "ON."
(6) Propeller Area -- Clear.
(7) Ignition Switch -- "START" (release when engine starts).
(8) Oil Pressure -- Check.

BEFORE TAKE-OFF.

(1) Parking Brake -- Set.
(2) Flight Controls -- Check for free and correct movement.
(3) Fuel Selector Valve -- "BOTH."
(4) Elevator Trim -- "TAKE-OFF" setting.
(5) Throttle Setting -- 1700 RPM.
(6) Engine Instruments and Ammeter -- Check.
(7) Suction Gage -- Check (4.6 to 5.4 inches of mercury).
(8) Magnetos -- Check (RPM drop should not exceed 125 RPM on either magneto or 50 RPM differential between magnetos).
(9) Carburetor Heat -- Check operation.
(10) Flight Instruments and Radios -- Set.
(11) Optional Autopilot or Wing Leveler -- Off.
(12) Cabin Doors and Window -- Closed and locked.

TAKE-OFF.

NORMAL TAKE-OFF.

(1) Wing Flaps -- 0°.
(2) Carburetor Heat -- Cold.
(3) Power -- Full throttle.
(4) Elevator Control -- Lift nose wheel at 60 MPH.
(5) Climb Speed -- 75 to 85 MPH.

MAXIMUM PERFORMANCE TAKE-OFF.

(1) Wing Flaps -- 0°.

(2) Carburetor Heat -- Cold.
(3) Brakes -- Apply.
(4) Power -- Full throttle.
(5) Brakes -- Release.
(6) Airplane Attitude -- Slightly tail low.
(7) Climb Speed -- 68 MPH until all obstacles are cleared.

CLIMB.

(1) Airspeed -- 80 to 90 MPH.

NOTE

If a maximum performance climb is necessary, use speeds shown in the Maximum Rate-Of-Climb Data chart in Section VI.

(2) Power -- Full throttle.
(3) Mixture -- Full rich (mixture may be leaned above 5000 feet).

CRUISING.

(1) Power -- 2200 to 2700 RPM.

NOTE

Maximum cruise RPM varies with altitude. For details, refer to Section IV.

(2) Trim Tab -- Adjust.
(3) Mixture -- Lean for maximum RPM.

LET-DOWN.

(1) Mixture -- Rich.
(2) Power -- As desired.
(3) Carburetor Heat -- As required to prevent carburetor icing.

BEFORE LANDING.

(1) Fuel Selector Valve -- "BOTH."
(2) Mixture -- Rich.
(3) Carburetor Heat -- Apply full heat before closing throttle.
(4) Wing Flaps -- As desired.
(5) Airspeed -- 70 to 80 MPH (flaps up), 65 to 75 MPH (flaps down).

BALKED LANDING (GO-AROUND).

(1) Power -- Full throttle.
(2) Carburetor Heat -- Cold.
(3) Wing Flaps -- Retract to 20°.
(4) Upon reaching an airspeed of approximately 65 MPH, retract flaps slowly.

NORMAL LANDING.

(1) Touchdown -- Main wheels first.
(2) Landing Roll -- Lower nose wheel gently.
(3) Braking -- Minimum required.

AFTER LANDING.

(1) Wing Flaps -- Up.
(2) Carburetor Heat -- Cold.

SECURING AIRCRAFT.

(1) Parking Brake -- Set.
(2) Radios and Electrical Equipment -- "OFF."
(3) Mixture -- Idle cut-off (pulled full out).
(4) Ignition and Master Switch -- "OFF."
(5) Control Lock -- Installed.

MODIFIED FUEL MANAGEMENT PROCEDURES

With a combination of highly volatile fuel, high fuel temperature, high operating altitude, and low fuel flow rate in the tank outlet lines, there is a remote possibility of accumulating fuel vapor and encountering power irregularities on some airplanes. To minimize this possibility, the following operating procedures are recommended:

(1) Take-off and climb to cruise altitude on "both" tanks. (This is consistent with current recommendations.)
(2) When reaching cruise altitude above 5000 feet MSL, promptly switch the fuel selector valve from "both" tanks to either the "right" or "left" tank.
(3) During cruise, use "left" and "right" tank as required.
(4) Select "both" tanks for landing as currently recommended.

POWER RECOVERY TECHNIQUES

In the remote event that vapor is present in sufficient amounts to cause a power irregularity, the following power recovery techniques should be followed:

OPERATION ON A SINGLE TANK

Should power irregularities occur when operating on a single tank, power can be restored immediately by switching to the opposite tank. In addition, the vapor accumulation in the tank on which the power irregularity occurred will rapidly dissipate itself such that that tank will also be available for normal operation after it has been unused for approximately one (1) minute.

OPERATION ON BOTH TANKS

Should power irregularities occur with the fuel selector on both tanks, the following steps are to be taken to restore power:

(1) Switch to a single tank for a period of 60 seconds.
(2) Then switch to the opposite tank and power will be restored.

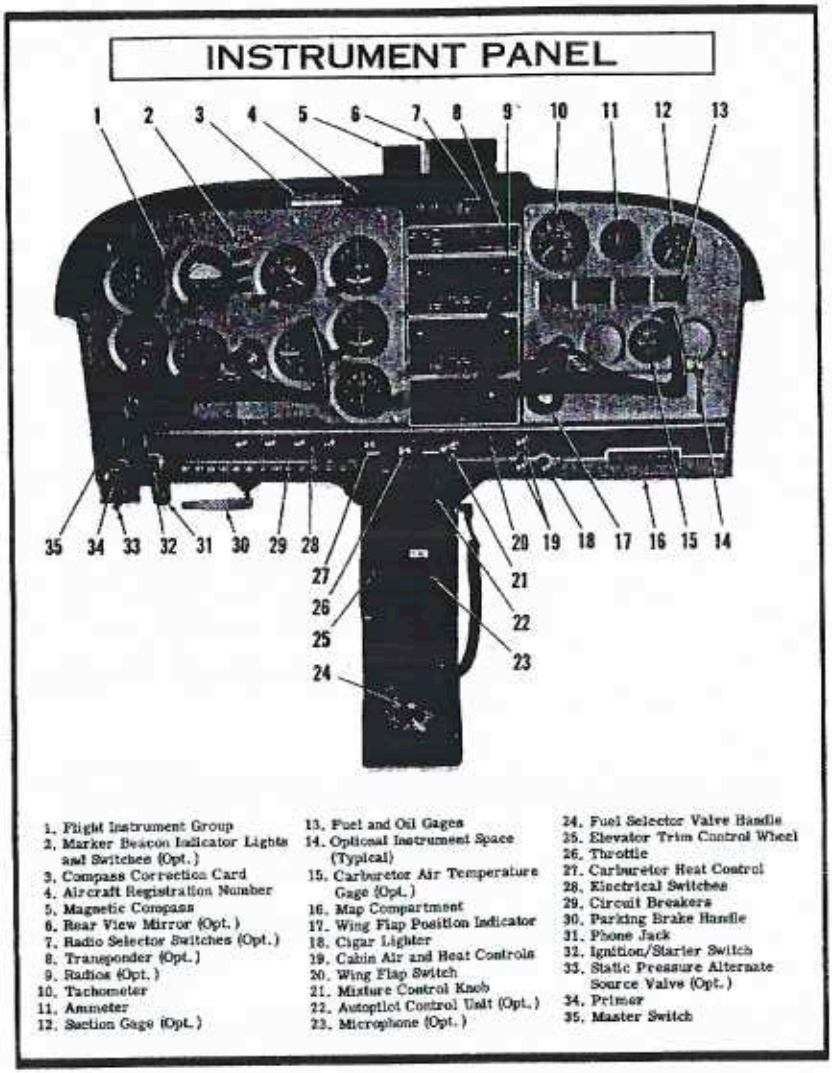

INSTRUMENT PANEL
1 2 3 4 5 6 7 8 9 10 11 12 13
35 34 33 32 31 30 29 28 27 26 25 24 20 19 18 17 16 15 14 21 22 23
1. Flight Instrument Group
2. Marker Beacon Indicator Lights and Switches (Opt.)
3. Compass Correction Card
4. Aircraft Registration Number
5. Magnetic Compass
6. Rear View Mirror (Opt.)
7. Radio Selector Switches (Opt.)
8. Transponder (Opt.)
9. Radios (Opt.)
10. Tachometer
11. Ammeter
12. Suction Gage (Opt.)
13. Fuel and Oil Gages
14. Optional Instrument Space (Typical)
15. Carburetor Air Temperature Gage (Opt.)
16. Map Compartment
17. Wing Flap Position Indicator
18. Cigar Lighter
19. Cabin Air and Heat Controls
20. Wing Flap Switch
21. Mixture Control Knob
22. Autopilot Control Unit (Opt.)
23. Microphone (Opt.)
24. Fuel Selector Valve Handle
25. Elevator Trim Control Wheel
26. Throttle
27. Carburetor Heat Control
28. Electrical Switches
29. Circuit Breakers
30. Parking Brake Handle
31. Phone Jack
32. Ignition/Starter Switch
33. Static Pressure Alternate Source Valve (Opt.)
34. Primer
35. Master Switch

Figure 2-1.

Section II

DESCRIPTION AND OPERATING DETAILS

The following paragraphs describe the systems and equipment whose function and operation is not obvious when sitting in the airplane. This section also covers in somewhat greater detail some of the items listed in Check List form in Section I that require further explanation.

FUEL SYSTEM.

Fuel is supplied to the engine from two tanks, one in each wing. With the fuel selector valve on "BOTH," the total usable fuel for all flight conditions is 38 gallons for the standard tanks and 48 gallons for the optional long range tanks.

Fuel from each wing tank flows by gravity to a selector valve. Depending upon the setting of the selector valve, fuel from the left, right, or both tanks flows through a fuel strainer and carburetor to the engine induction system.

IMPORTANT

The fuel selector valve should be in the "BOTH" position for take-off, climb, landing, and maneuvers that involve prolonged slips or skids. Operation from either "LEFT" or "RIGHT" tank is reserved for cruising flight.

NOTE

When the fuel selector valve handle is in the "BOTH" position in cruising flight, unequal fuel flow from each tank may occur if the wings are not maintained exactly level. Resulting wing heaviness can be alleviated gradually by turning the selector valve handle to the tank in the "heavy" wing.

For fuel system servicing information, refer to Lubrication and Servicing Procedures in Section V.

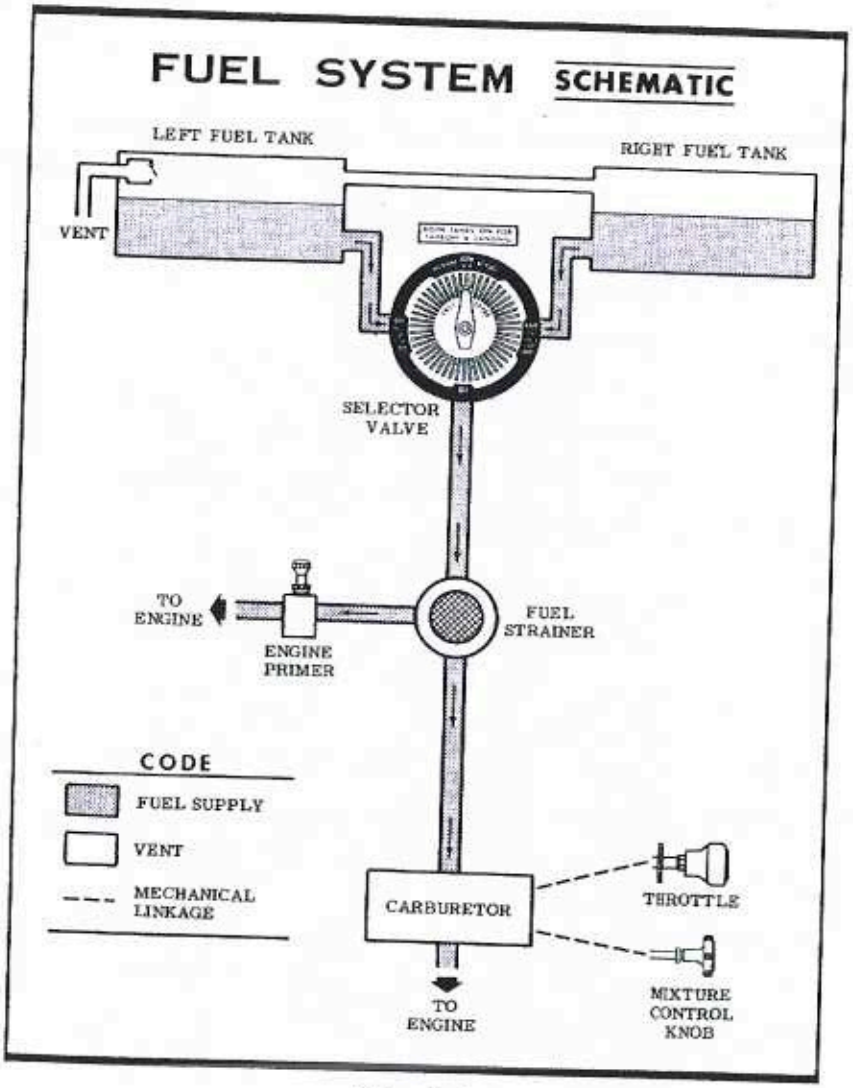

FUEL SYSTEM SCHEMATIC
LEFT FUEL TANK
RIGHT FUEL TANK
VENT
SELECTOR VALVE
TO ENGINE
ENGINE PRIMER
FUEL STRAINER
CODE
FUEL SUPPLY
VENT
MECHANICAL LINKAGE
CARBURETOR
THROTTLE
MIXTURE CONTROL KNOB
TO ENGINE

Figure 2-2.

ELECTRICAL SYSTEM.

Electrical energy is supplied by a 14-volt, direct-current system powered by an engine-driven alternator (see figure 2-3). A 12- volt battery is located on the left-hand forward portion of the firewall. Power is supplied to all electrical circuits through a split bus bar, one side containing electronic systems and the other side having general electrical systems. Both sides of the bus are on at all times except when either an external power source is connected or the ignition/starter switch is turned on; then a power contactor is automatically activated to open the circuit to the electronic bus. Isolating the electronic circuits in this manner prevents harmful transient voltages from damaging the transistors in the electronic equipment.

MASTER SWITCH.

The master switch is a split-rocker type switch labeled "MASTER," and is "ON" in the up position and "OFF" in the down position. The right half of the switch, labeled "BAT," controls all electrical power to the airplane. The left half, labeled "ALT" controls the alternator.

Normally, both sides of the master switch should be used simultaneously, however, the "BAT" side of the switch could be turned "ON" separately to check equipment while on the ground. The "ALT" side of the switch, when placed in the "OFF" position, removes the alternator from the electrical system. With this switch in the "OFF" position, the entire electrical load is placed on the battery, and all non-essential electrical equipment should be turned off for the remainder of the flight.

AMMETER.

The ammeter indicates the flow of current, in amperes, from the alternator to the battery or from the battery to the aircraft electrical system. When the engine is operating and the master switch is "ON," the ammeter indicates the charging rate applied to the battery. In the event the alternator is not functioning or the electrical load exceeds the output of the alternator, the ammeter indicates the discharge rate of the battery.

CIRCUIT BREAKERS AND FUSES.

The majority of electrical circuits in the airplane are protected by "push-to-reset" circuit breakers mounted on the instrument panel. Exceptions to this are the optional clock, flight hour recorder, and battery

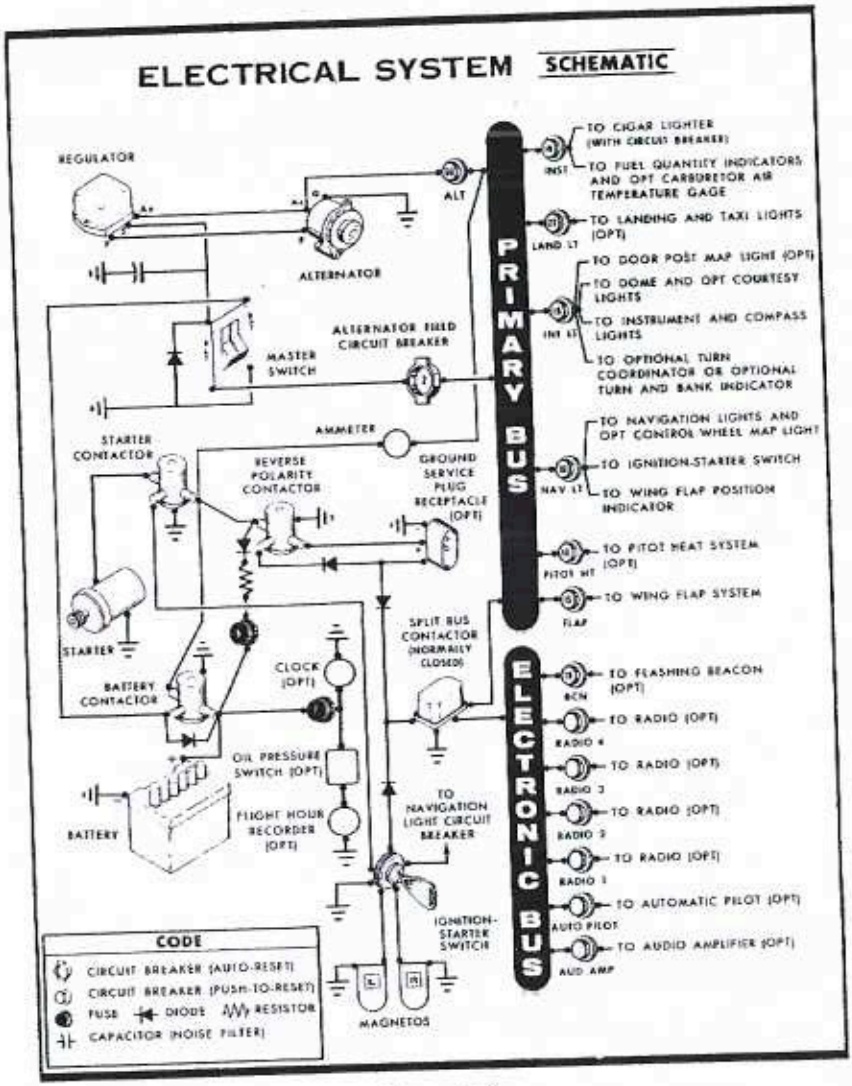

ELECTRICAL SYSTEM SCHEMATIC
REGULATOR
ALT
ALTERNATOR
MASTER SWITCH
ALTERNATOR FIELD CIRCUIT BREAKER
AMMETER
STARTER CONTACTOR
REVERSE POLARITY CONTACTOR
GROUND SERVICE PLUG RECEPTACLE (OPT)
STARTER
BATTERY CONTACTOR
CLOCK (OPT)
OIL PRESSURE SWITCH (OPT)
FLIGHT HOUR RECORDER (OPT)
BATTERY
SPLIT BUS CONTACTOR (NORMALLY CLOSED)
TO NAVIGATION LIGHT CIRCUIT BREAKER
IGNITION-STARTER SWITCH
MAGNETOS
PRIMARY BUS
TO CIGAR LIGHTER (WITH CIRCUIT BREAKER)
INST
TO FUEL QUANTITY INDICATORS AND OPT CARBURETOR AIR TEMPERATURE GAGE
TO LANDING AND TAXI LIGHTS (OPT)
LAND LT
TO DOOR POST MAP LIGHT (OPT)
TO DOME AND OPT COURTESY LIGHTS
TO INSTRUMENT AND COMPASS LIGHTS
INST LT
TO OPTIONAL TURN COORDINATOR OR OPTIONAL TURN AND BANK INDICATOR
TO NAVIGATION LIGHTS AND OPT CONTROL WHEEL MAP LIGHT
TO IGNITION-STARTER SWITCH
NAV LT
TO WING FLAP POSITION INDICATOR
TO PITOT HEAT SYSTEM (OPT)
PITOT HT
TO WING FLAP SYSTEM
FLAP
ELECTRONIC BUS
TO FLASHING BEACON (OPT)
BCN
TO RADIO (OPT)
RADIO 4
TO RADIO (OPT)
RADIO 3
TO RADIO (OPT)
RADIO 2
TO RADIO (OPT)
RADIO 1
TO AUTOMATIC PILOT (OPT)
AUTO PILOT
TO AUDIO AMPLIFIER (OPT)
AUD AMP
CODE
CIRCUIT BREAKER (AUTO-RESET)
CIRCUIT BREAKER (PUSH-TO-RESET)
FUSE
DIODE
RESISTOR
CAPACITOR (NOISE FILTER)

Figure 2-3.

contactor closing (external power) circuits which have fuses mounted adjacent to the battery. Also, the cigar lighter is protected by a manually reset type circuit breaker mounted directly on the back of the lighter behind the instrument panel. The alternator field and wiring is protected by an automatically resetting circuit breaker.

LANDING LIGHTS (OPT).

A three-position, push-pull switch controls the optional landing lights. To turn one lamp on for taxiing, pull the switch out to the first stop. To turn both lamps on for landing, pull the switch out to the second stop. To turn both lamps off, push the switch full in.

CONTROL WHEEL MAP LIGHT (OPT).

A map light may be mounted on the bottom of the pilot's control wheel. The light illuminates the lower portion of the cabin just forward of the pilot and is helpful when checking maps and other flight data during night operations. To operate the light, first turn the "NAV LIGHTS" switch on, then adjust the map light's intensity with the knurled rheostat knob located at the bottom of the control wheel.

FLASHING BEACON (OPT).

The flashing beacon should not be used when flying through clouds or overcast; the flashing light reflected from water droplets or particles in the atmosphere, particularly at night, can produce vertigo and loss of orientation.

CABIN HEATING, VENTILATING AND DEFROSTING SYSTEM.

For cabin ventilation, pull the "CABIN AIR" knob out. To raise the air temperature, pull the "CABIN HT" knob out approximately 1/4" to 1/2" for a small amount of cabin heat. Additional heat is available by pulling the knob out farther; maximum heat is available with the "CABIN HT" knob pulled full out and the "CABIN AIR" knob pushed full in. When no heat is desired in the cabin, the "CABIN HT" knob is pushed full in.

Front cabin heat and ventilating air is supplied by outlet holes spaced across a cabin manifold just forward of the pilot's and copilot's feet.

Rear cabin heat and air is supplied by two ducts from the manifold, one extending down each side of the cabin to an outlet at the front door post at floor level. Windshield defrost air is also supplied by a duct leading from the cabin manifold.

Separate adjustable ventilators supply additional air; one near each upper corner of the windshield supplies air for the pilot and copilot, and two optional ventilators in the rear cabin ceiling supply air to the rear seat passengers.

STARTING ENGINE.

During engine starting, open the throttle approximately 1/8 inch. In warm temperatures, one or two strokes of the primer should be sufficient. In cold weather, up to six strokes of the primer may be necessary. If the engine is warm, no priming will be required. In extremely cold temperatures, it may be necessary to continue priming while cranking the engine.

Weak intermittent firing followed by puffs of black smoke from the exhaust stack indicates overpriming or flooding. Excess fuel can be cleared from the combustion chambers by the following procedure: Set the mixture control full lean and the throttle full open; then crank the engine through several revolutions with the starter. Repeat the starting procedure without any additional priming.

If the engine is underprimed (most likely in cold weather with a cold engine) it will not fire at all, and additional priming will be necessary. As soon as the cylinders begin to fire, open the throttle slightly to keep it running.

After starting, if the oil gage does not begin to show pressure within 30 seconds in the summertime and about twice that long in very cold weather, stop engine and investigate. Lack of oil pressure can cause serious engine damage. After starting, avoid the use of carburetor heat unless icing conditions prevail.

NOTE

Additional details for cold weather starting and operation may be found under Cold Weather Operation in this section.

TAXIING.

When taxiing, it is important that speed and use of brakes be held to a minimum and that all controls be utilized (see Taxiing Diagram, figure 2-4) to maintain directional control and balance.

The carburetor heat control knob should be pushed full in during all ground operations unless heat is absolutely necessary. When the knob is pulled out to the heat position, air entering the engine is not filtered.

Taxiing over loose gravel or cinders should be done at low engine speed to avoid abrasion and stone damage to the propeller tips.

BEFORE TAKE-OFF.

WARM-UP.

If the engine accelerates smoothly, the airplane is ready for take-off. Since the engine is closely cowled for efficient in-flight engine cooling, precautions should be taken to avoid overheating during prolonged engine operation on the ground. Also, long periods of idling may cause fouled spark plugs.

MAGNETO CHECK.

The magneto check should be made at 1700 RPM as follows: Move ignition switch first to "R" position, and note RPM. Next move switch back to "BOTH" to clear the other set of plugs. Then move switch to the "L" position and note RPM. RPM drop should not exceed 125 RPM on either magneto or show greater than 50 RPM differential between magnetos. If there is a doubt concerning operation of the ignition system, RPM checks at higher engine speeds will usually confirm whether a deficiency exists.

An absence of RPM drop may be an indication of faulty grounding of one side of the ignition system or should be cause for suspicion that the magneto timing is set in advance of the setting specified.

ALTERNATOR CHECK.

Prior to flights where verification of proper alternator and voltage regulator operation is essential (such as night or instrument flights), a

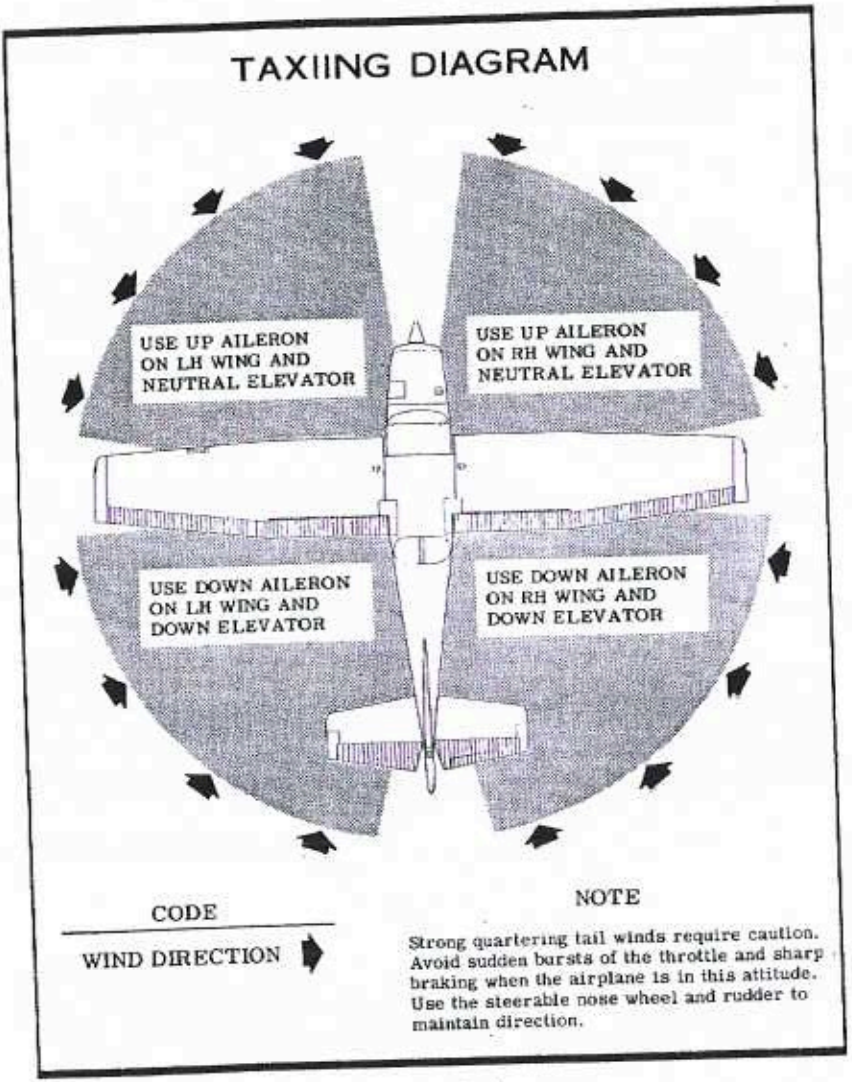

TAXIING DIAGRAM
USE UP AILERON ON LH WING AND NEUTRAL ELEVATOR
USE UP AILERON ON RH WING AND NEUTRAL ELEVATOR
USE DOWN AILERON ON LH WING AND DOWN ELEVATOR
USE DOWN AILERON ON RH WING AND DOWN ELEVATOR
CODE
WIND DIRECTION
NOTE
Strong quartering tail winds require caution. Avoid sudden bursts of the throttle and sharp braking when the airplane is in this attitude. Use the steerable nose wheel and rudder to maintain direction.

Figure 2-4.

positive verification can be made by loading the electrical system momentarily (3 to 5 seconds) with the optional landing light (if so equipped), or by operating the wing flaps during the engine runup (1700 RPM). The ammeter will remain within a needle width of zero if the alternator and voltage regulator are operating properly.

TAKE-OFF.

POWER CHECK.

It is important to check full-throttle engine operation early in the take-off run. Any signs of rough engine operation or sluggish engine acceleration is good cause for discontinuing the take-off. If this occurs, you are justified in making a thorough full-throttle, static runup before another take-off is attempted. The engine should run smoothly and turn approximately 2260 to 2360 RPM with carburetor heat off.

NOTE

Carburetor heat should not be used during take-off unless it is absolutely necessary for obtaining smooth engine acceleration.

Full-throttle runups over loose gravel are especially harmful to propeller tips. When take-offs must be made over a gravel surface, it is very important that the throttle be advanced slowly. This allows the airplane to start rolling before high RPM is developed, and the gravel will be blown back of the propeller rather than pulled into it. When unavoidable small dents appear in the propeller blades, they should be immediately corrected as described in Section V under propeller care.

Prior to take-off from fields above 5000 feet elevation, the mixture should be leaned to give maximum RPM in a full-throttle, static runup.

WING FLAP SETTINGS.

Normal and obstacle clearance take-offs are performed with wing flaps up. The use of 10° flaps will shorten the ground run approximately 10%, but this advantage is lost in the climb to a 50-foot obstacle. Therefore, the use of 10° flaps is reserved for minimum ground runs or for take-off from soft or rough fields. If 10° of flaps are used for minimum

ground runs, it is preferable to leave them extended rather than retract them in the climb to the obstacle. In this case, use an obstacle clearance speed of 65 MPH. As soon as the obstacle is cleared, the flaps may be retracted as the airplane accelerates to the normal flaps-up climb speed of 80 to 90 MPH.

During a high altitude take-off in hot weather where climb would be marginal with 10° flaps, it is recommended that the flaps not be used for take-off. Flap settings of 30° to 40° are not recommended at any time for take-off.

PERFORMANCE CHARTS.

Consult the Take-Off Data chart in Section VI for take-off distances under various gross weight, altitude, headwind, temperature, and runway surface conditions.

CROSSWIND TAKE-OFFS.

Take-offs into strong crosswinds normally are performed with the minimum flap setting necessary for the field length to minimize the drift angle immediately after take-off. The airplane is accelerated to a speed slightly higher than normal, then pulled off abruptly to prevent possible settling back to the runway while drifting. When clear of the ground, make a coordinated turn into the wind to correct for drift.

CLIMB.

CLIMB DATA.

For detailed data, refer to the Maximum Rate-Of-Climb Data chart in Section VI.

CLIMB SPEEDS.

Normal climbs are performed at 80 to 90 MPH with flaps up and full throttle for best engine cooling. The mixture should be full rich below 5000 feet and may be leaned above 5000 feet for smoother engine operation. The maximum rate-of-climb speeds range from 82 MPH at sea level to 79 MPH at 10,000 feet. If an obstruction dictates the use of a steep climb angle, climb at 68 MPH with flaps retracted.

NOTE

Steep climbs at low speeds should be of short duration to improve engine cooling.

CRUISE.

Normal cruising is done between 65% and 75% power. The power settings required to obtain these powers at various altitudes and outside air temperatures can be determined by using your Cessna Power Computer or the OPERATIONAL DATA, Section VI.

Cruising can be done more efficiently at high altitudes because of lower air density and therefore higher true airspeeds for the same power. This is illustrated in the table below, which shows performance at 75% power at various altitudes. All figures are based on lean mixture, 38 gallons of fuel (no reserve), zero wind, standard atmospheric conditions, and 2300 pounds gross weight.

To achieve the lean mixture fuel consumption figures shown in Section VI, the mixture should be leaned as follows: pull mixture control out until engine RPM peaks and begins to fall off, then enrichen slightly back to peak RPM.

Carburetor ice, as evidenced by an unexplained drop in RPM, can be removed by application of full carburetor heat. Upon regaining the original RPM (with heat off), use the minimum amount of heat (by trial and error) to prevent ice from forming. Since the heated air causes a richer mixture, readjust the mixture setting when carburetor heat is to be used continuously in cruise flight.

The use of full carburetor heat is recommended during flight in heavy

OPTIMUM CRUISE PERFORMANCE

ALTITUDE	RPM	TRUE AIRSPEED	RANGE
Sea Level	2490	123	575
5000 ft.	2600	128	600
9000 ft.	Full Throttle	132	620

rain to avoid the possibility of engine stoppage due to excessive water ingestion or carburetor ice. The mixture setting should be readjusted for smoothest operation.

In extremely heavy rain, the use of partial carburetor heat (control approximately 2/3 out), and part throttle (closed at least one inch), may be necessary to retain adequate power. Power changes should be made cautiously followed by prompt adjustment of the mixture for smoothest operation.

STALLS.

The stall characteristics are conventional and aural warning is provided by a stall warning horn which sounds between 5 and 10 MPH above the stall in all configurations.

Power-off stall speeds at maximum gross weight and aft c.g. position are presented on page 6-2 as calibrated airspeeds since indicated airspeeds are unreliable near the stall.

LANDINGS.

Normal landings are made power-off with any flap setting desired. Slips should be avoided with flap settings greater than 30° due to a downward pitch encountered under certain combinations of airspeed, side slip angle, and center of gravity loadings.

NOTE

Carburetor heat should be applied prior to any significant reduction or closing of the throttle.

NORMAL LANDING.

Landings should be made on the main wheels first to reduce the landing speed and subsequent need for braking in the landing roll. The nose wheel is lowered to the runway gently after the speed has diminished to avoid unnecessary nose gear loads. This procedure is especially important in rough or soft field landings.

SHORT FIELD LANDING.

For short field landings, make a power-off approach at approximately

69 MPH indicated airspeed with 40° of flaps. Touchdown should be made on the main wheels first. Immediately after touchdown, lower the nose gear to the ground and apply heavy braking as required. For maximum brake effectiveness after all three wheels are on the ground, retract the flaps, hold full nose up elevator and apply maximum possible brake pressure without sliding the tires.

CROSSWIND LANDING.

When landing in a strong crosswind, use the minimum flap setting required for the field length. Although the crab or combination method of drift correction may be used, the wing-low method gives the best control. After touchdown, hold a straight course with the steerable nose wheel and occasional braking if necessary.

The maximum allowable crosswind velocity is dependent upon pilot capability rather than airplane limitations. With average pilot technique, direct crosswinds of 15 MPH can be handled with safety.

BALKED LANDING (GO-AROUND).

In a balked landing (go-around) climb, reduce the wing flap setting to 20° immediately after full power is applied. If obstacles must be cleared during the go-around climb, leave the wing flaps in the 10° to 20° range until the obstacles are cleared. After clearing any obstacles the flaps may be retracted as the airplane accelerates to the normal flaps-up climb speed of 80 to 90 MPH.

COLD WEATHER OPERATION.

STARTING.

Prior to starting on a cold morning, it is advisable to pull the propeller through several times by hand to "break loose" or "limber" the oil, thus conserving battery energy. In extremely cold (0°F and lower) weather, the use of an external pre-heater and an external power source are recommended whenever possible to obtain positive starting and to reduce wear and abuse to the engine and electrical system. Pre-heat will thaw the oil trapped in the oil cooler, which probably will be congealed prior to starting in extremely cold temperatures. When using an external power source, the position of the master switch is important.

Refer to Section VII under Ground Service Plug Receptacle for operating details.

Cold weather starting procedures are as follows:

With Preheat:

(1) With ignition switch "OFF" and throttle closed, prime the engine four to eight strokes as the propeller is being turned over by hand.

NOTE

Use heavy strokes of primer for best atomization of fuel. After priming, push primer all the way in and turn to locked position to avoid possibility of engine drawing fuel through the primer.

(2) Propeller Area -- Clear.
(3) Master Switch -- "ON."
(4) Throttle -- Open 1/8".
(5) Ignition Switch -- "START."
(6) Release ignition switch to "BOTH" when engine starts.
(7) Oil Pressure -- Check.

Without Preheat:

(1) Prime the engine six to ten strokes while the propeller is being turned by hand with throttle closed. Leave primer charged and ready for stroke.
(2) Propeller Area -- Clear.
(3) Master Switch -- "ON."
(4) Pump throttle rapidly to full open twice. Return to 1/8" open position.
(5) Ignition Switch -- "START."
(6) Release ignition switch to "BOTH" when engine starts.
(7) Continue to prime engine until it is running smoothly, or alternately pump throttle rapidly over first 1/4 to total travel.
(8) Oil Pressure -- Check.
(9) Pull carburetor heat knob full on after engine has started. Leave on until engine is running smoothly.
(10) Lock Primer.

NOTE

If the engine does not start during the first few attempts, or if engine firing diminishes in strength, it is probable that the spark plugs have been frosted over. Preheat must be used before another start is attempted.

IMPORTANT

Pumping the throttle may cause raw fuel to accumulate in the intake air duct, creating a fire hazard in the event of a backfire. If this occurs, maintain a cranking action to suck flames into the engine. An outside attendant with a fire extinguisher is advised for cold starts without preheat.

During cold weather operations, no indication will be apparent on the oil temperature gage prior to take-off if outside air temperatures are very cold. After a suitable warm-up period (2 to 5 minutes at 1000 RPM), accelerate the engine several times to higher engine RPM. If the engine accelerates smoothly and the oil pressure remains normal and steady, the airplane is ready for take-off.

FLIGHT OPERATIONS.

Take-off is made normally with carburetor heat off. Avoid excessive leaning in cruise.

Carburetor heat may be used to overcome any occasional engine roughness due to ice.

When operating in sub-zero temperature, avoid using partial carburetor heat. Partial heat may increase the carburetor air temperature to the 32° to 70°F range, where icing is critical under certain atmospheric conditions.

Refer to Section VII for cold weather equipment.

HOT WEATHER OPERATION.

Refer to the general warm temperature starting information under Starting Engine in this section. Avoid prolonged engine operation on the ground.

Section III

EMERGENCY PROCEDURES

Emergencies caused by aircraft or engine malfunctions are extremely rare if proper pre-flight inspections and maintenance are practiced. Enroute weather emergencies can be minimized or eliminated by careful flight planning and good judgement when unexpected weather is encountered. However, should an emergency arise the basic guidelines described in this section should be considered and applied as necessary to correct the problem.

ELECTRICAL POWER SUPPLY SYSTEM MALFUNCTIONS.

Malfunctions in the electrical power supply system can be detected by periodic monitoring of the ammeter; however, the cause of these malfunctions is usually difficult to determine. A broken alternator drive belt or wiring is most likely the cause of alternator failures, although other factors could cause the problem. A damaged or improperly adjusted voltage regulator can also cause malfunctions. All electrical problems of this nature constitute an electrical emergency and should be dealt with immediately. Electrical power malfunctions usually fall into two categories, excessive rate of charge and insufficient rate of charge. The paragraphs below describe the recommended remedy for each situation.

EXCESSIVE RATE OF CHARGE.

After periods of engine starting and heavy electrical usage at low engine speeds (such as extended taxiing) the battery condition will be low enough to accept above normal charging during the initial part of a flight. However, after thirty minutes of cruising flight, the ammeter should be indicating less than two needle widths of charging current. If the charging rate remains above this value on a long flight, it is possible that the battery will overheat and evaporate the electrolyte at an excessive rate. In addition, electronic components in the electrical system could be adversely affected by the higher than normal voltage if a faulty voltage regulator setting is causing the overcharging.

To preclude these possibilities, the alternator side of the split master switch should be turned "OFF." The flight should be terminated and/or the current drain on the battery minimized as soon as practical because the battery can supply the electrical system for only a limited period of time. If it becomes apparent that the battery voltage is getting too low to operate the electrical system, the alternator switch can be turned back on for several minutes at a time until the battery is partially recharged. If the emergency occurs at night, the alternator switch should be returned to the "ON" position just before landing lights and flaps will be required for landing.

INSUFFICIENT RATE OF CHARGE.

If the ammeter indicates a continuous discharge rate in flight, the alternator is not supplying power to the system and should be shut down since the alternator field circuit may be placing an unnecessary load on the system. All non-essential equipment should be turned "OFF" and the flight terminated as soon as practical.

ROUGH ENGINE OPERATION OR LOSS OF POWER.

SPARK PLUG FOULING.

An engine roughness in flight may be caused by one or more spark plugs becoming fouled by carbon or lead deposits. This may be verified by turning the ignition switch momentarily from "BOTH" to either "LEFT" or "RIGHT" position. An obvious power loss in single ignition operation is evidence of spark plug or magneto trouble. Assuming that spark plugs are the more likely cause, lean the mixture to the normal lean setting for cruising flight. If the problem does not clear up in several minutes, determine if a richer mixture setting will produce smoother operation. If not, proceed to the nearest airport for repairs using the "BOTH" position of the ignition switch unless extreme roughness dictates the use of a single ignition position.

MAGNETO MALFUNCTION.

A sudden engine roughness or misfiring is usually evidence of magneto problems. Switching from "BOTH" to either "LEFT" or "RIGHT" ignition switch position will identify which magneto is malfunctioning. Select different power settings and enrichen the mixture to determine if

continued operation on "BOTH" magnetos is practicable. If not, switch to the good magneto and proceed to the nearest airport for repairs.

LOW OIL PRESSURE.

If low oil pressure is accompanied by normal oil temperature, there is a possibility the oil pressure gage or relief valve is malfunctioning. A leak in the line to the gage is not cause for immediate concern because an orifice in this line will prevent a sudden loss of oil from the engine sump. However, a landing at the nearest airport is advisable.

If a total loss of oil pressure is accompanied by a sudden rise in oil temperature, there is reason to suspect an engine failure is imminent. Reduce engine power immediately and select a suitable forced landing field. Leave the engine running at low power during the approach, using only the minimum power required to reach the desired touchdown spot.

FORCED LANDINGS.

PRECAUTIONARY LANDING WITH ENGINE POWER.

Before attempting an "off airport" landing, one should drag the landing area at a safe but low altitude to inspect the terrain for obstructions and surface conditions, proceeding as follows:

(1) Drag over selected field with flaps 20° and 70 MPH airspeed, noting the preferred area for touchdown for the next landing approach. Then retract flaps after well clear of all obstacles.
(2) On downwind leg, turn off all switches except the ignition and master switches.
(3) Approach with flaps 40° at 70 MPH.
(4) Unlatch cabin doors prior to final approach.
(5) Before touchdown, turn ignition and master switches "OFF."
(6) Land in a slightly tail-low attitude.

EMERGENCY LANDING WITHOUT ENGINE POWER.

If an engine stoppage occurs, establish a flaps up glide at 80 MPH. If time permits, attempt to restart the engine by checking for fuel quantity, proper fuel selector valve position, and mixture control setting. Also check that engine primer is full in and locked and ignition switch is properly positioned.

If all attempts to restart the engine fail, and a forced landing is imminent, select a suitable field and prepare for the landing as follows:

(1) Pull mixture control to idle cut-off position.
(2) Turn fuel selector valve handle to "OFF."
(3) Turn all switches "OFF" except master switch.
(4) Airspeed -- 70 to 80 MPH (flaps up).
(5) Extend wing flaps as necessary within gliding distance of field.
(6) Airspeed -- 65 to 75 MPH (flaps down).
(7) Turn master switch "OFF."
(8) Unlatch cabin doors prior to final approach.
(9) Land in a slightly tail-low attitude.
(10) Apply heavy braking while holding full up elevator.

DITCHING.

Prepare for ditching by securing or jettisoning heavy objects located in the baggage area, and collect folded coats or cushions for protection of occupant's face at touchdown. Transmit Mayday message on 121.5 MHz., giving location and intentions.

(1) Plan approach into wind if winds are high and seas are heavy. With heavy swells and light wind, land parallel to swells.
(2) Approach with flaps 40° and sufficient power for a 300 ft./min. rate of descent at 70 MPH.
(3) Unlatch the cabin doors.
(4) Maintain a continuous descent until touchdown in level attitude. Avoid a landing flare because of difficulty in judging airplane height over a water surface.
(5) Place folded coat or cushion in front of face at time of touchdown.
(6) Expect a second impact for the airplane may skip after touchdown.
(7) Evacuate airplane through cabin doors. If necessary, open window to flood cabin compartment for equalizing pressure so that door can be opened.
(8) Inflate life vests and raft (if available) after evacuation of cabin. The aircraft can not be depended on for floatation for more than a few minutes.

DISORIENTATION IN CLOUDS.

When flying in marginal weather, the pilot should make sure that the

Wing Leveler control knob (if installed) is "ON." However, if the airplane is not equipped with this device or gyro horizon and directional gyro instruments, the pilot will have to rely on the turn coordinator (or turn and bank indicator) if he inadvertently flies into clouds. The following instructions assume that only one of the latter two instruments is available.

EXECUTING A 180° TURN IN CLOUDS.

Upon entering the clouds, an immediate plan should be made to turn back as follows:

(1) Note the time of the minute hand and observe the position of the sweep second hand on the clock.
(2) When the sweep second hand indicates the nearest half-minute, initiate a standard rate left turn, holding the turn coordinator symbolic airplane wing opposite the lower left index mark for 60 seconds. Then roll back to level flight by leveling the miniature airplane.
(3) Check accuracy of the turn by observing the compass heading which should be the reciprocal of the original heading.
(4) If necessary, adjust heading primarily with skidding motions rather than rolling motions so that the compass will read more accurately.
(5) Maintain altitude and airspeed by cautious application of elevator control. Avoid overcontrolling by keeping the hands off the control wheel and steering only with rudder.

EMERGENCY LET-DOWNS THROUGH CLOUDS.

If possible, obtain radio clearance for an emergency descent through clouds. To guard against a spiral dive, choose an easterly or westerly heading to minimize compass card swings due to changing bank angles. In addition, keep hands off the control wheel and steer a straight course with rudder control by monitoring the turn coordinator. Occasionally check the compass heading and make minor corrections to hold an approximate course. Before descending into the clouds, set up a stabilized let-down condition as follows:

(1) Apply full rich mixture.
(2) Use full carburetor heat.
(3) Reduce power to set up a 500 to 800 ft./min. rate of descent.
(4) Adjust the elevator trim tab for a stabilized descent at 90 MPH.
(5) Keep hands off the control wheel.
(6) Monitor turn coordinator and make corrections by rudder alone.

(7) Check trend of compass card movement and make cautious corrections with rudder to stop the turn.
(8) Upon breaking out of clouds resume normal cruising flight.

RECOVERY FROM A SPIRAL DIVE.

If a spiral is encountered, proceed as follows:

(1) Close the throttle.
(2) Stop the turn by using coordinated aileron and rudder control to align the symbolic airplane in the turn coordinator with the horizon reference line.
(3) Cautiously apply elevator back pressure to slowly reduce the indicated airspeed to 90 MPH.
(4) Adjust the elevator trim control to maintain a 90 MPH glide.
(5) Keep hands off the control wheel, using rudder control to hold a straight heading.
(6) Apply carburetor heat.
(7) Clear engine occasionally, but avoid using enough power to disturb the trimmed glide.
(8) Upon breaking out of clouds, apply normal cruising power and resume flight.

FIRES.

ENGINE FIRE DURING START ON GROUND.

Improper starting procedures such as pumping the throttle during a difficult cold weather start can cause a backfire which could ignite fuel that has accumulated in the intake duct. In this event, proceed as follows:

(1) Continue cranking in an attempt to get a start which would suck the flames and accumulated fuel through the carburetor and into the engine.
(2) If the start is successful, run the engine at 1700 RPM for a few minutes before shutting it down to inspect the damage.
(3) If engine start is unsuccessful, continue cranking for two or three minutes with throttle full open while ground attendants obtain fire extinguishers.
(4) When ready to extinguish fire, release the starter switch and turn off master switch, ignition switch, and fuel selector valve handle.

(5) Smother flames with fire extinguisher, seat cushion, wool blanket, or loose dirt. If practical try to remove carburetor air filter if it is ablaze.
(6) Make a thorough inspection of fire damage, and repair or replace damaged components before conducting another flight.

ENGINE FIRE IN FLIGHT.

Although engine fires are extremely rare in flight, the following steps should be taken if one is encountered:

(1) Pull mixture control to idle cut-off.
(2) Turn fuel selector valve handle "OFF."
(3) Turn master switch "OFF."
(4) Establish a 120 MPH glide.
(5) Close cabin heat control.
(6) Select a field suitable for a forced landing.
(7) If fire is not extinguished, increase glide speed in an attempt to find an airspeed that will provide an incombustible mixture.
(8) Execute a forced landing as described in paragraph Emergency Landing Without Engine Power. Do not attempt to restart the engine.

ELECTRICAL FIRE IN FLIGHT.

The initial indication of an electrical fire is the odor of burning insulation. The immediate response should be to turn the master switch "OFF." Then close off ventilating air as much as practicable to reduce the chances of a sustained fire.

If electrical power is indispensable for the flight, an attempt may be made to identify and cut off the defective circuit as follows:

(1) Master Switch -- "OFF."
(2) All other switches (except ignition switch) -- "OFF."
(3) Check condition of circuit breakers to identify faulty circuit if possible. Leave faulty circuit deactivated.
(4) Master Switch -- "ON."
(5) Select switches "ON" successively, permitting a short time delay to elapse after each switch is turned on until the short circuit is localized.
(6) Make sure fire is completely extinguished before opening ventilators.

FLIGHT IN ICING CONDITIONS.

Although flying in known icing conditions is prohibited, an unexpected icing encounter should be handled as follows:

(1) Turn pitot heat switch "ON" (if installed).
(2) Turn back or change altitude to obtain an outside air temperature that is less conducive to icing.
(3) Pull cabin heat control full out to obtain windshield defroster airflow. Adjust cabin air control to get maximum defroster heat and airflow.
(4) Open the throttle to increase engine speed and determine if ice is soft enough to be thrown off the propeller blades.
(5) Watch for signs of carburetor air filter ice and apply carburetor heat as required. An unexplained loss in engine speed could be caused by carburetor ice or air intake filter ice.
(6) Plan a landing at the nearest airport. With an extremely rapid ice build-up, select a suitable "off airport" landing site.
(7) With an ice accumulation of one inch or more on the wing leading edges, be prepared for significantly higher stall speed.
(8) Leave wing flaps retracted. With a severe ice build-up on the horizontal tail, the change in wing wake airflow direction caused by wing flap extension could result in a loss of elevator effectiveness.
(9) Open left window and scrape ice from a portion of the windshield for visibility in the landing approach. The metal control lock shield may be used as a scraper.
(10) Perform a landing approach using a forward slip, if necessary, for improved visibility.
(11) Approach at 75 to 85 MPH, depending upon the amount of ice accumulation.
(12) Avoid steep turns during the landing approach.
(13) Perform a landing in level attitude.

Section IV

OPERATING LIMITATIONS

OPERATIONS AUTHORIZED.

Your Cessna exceeds the requirements of airworthiness as set forth by the United States Government, and is certificated under FAA Type Certificate No. 3A12 as Cessna Model No. 172K.

With standard equipment, the airplane is approved for day and night operations under VFR. Additional optional equipment is available to increase its utility and to make it authorized for use under IFR day and night. An owner of a properly equipped Cessna is eligible to obtain approval for its operation on single-engine scheduled airline service. Your Cessna Dealer will be happy to assist you in selecting equipment best suited to your needs.

MANEUVERS - NORMAL CATEGORY.

This airplane is certificated in both the normal and utility category. The normal category is applicable to airplanes intended for non-aerobatic operations. These include any maneuvers incidental to normal flying, stalls (except whip stalls) and turns in which the angle of bank is not more than 60°. In connection with the foregoing, the following gross weight and flight load factors apply:

Gross Weight	2300 lbs	
Flight Load Factor		
*Flaps Up	+3.8	-1.52
*Flaps Down	+3.5	

*The design load factors are 150% of the above, and in all cases, the structure meets or exceeds design loads.

Your airplane must be operated in accordance with all FAA-approved markings, placards and check lists in the airplane. If there is any information in this section which contradicts the FAA-approved markings, placards and check lists, it is to be disregarded.

MANEUVERS - UTILITY CATEGORY.

This airplane is not designed for purely aerobatic flight. However, in the acquisition of various certificates such as commercial pilot, instrument pilot and flight instructor, certain maneuvers are required by the FAA. All of these maneuvers are permitted in this airplane when operated in the utility category. In connection with the utility category, the following gross weight and flight load factors apply, with maximum entry speeds for maneuvers as shown:

Gross Weight	2000 lbs	
Flight Maneuvering Load Factor		
Flaps Up	+4.4	-1.76
Flaps Down	+3.5	

In the utility category, the baggage compartment and rear seat must not be occupied. No aerobatic maneuvers are approved except those listed below:

MANEUVER	MAXIMUM ENTRY SPEED*
Chandelles	122 mph (106 knots)
Lazy Eights	122 mph (106 knots)
Steep Turns	122 mph (106 knots)
Spins	122 mph (106 knots)
Stalls (Except Whip Stalls)	Slow Deceleration
	Slow Deceleration

*Higher speeds can be used if abrupt use of the controls is avoided.

For spin recovery, apply opposite rudder followed by forward pressure on the control wheel. When airplane rotation has stopped, use moderate back pressure on the control wheel to avoid excessive loads while recovering from the resulting dive.

Aerobatics that may impose high inverted loads should not be attempted. The important thing to bear in mind in flight maneuvers is that the airplane is clean in aerodynamic design and will build up speed quickly with the nose down. Proper speed control is an essential requirement for execution of any maneuver, and care should always be exercised to avoid excessive speed which in turn can impose excessive loads. In the execution of all maneuvers, avoid abrupt use of controls.

AIRSPEED LIMITATIONS (CAS).

The following is a list of the certificated calibrated airspeed (CAS) limitations for the airplane.

Never Exceed Speed (glide or dive, smooth air)	174 MPH
Maximum Structural Cruising Speed	140 MPH
Maximum Speed, Flaps Extended	100 MPH
*Maneuvering Speed	122 MPH

*The speed at which abrupt control travel can be used without exceeding the specified load factor.

AIRSPEED INDICATOR MARKINGS.

The following is a list of the certificated calibrated airspeed markings (CAS) for the airplane.

Never Exceed (glide or dive, smooth air)	.174 MPH (red line)
Caution Range	140-174 MPH (yellow arc)
Normal Operating Range.	59-140 MPH (green arc)
Flap Operating Range	52-100 MPH (white arc)

ENGINE OPERATION LIMITATIONS.

Power and Speed	150 BHP at 2700 RPM

ENGINE INSTRUMENT MARKINGS.

OIL TEMPERATURE GAGE.

Normal Operating Range	Green Arc
Maximum Allowable	245°F (red line)

OIL PRESSURE GAGE.

Minimum Idling	25 psi (red line)
Normal Operating Range	60-90 psi (green arc)
Maximum	100 psi (red line)

FUEL QUANTITY INDICATORS.

Empty (2.0 gallons unusable each tank) . . . . . E (red line)

TACHOMETER.

Normal Operating Range:

At sea level . . . . . 2200-2500 (inner green arc)

At 5000 feet . . . . . 2200-2600 (middle green arc)

At 10,000 feet . . . . . 2200-2700 (outer green arc)

Maximum Allowable. . . . . 2700 (red line)

CARBURETOR AIR TEMPERATURE GAGE (OPT).

Icing Range . . . . . -15° to 5°C (yellow arc)

WEIGHT AND BALANCE.

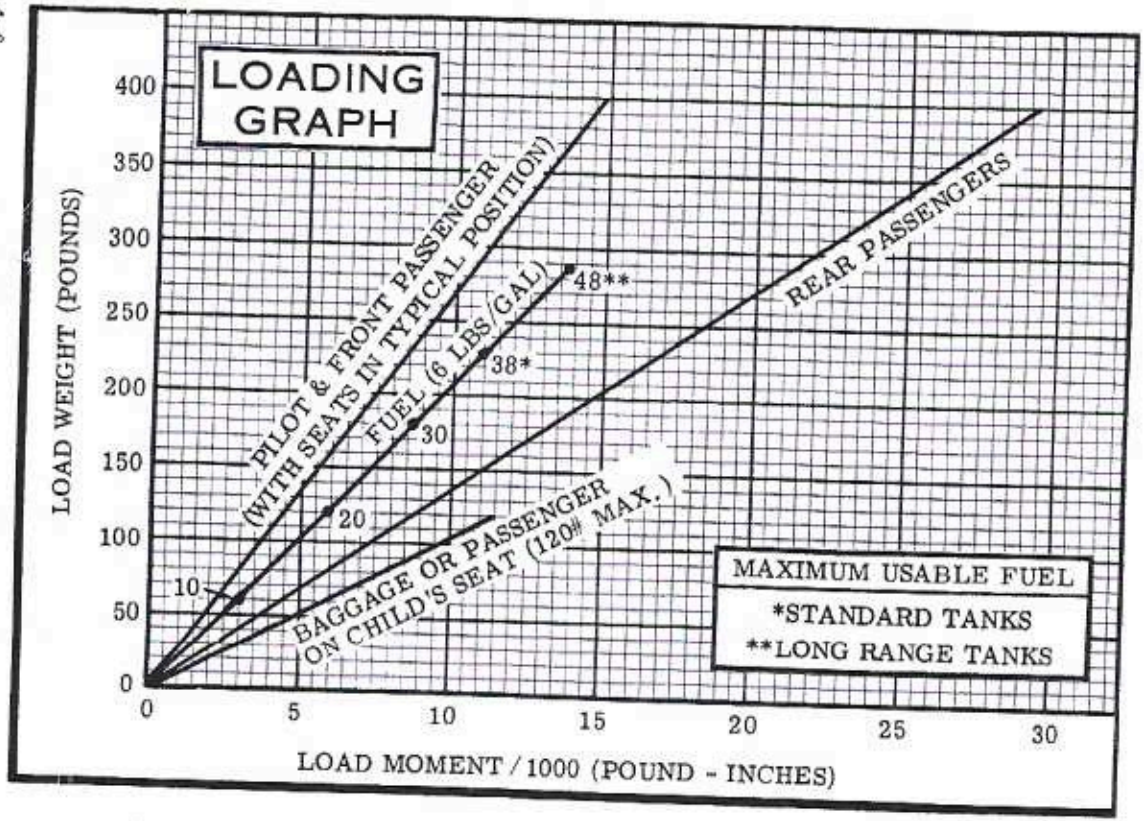
The following information will enable you to operate your Cessna within the prescribed weight and center of gravity limitations. To figure the weight and balance for your particular airplane, use the Sample Problem, Loading Graph, and Center of Gravity Moment Envelope as follows:

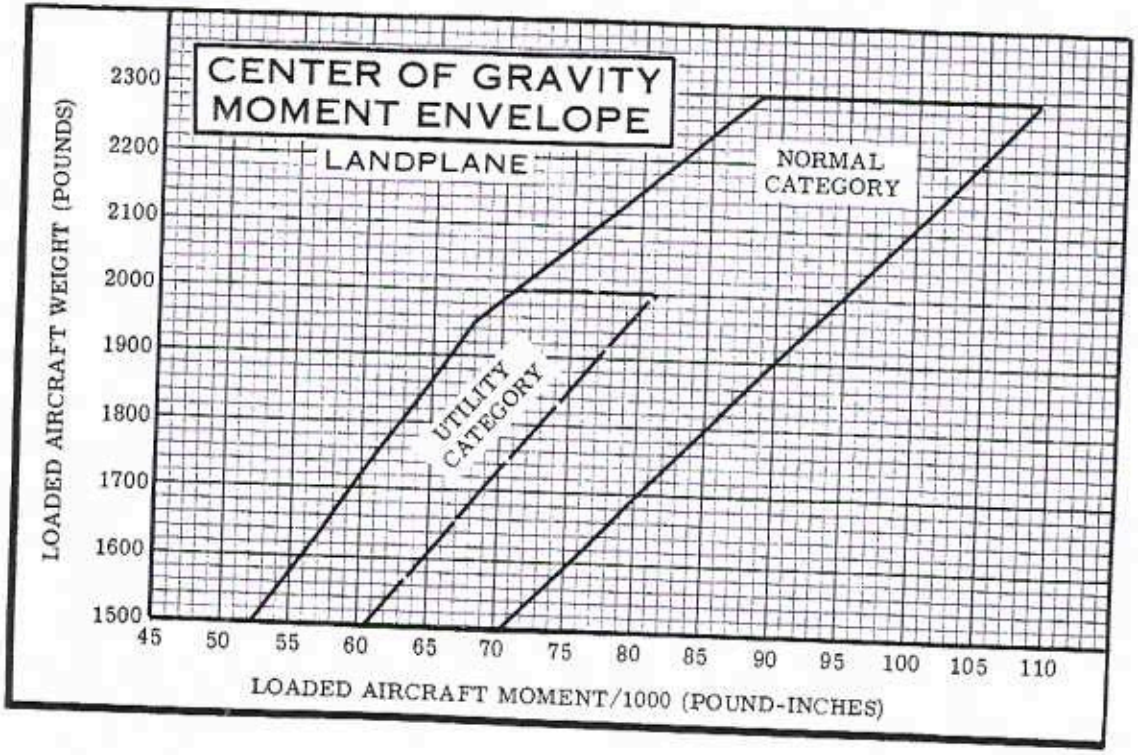
Take the licensed Empty Weight and Moment/1000 from the Weight and Balance Data sheet, plus any change noted on forms FAA-337, carried in your airplane, and write them down in the proper columns. Using the Loading Graph, determine the moment/1000 of each item to be carried. Total the weights and moments/1000 and use the Center of Gravity Moment Envelope to determine whether the point falls within the envelope, and if the loading is acceptable.

NOTE

The Weight and Balance Data Sheet noted above is included in the aircraft file. The Loading Graph and Center of Gravity Moment Envelope shown in this section are also on the sheet titled Loading/Center of Gravity Charts and Weighing Procedures which is provided in the aircraft file.

SAMPLE LOADING PROBLEM	SAMPLE AIRPLANE		YOUR AIRPLANE	
	Weight (lbs.)	Moment (lb.-ins. /1000)	Weight (lbs.)	Moment (lb.-ins. /1000)
1. Licensed Empty Weight (Sample Airplane) . . .	1306	47.6		
2. Oil (8 qts. - Full oil may be assumed for all flights)	15	-0.2	15	-0.2
3. Fuel (Standard - 38 Gal at 6#/Gal)	228	10.9		
Fuel (Long Range - 48 Gal at 6#/Gal)				
4. Pilot and Front Passenger	340	12.6		
5. Rear Passengers	340	24.8		
6. Baggage (or Passenger on Child's Seat)	71	6.7		
7. TOTAL WEIGHT AND MOMENT	2300	102.4		
8. Locate this point (2300 at 102.4) on the center of gravity moment envelope, and since this point falls within the envelope, the loading is acceptable.				


LOADING GRAPH
LOAD WEIGHT (POUNDS)
400
350
300
250
200
150
100
50
0
0
5
10
15
20
25
30
LOAD MOMENT / 1000 (POUND - INCHES)
PILOT & FRONT PASSENGER (WITH SEATS IN TYPICAL POSITION)
FUEL (6 LBS/GAL)
REAR PASSENGERS
BAGGAGE OR PASSENGER ON CHILD'S SEAT (120# MAX.)
10
20
30
38*
48**
MAXIMUM USABLE FUEL
*STANDARD TANKS
**LONG RANGE TANKS


CENTER OF GRAVITY
MOMENT ENVELOPE
LANDPLANE
NORMAL
CATEGORY
UTILITY
CATEGORY
LOADED AIRCRAFT WEIGHT (POUNDS)
2300
2200
2100
2000
1900
1800
1700
1600
1500
45
50
55
60
65
70
75
80
85
90
95
100
105
110
LOADED AIRCRAFT MOMENT/1000 (POUND-INCHES)

Section V

CARE OF THE AIRPLANE

If your airplane is to retain that new plane performance and dependability, certain inspection and maintenance requirements must be followed. It is wise to follow a planned schedule of lubrication and preventative maintenance based on climatic and flying conditions encountered in your locality.

Keep in touch with your Cessna Dealer and take advantage of his knowledge and experience. He knows your airplane and how to maintain it. He will remind you when lubrications and oil changes are necessary, and about other seasonal and periodic services.

GROUND HANDLING.

The airplane is most easily and safely maneuvered by hand with the tow-bar attached to the nose wheel.

NOTE

When using the tow-bar, never exceed the turning angle of 30°, either side of center, or damage to the gear will result.

MOORING YOUR AIRPLANE.

Proper tie-down procedure is your best precaution against damage to your parked airplane by gusty or strong winds. To tie down your airplane securely, proceed as follows:

(1) Set the parking brake and install the control wheel lock.
(2) Tie sufficiently strong ropes or chains (700 pounds tensile strength) to wing, tail and nose tie-down rings and secure each rope to a ramp tie-down.

(3) Install a surface control lock over the fin and rudder.
(4) Install a pitot tube cover.

WINDSHIELD - WINDOWS.

The plastic windshield and windows should be cleaned with an aircraft windshield cleaner. Apply the cleaner sparingly with soft cloths, and rub with moderate pressure until all dirt, oil scum and bug stains are removed. Allow the cleaner to dry, then wipe it off with soft flannel cloths.

If a windshield cleaner is not available, the plastic can be cleaned with soft cloths moistened with Stoddard solvent to remove oil and grease.

NOTE

Never use gasoline, benzine, alcohol, acetone, carbon tetrachloride, fire extinguisher or anti-ice fluid, lacquer thinner or glass cleaner to clean the plastic. These materials will attack the plastic and may cause it to craze.

Follow by carefully washing with a mild detergent and plenty of water. Rinse thoroughly, then dry with a clean moist chamois. Do not rub the plastic with a dry cloth since this builds up an electrostatic charge which attracts dust. Waxing with a good commercial wax will finish the cleaning job. A thin, even coat of wax, polished out by hand with clean soft flannel cloths, will fill in minor scratches and help prevent further scratching.

Do not use a canvas cover on the windshield unless freezing rain or sleet is anticipated since the cover may scratch the plastic surface.

PAINTED SURFACES.

The painted exterior surfaces of your new Cessna have a durable, long lasting finish and, under normal conditions, require no polishing or buffing. Approximately 15 days are required for the paint to cure completely; in most cases, the curing period will have been completed prior to delivery of the airplane. In the event that polishing or buffing is required within the curing period, it is recommended that the work be done by someone experienced in handling uncured paint. Any Cessna Dealer can accomplish this work.

Generally, the painted surfaces can be kept bright by washing with water and mild soap, followed by a rinse with water and drying with cloths or a chamois. Harsh or abrasive soaps or detergents which cause corrosion or scratches should never be used. Remove stubborn oil and grease with a cloth moistened with Stoddard solvent.

Waxing is unnecessary to keep the painted surfaces bright. However, if desired, the airplane may be waxed with a good automotive wax. A heavier coating of wax on the leading edges of the wings and tail and on the engine nose cap and propeller spinner will help reduce the abrasion encountered in these areas.

When the airplane is parked outside in cold climates and it is necessary to remove ice before flight, care should be taken to protect the painted surfaces during ice removal with chemical liquids. A 50-50 solution of isopropyl alcohol and water will satisfactorily remove ice accumulations without damaging the paint. A solution with more than 50% alcohol is harmful and should be avoided. While applying the de-icing solution, keep it away from the windshield and cabin windows since the alcohol will attack the plastic and may cause it to craze.

ALUMINUM SURFACES.

The clad aluminum surfaces of your Cessna may be washed with clear water to remove dirt; oil and grease may be removed with gasoline, naptha, carbon tetrachloride or other non-alkaline solvents. Dulled aluminum surfaces may be cleaned effectively with an aircraft aluminum polish.

After cleaning, and periodically thereafter, waxing with a good automotive wax will preserve the bright appearance and retard corrosion. Regular waxing is especially recommended for airplanes operated in salt water areas as a protection against corrosion.

PROPELLER CARE.

Preflight inspection of propeller blades for nicks, and wiping them occasionally with an oily cloth to clean off grass and bug stains will assure long, trouble-free service. It is vital that small nicks on the propeller, particularly near the tips and on the leading edges, are dressed out as soon as possible since these nicks produce stress concentrations,

and if ignored, may result in cracks. Never use an alkaline cleaner on the blades; remove grease and dirt with carbon tetrachloride or Stoddard solvent.

INTERIOR CARE.

To remove dust and loose dirt from the upholstery and carpet, clean the interior regularly with a vacuum cleaner.

Blot up any spilled liquid promptly, with cleansing tissue or rags. Don't pat the spot; press the blotting material firmly and hold it for several seconds. Continue blotting until no more liquid is taken up. Scrape off sticky materials with a dull knife, then spot-clean the area.

Oily spots may be cleaned with household spot removers, used sparingly. Before using any solvent, read the instructions on the container and test it on an obscure place on the fabric to be cleaned. Never saturate the fabric with a volatile solvent; it may damage the padding and backing materials.

Soiled upholstery and carpet may be cleaned with foam-type detergent, used according to the manufacturer's instructions. To minimize wetting the fabric, keep the foam as dry as possible and remove it with a vacuum cleaner.

The plastic trim, headliner, instrument panel and control knobs need only be wiped off with a damp cloth. Oil and grease on the control wheel and control knobs can be removed with a cloth moistened with Stoddard solvent. Volatile solvents, such as mentioned in paragraphs on care of the windshield, must never be used since they soften and craze the plastic.

Radio and autopilot face plates are finished with a suede coating which produces a soft, rich appearance and warm feel comparable to suede. Unlike suede leather, dust and dirt marks can be removed easily with a damp sponge. Remove non-greasy stains with a liquid cleaner such as "Mr. Clean", "Handy Andy", "Lestoil", "Liquid Ajax", or "Cinch". Greasy stains can be removed with a naptha-dampened sponge, scrub brush or lint-free cloth.

FLYABLE STORAGE.

Aircraft which are not in daily flight should have the engine started

and warmed up at least once each week. In damp climates and in storage areas where the daily temperature variation can cause condensation, the warm-up operation should be accomplished more frequently. Warming up the engine replaces oil which has drained from surfaces of internal parts while standing idle. Warm-up should be accomplished at a throttle setting necessary to produce a minimum oil temperature of 100°F.

NOTE

Excessive ground run-up should be avoided. Run-up should not exceed 10 minutes duration.

Engine warm-up also helps to eliminate excessive accumulations of water in the fuel system and other air spaces in the engine. Keep fuel tanks full to minimize condensation in the tanks. Keep the battery fully charged to prevent the electrolyte from freezing in cold weather. If the aircraft is to be stored temporarily, or indefinitely, refer to the Service Manual for proper storage procedures.

INSPECTION SERVICE AND INSPECTION PERIODS.

With your airplane you will receive an Owner's Service Policy. Coupons attached to the policy entitle you to an initial inspection and the first 100-hour inspection at no charge. If you take delivery from your Dealer, he will perform the initial inspection before delivery of the airplane to you. If you pick up the airplane at the factory, plan to take it to your Dealer reasonably soon after you take delivery on it. This will permit him to check it over and to make any minor adjustments that may appear necessary. Also, plan an inspection by your Dealer at 100 hours or 180 days, whichever comes first. This inspection also is performed by your Dealer for you at no charge. While these important inspections will be performed for you by any Cessna Dealer, in most cases you will prefer to have the Dealer from whom you purchased the airplane accomplish this work.

Federal Aviation Regulations require that all airplanes have a periodic (annual) inspection as prescribed by the administrator, and performed by a person designated by the administrator. In addition, 100-hour periodic inspections made by an "appropriately-rated mechanic" are required if the airplane is flown for hire. The Cessna Aircraft Company recommends the 100-hour periodic inspection for your airplane. The procedure for this 100-hour inspection has been carefully

worked out by the factory and is followed by the Cessna Dealer Organization. The complete familiarity of the Cessna Dealer Organization with Cessna equipment and with factory-approved procedures provides the highest type of service possible at lower cost.

AIRCRAFT FILE.

There are miscellaneous data, information and licenses that are a part of the aircraft file. The following is a check list for that file. In addition, a periodic check should be made of the latest Federal Aviation Regulations to insure that all data requirements are met.

A. To be displayed in the aircraft at all times:

(1) Aircraft Airworthiness Certificate (Form FAA-1362B).
(2) Aircraft Registration Certificate (Form FAA-500A).
(3) Aircraft Radio Station License (Form FCC-404, if transmitter installed).

B. To be carried in the aircraft at all times:

(1) Weight and Balance, and associated papers (latest copy of the Repair and Alteration Form, Form FAA-337, if applicable).
(2) Aircraft Equipment List.

C. To be made available upon request:

(1) Aircraft Log Book.
(2) Engine Log Book.

NOTE

Cessna recommends that these items, plus the Owner's Manual, "Cessna Flight Guide" (Flight Computer), and Service Policies, be carried in the aircraft at all times.

Most of the items listed are required by the United States Federal Aviation Regulations. Since the regulations of other nations may require other documents and data, owners of exported aircraft should check with their own aviation officials to determine their individual requirements.

LUBRICATION AND SERVICING PROCEDURES

Specific servicing information is provided here for items requiring daily attention. A Servicing Intervals Check List is included to inform the pilot when to have other items checked and serviced.

DAILY

FUEL TANK FILLERS:

Service after each flight with 80/87 minimum grade fuel. The capacity of each tank is 21 gallons. When optional long range tanks are installed, the capacity of each tank is 26 gallons.

FUEL STRAINER:

Before the first flight of the day and after each refueling, pull out fuel strainer drain knob for about four seconds, to clear fuel strainer of possible water and sediment. Release drain knob, then check that strainer drain is closed after draining. If water is observed, there is a possibility that the fuel tank sumps contain water. Thus, the fuel tank sump drain plugs and fuel selector valve drain plug should be removed to check for the presence of water.

OIL DIPSTICK:

Check oil level before each flight. Do not operate on less than 6 quarts. To minimize loss of oil through breather, fill to 7 quart level for normal flights of less than 3 hours. For extended flight, fill to 8 quarts. If optional oil filter is installed, one additional quart is required when the filter element is changed.

OIL FILLER:

When preflight check shows low oil level, service with aviation grade engine oil; SAE 50 above 60°F, SAE 10W30 or SAE 30 at temperatures from 0° to 70°F, and SAE 10W30 or SAE 20 at temperatures below 10°F. (Multi-viscosity oil with a range of SAE 10W30 is recommended for improved starting and lubrication during warm-up in cold weather.) Detergent or dispersant oil, conforming to Lycoming Specification No. 301E, must be used. Your Cessna Dealer can supply approved brands of oil.

NOTE

To promote faster ring seating and improved oil control, your Cessna was delivered from the factory with straight mineral oil (non-detergent). This "break-in" oil should be used only for the first 50 hours of operation, or until oil consumption has stabilized at which time it must be replaced with detergent oil.

SERVICING INTERVALS CHECK LIST

FIRST 25 HOURS

ENGINE OIL SUMP, OIL COOLER AND OIL FILTER -- After first 25 hours of operation, drain engine oil sump and oil cooler and clean both the oil suction strainer and oil pressure screen. If an optional oil filter is installed, change filter element at this time. Refill sump with straight mineral oil (non-detergent) and use until a total of 50 hours have accumulated or oil consumption has stabilized, then change to detergent oil.

EACH 50 HOURS

BATTERY -- Check and service. Check oftener (at least every 30 days) if operating in hot weather.
ENGINE OIL SUMP, OIL COOLER AND OIL FILTER -- On airplanes not equipped with an optional oil filter, drain the engine oil sump and oil cooler and clean both the oil suction strainer and oil pressure screen. On the airplanes which have an optional oil filter, the oil change interval may be extended to 100-hour intervals providing the oil filter element is changed at 50-hour intervals. Change engine oil at least every four months even though less than 50 hours have accumulated. Reduce intervals for prolonged operation in dusty areas, cold climates, or when short flights and long idle periods result in sludging conditions.
CARBURETOR AIR FILTER -- Clean or replace. Under extremely dusty conditions, daily maintenance of the filter is recommended.
NOSE GEAR TORQUE LINKS -- Lubricate. When operating under dusty conditions, more frequent lubrication is recommended.

EACH 100 HOURS

SPARK PLUGS -- Clean, test and regap.
BRAKE MASTER CYLINDERS -- Check and fill.
SHIMMY DAMPENER -- Check and fill.
FUEL STRAINER -- Disassemble and clean.
FUEL TANK SUMP DRAINS -- Drain water and sediment.
FUEL SELECTOR VALVE DRAIN PLUG -- Drain water and sediment.
SUCTION RELIEF VALVE INLET SCREEN (OPT) -- Clean.

SERVICING INTERVALS CHECK LIST

(Continued)

EACH 500 HOURS

VACUUM SYSTEM AIR FILTER (OPT) -- Replace filter element. Replace sooner if suction gage reading drops to 4.6 in. Hg.
WHEEL BEARINGS -- Lubricate at first 100 hours and at 500 hours thereafter. Reduce lubrication interval to 100 hours when operating in dusty or seacoast areas, during periods of extensive taxiing, or when numerous take-offs and landings are made.

AS REQUIRED

NOSE GEAR SHOCK STRUT -- Keep filled with fluid and inflated to 45 psi.

OWNER FOLLOW-UP SYSTEM

Your Cessna Dealer has an owner follow-up system to notify you when he receives information that applies to your Cessna. In addition, if you wish, you may choose to receive similar notification directly from the Cessna Service Department. A subscription card is supplied in your aircraft file for your use, should you choose to request this service. Your Cessna Dealer will be glad to supply you with details concerning these follow-up programs, and stands ready through his Service Department to supply you with fast, efficient, low cost service.

PUBLICATIONS

Various publications and flight operation aids are furnished in the aircraft when delivered from the factory. These items are listed below.

- OWNER'S MANUALS FOR YOUR
  AIRCRAFT
  ELECTRONICS AND AUTOPILOT
- CESSNA FLIGHT GUIDE (FLIGHT COMPUTER)
- SALES AND SERVICE DEALER DIRECTORY

The following additional publications, plus many other supplies that are applicable to your aircraft, are available from your Cessna Dealer.

- SERVICE MANUALS AND PARTS CATALOGS FOR YOUR
  AIRCRAFT
  ENGINE AND ACCESSORIES
  ELECTRONICS AND AUTOPILOT

Your Cessna Dealer has a current catalog of all available Customer Services Supplies, many of which he keeps on hand. If supplies are not in stock, your Cessna Dealer will be happy to order for you.

Section VI

OPERATIONAL DATA

The operational data shown on the following pages are compiled from actual tests with the airplane and engine in good condition and using average piloting technique and best power mixture. You will find this data a valuable aid when planning your flights.

A power setting selected from the range chart usually will be more efficient than a random setting, since it will permit you to estimate your fuel consumption more accurately. You will find that using the charts and your Power Computer will pay dividends in overall efficiency.

Cruise and range performance shown in this section is based on flight tests using a McCauley 1C172/MTM 7653 propeller. Other conditions of the tests are shown in the chart headings. Allowances for fuel reserve, headwinds, take-offs, and climb, and variations in mixture leaning technique should be made and are in addition to those shown on the chart. Other indeterminate variables such as carburetor metering-characteristics, engine and propeller conditions, and turbulence of the atmosphere may account for variations of 10% or more in maximum range.

Remember that the charts contained herein are based on standard day conditions. For more precise power, fuel consumption, and endurance information, consult the Cessna Flight Guide (Power Computer) supplied with your aircraft. With the Flight Guide, you can easily take into account temperature variations from standard at any flight altitude.

AIRSPEED CORRECTION TABLE

	IAS	40	50	60	70	80	90	100	110	120	130	140
FLAPS UP	CAS	55	58	65	72	82	91	101	110	120	129	139
FLAPS DOWN	CAS	48	54	63	72	82	93	105	•	•	•	•

Figure 6-1.

STALL SPEEDS, POWER OFF

	CONDITION	ANGLE OF BANK 0°	20°	40°	60°
2300 LBS. GROSS WEIGHT	FLAPS UP	57	59	65	81
	FLAPS 10°	52	54	59	74
	FLAPS 40°	49	51	56	69

SPEEDS ARE MPH, CAS

Figure 6-2.

TAKE-OFF DATA

TAKE-OFF DISTANCE FROM HARD SURFACE RUNWAY WITH FLAPS UP

GROSS WEIGHT POUNDS	IAS AT 50' MPH	HEAD WIND KNOTS	AT SEA LEVEL & 59°		AT 2500 FT. & 50°F		AT 5000 FT. & 41°F		AT 7500 FT. & 32°F	
			GROUND RUN	TOTAL TO CLEAR 50 FT OBS	GROUND RUN	TOTAL TO CLEAR 50 FT OBS	GROUND RUN	TOTAL TO CLEAR 50 FT OBS	GROUND RUN	TOTAL TO CLEAR 50 FT OBS
2300	68	0	865	1525	1040	1910	1255	2430	1565	3855
		10	615	1170	750	1465	920	1955	1160	3110
		20	405	850	505	1100	630	1480	810	2425
2000	63	0	630	1095	755	1325	905	1625	1120	2155
		10	435	820	530	1005	645	1250	810	1685
		20	275	580	340	720	425	910	595	1255
1700	56	0	435	780	520	920	625	1095	765	1570
		10	290	570	365	660	430	820	535	1040
		20	175	385	215	470	270	575	345	745

NOTES:
1. Increase distance 10% for each 25°F above standard temperature for particular altitude.
2. For operation on a dry, grass runway, increase distances (both "ground run" and "total to clear 50 ft. obstacle") by 7% of the "total to clear 50 ft. obstacle" figure.

MAXIMUM RATE-OF-CLIMB DATA

GROSS WEIGHT POUNDS	AT SEA LEVEL & 59°F			AT 5000 FT. & 41°F			AT 10,000 FT. & 23°F			AT 15,000 FT. & 5°F		
	IAS MPH	RATE OF CLIMB FT/MIN	GAL. OF FUEL USED	IAS MPH	RATE OF CLIMB FT/MIN	FROM S.L. FUEL USED	IAS MPH	RATE OF CLIMB FT/MIN	FROM S.L. FUEL USED	IAS MPH	RATE OF CLIMB FT/MIN	FROM S.L. FUEL USED
2300	82	645	1.0	81	435	2.6	79	230	4.8	78	22	11.5
2000	79	840	1.0	78	610	2.2	76	380	3.6	75	155	6.3
1700	77	1085	1.0	76	825	1.9	73	570	2.9	72	335	4.4

NOTES:
1. Flaps up, full throttle, mixture leaned for smooth operation above 5000 ft.
2. Fuel used includes warm up and take-off allowance.
3. For hot weather, decrease rate of climb 20 ft. /min. for each 10°F above standard day temperature for particular altitude.

Figure 6-3.

CRUISE & RANGE PERFORMANCE

SKYHAWK

Gross Weight- 2300 Lbs.
Standard Conditions
Zero Wind Lean Mixture

NOTE: Maximum cruise is normally limited to 75% power. Cruise speed for the standard Model 172 is approximately one MPH less than shown below for the Skyhawk configuration.

ALT.	RPM	% BHP	TAS MPH	GAL / HOUR	38 GAL (NO RESERVE) ENDR. HOURS	38 GAL (NO RESERVE) RANGE MILES	48 GAL (NO RESERVE) ENDR. HOURS	48 GAL (NO RESERVE) RANGE MILES
2500	2700	86	134	9.7	3.9	525	4.9	660
	2600	79	129	8.6	4.4	570	5.6	720
	2500	72	123	7.8	4.9	600	6.2	760
	2400	65	117	7.2	5.3	620	6.7	780
	2300	58	111	6.7	5.7	630	7.2	795
	2200	52	103	6.3	6.1	625	7.7	790
5000	2700	82	134	9.0	4.2	565	5.3	710
	2600	75	128	8.1	4.7	600	5.9	760
	2500	68	122	7.4	5.1	625	6.4	790
	2400	61	116	6.9	5.5	635	6.9	805
	2300	55	108	6.5	5.9	635	7.4	805
	2200	49	100	6.0	6.3	630	7.9	795
7500	2700	78	133	8.4	4.5	600	5.7	755
	2600	71	127	7.7	4.9	625	6.2	790
	2500	64	121	7.1	5.3	645	6.7	810
	2400	58	113	6.7	5.7	645	7.2	820
	2300	52	105	6.2	6.1	640	7.7	810
10,000	2650	70	129	7.6	5.0	640	6.3	810
	2600	67	125	7.3	5.2	650	6.5	820
	2500	61	118	6.9	5.5	655	7.0	830
	2400	55	110	6.4	5.9	650	7.5	825
	2300	49	100	6.0	6.3	635	8.0	800
12,500	2600	63	123	7.0	5.4	665	6.8	840
	2500	57	115	6.6	5.8	665	7.3	835
	2400	51	105	6.2	6.1	645	7.8	815

Figure 6-4.

LANDING DATA

LANDING DISTANCE ON HARD SURFACE RUNWAY
NO WIND – 40° FLAPS – POWER OFF

GROSS WEIGHT LBS.	APPROACH IAS MPH	@ S.L. & 59° F		@ 2500 ft. & 50° F		@ 5000 ft. & 41° F		@ 7500 ft. & 32° F	
		GROUND ROLL	TOTAL TO CLEAR 50' OBS.	GROUND ROLL	TOTAL TO CLEAR 50' OBS.	GROUND ROLL	TOTAL TO CLEAR 50' OBS.	GROUND ROLL	TOTAL TO CLEAR 50' OBS.
2300	69	520	1250	560	1310	605	1385	650	1455

NOTES: 1. Reduce landing distance 10% for each 5 knot headwind.
2. For operation on a dry, grass runway, increase distances (both "ground roll" and "total to clear 50 ft. obstacle") by 20% of the "total to clear 50 ft. obstacle" figure.

Figure 6-5.

LANDING.

(1) Before landing, push "WING LVLR" control knob full in to the off position.

EMERGENCY PROCEDURES

If a malfunction should occur, the system is easily overpowered with pressure on the control wheel. The system should then be turned off. In the event of partial or complete vacuum failure, the wing leveler will automatically become inoperative. However, the Turn Coordinator used with the wing leveler system will not be affected by loss of vacuum since it is designed with a "back-up" system enabling it to operate from either vacuum or electrical power in the event of failure of one of these sources.

OPERATING NOTES

(1) The wing leveler system may be overpowered at any time without damage or wear. However, for extended periods of maneuvering it may be desirable to turn the system off.

(2) It is recommended that the system not be engaged during take-off and landing. Although the system can be easily overpowered, servo forces could significantly alter the manual "feel" of the aileron control, especially should a malfunction occur.

TRUE AIRSPEED INDICATOR

A true airspeed indicator is available to replace the standard airspeed indicator in your airplane. The true airspeed indicator has a calibrated rotatable ring which works in conjunction with the airspeed indicator dial in a manner similar to the operation of a flight computer.

TO OBTAIN TRUE AIRSPEED, rotate ring until pressure altitude is aligned with outside air temperature in degrees Fahrenheit. Then read true airspeed on rotatable ring opposite airspeed needle.

NOTE

Pressure altitude should not be confused with indicated altitude. To obtain pressure altitude, set barometric scale on altimeter to "29.92" and read pressure altitude on altimeter. Be sure to return altimeter barometric scale to original barometric setting after pressure altitude has been obtained.

FUEL TANK QUICK-DRAIN VALVE KIT

Two fuel tank quick-drain valves and a fuel sampler cup are available as a kit to facilitate daily draining and inspection of fuel in the main tanks for the presence of water and sediment. The valves replace existing fuel tank drain plugs located at the lower inboard area of the wing. The fuel sampler cup, which may be stowed in the map compartment, is used to drain the valves. The sampler cup has a probe in the center of the cup. When the probe is inserted into the hole in the bottom of the drain valve and pushed upward, fuel flows into the cup to facilitate visual inspection of the fuel. As the cup is removed, the drain valve seats, stopping the flow of fuel.

OIL QUICK-DRAIN VALVE

An oil quick-drain valve is optionally offered to replace the drain plug in the oil sump drain port. The valve provides a quicker and cleaner method of draining engine oil. To drain the oil with this valve installed, slip a hose over the end of the valve, route the hose to a suitable container, then push upward on the end of the valve until it snaps into the open position. Spring clips will hold the valve open. After draining, use a screwdriver or suitable tool to snap the valve into the extended (closed) position and remove the drain hose.

CARBURETOR AIR TEMPERATURE GAGE

A carburetor air temperature gage may be installed in the airplane to help detect carburetor icing conditions. The gage is marked with a yellow arc between -15° and +5°C. The yellow arc indicates the carburetor temperature range where carburetor icing can occur; a placard on the gage reads "KEEP NEEDLE OUT OF YELLOW ARC DURING POSSIBLE ICING CONDITIONS."

Visible moisture or high humidity can cause carburetor ice formation, especially in idle or low power conditions. Under cruising conditions, the formation of ice is usually slow, providing time to detect the loss of RPM caused by the ice. Carburetor icing during take-off is rare since the full-open throttle condition is less susceptible to ice obstruction.

If the carburetor air temperature gage needle moves into the yellow arc during potential carburetor icing conditions, or there is an unexplained drop in RPM, apply full carburetor heat. Upon regaining the original RPM (with heat off), determine by trial and error the minimum amount of carburetor heat required for ice-free operation.

NOTE

Carburetor heat should not be applied during take-off unless absolutely necessary to obtain smooth engine acceleration (usually in sub-zero temperatures).

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WARRANTY

The Cessna Aircraft Company ("Cessna") warrants each new aircraft manufactured by it and such new aircraft equipment, accessories and service parts as are sold through its Commercial Aircraft Marketing Division to be free from defects in material and workmanship under normal use and service for a period of six (6) months after delivery to the original retail purchaser or first user in the case of aircraft, aircraft equipment and accessories (except Cessna-Crafted Electronics as herein defined) and service parts therefor, and for a period of one (1) year after such delivery in the case of Cessna-Crafted Electronics (which term includes all communication, navigation and autopilot systems bearing the name "Cessna", beginning at the connection to the aircraft electrical system (bus bar) and including "black boxes", antennas, microphones, speakers and other components and associated wiring but excluding gyro instruments used in connection with autopilot and navigation systems) and service parts therefor.

Cessna's obligation under this warranty is limited to repairing or replacing, at its option, any part or parts which, within the applicable six (6) or twelve (12) months period as above set forth, shall be returned transportation charges prepaid to Cessna at Wichita, Kansas, or to any Cessna appointed or Cessna Distributor appointed dealer authorized by such appointment to sell the aircraft, equipment, accessories and service parts of the type involved and which upon examination shall disclose to Cessna's satisfaction to have been thus defective. (A new warranty period is not established for replacements. Replacements are warranted for the remainder of the applicable six (6) or twelve (12) months original warranty period.) The repair or replacement of defective parts under this warranty will be made by Cessna or the dealer without charge for parts, or labor for removal, installation and/or actual repair of such defective parts. (Locations of such dealers will be furnished by Cessna on request.)

The provisions of this warranty do not apply to any aircraft, equipment, accessories (including Cessna-Crafted Electronics) or service parts therefor manufactured or sold by Cessna which have been subject to misuse, negligence, or accident, or which shall have been repaired or altered outside of Cessna's factory in any way so as in the judgment of Cessna to affect adversely its performance, stability and reliability, nor to normal maintenance services (such as engine tune up, cleaning, control rigging, brake and other mechanical adjustments, maintenance inspections, etc.) and the replacement of service items (such as spark plugs, brake linings, filters, hoses, belts, tires, etc.) made in connection with such services or required as maintenance, nor to normal deterioration of soft trim and appearance items (such as paint, upholstery, rubber-like items, etc.) due to wear and exposure.

THIS WARRANTY IS EXPRESSLY IN LIEU OF ANY OTHER WARRANTIES, EXPRESSED OR IMPLIED IN FACT OR BY LAW, INCLUDING ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, AND OF ANY OTHER OBLIGATION OR LIABILITY ON THE PART OF CESSNA TO ANYONE OF ANY NATURE WHATSOEVER BY REASON OF THE MANUFACTURE AND/OR SALE OR THE USE OF SUCH AIRCRAFT PRODUCTS, INCLUDING LIABILITY FOR CONSEQUENTIAL OR SPECIAL DAMAGES, AND CESSNA NEITHER ASSUMES NOR AUTHORIZES ANYONE TO ASSUME FOR IT ANY OTHER OBLIGATION OR LIABILITY IN CONNECTION WITH SUCH AIRCRAFT PRODUCTS.

SERVICING REQUIREMENTS

FUEL:

AVIATION GRADE -- 80/87 MINIMUM GRADE
CAPACITY EACH STANDARD TANK -- 21 GALLONS
CAPACITY EACH LONG RANGE TANK -- 26 GALLONS

ENGINE OIL:

AVIATION GRADE -- SAE 50 ABOVE 60°F
SAE 10W30 or SAE 30 BETWEEN 0° and 70°F
SAE 10W30 OR SAE 20 BELOW 10°F

(MULTI-VISCOSITY OIL WITH A RANGE OF SAE 10W30 IS RECOMMENDED FOR IMPROVED STARTING AND LUBRICATION DURING WARM-UP IN COLD WEATHER. DETERGENT OR DISPERSANT OIL, CONFORMING TO LYCOMING SPECIFICATION NO. 301E, MUST BE USED.)

CAPACITY OF ENGINE SUMP -- 8 QUARTS

(DO NOT OPERATE ON LESS THAN 6 QUARTS. TO MINIMIZE LOSS OF OIL THROUGH BREATHER, FILL TO 7 QUART LEVEL FOR NORMAL FLIGHTS OF LESS THAN 3 HOURS. FOR EXTENDED FLIGHT, FILL TO 8 QUARTS. IF OPTIONAL OIL FILTER IS INSTALLED, ONE ADDITIONAL QUART IS REQUIRED WHEN THE FILTER ELEMENT IS CHANGED.)

HYDRAULIC FLUID:

MIL-H-5606 HYDRAULIC FLUID

TIRE PRESSURES:

NOSE WHEEL ---------- 26 PSI ON 5.00 X 5 TIRE
26 PSI ON 6.00 X 6 TIRE
MAIN WHEELS ---------- 24 PSI ON 6.00 X 6 TIRES

NOSE GEAR SHOCK STRUT:

KEEP FILLED WITH FLUID AND INFLATED TO 45 PSI.

CESSNA

"TAKE YOUR CESSNA HOME FOR SERVICE AT THE SIGN OF THE CESSNA SHIELD".

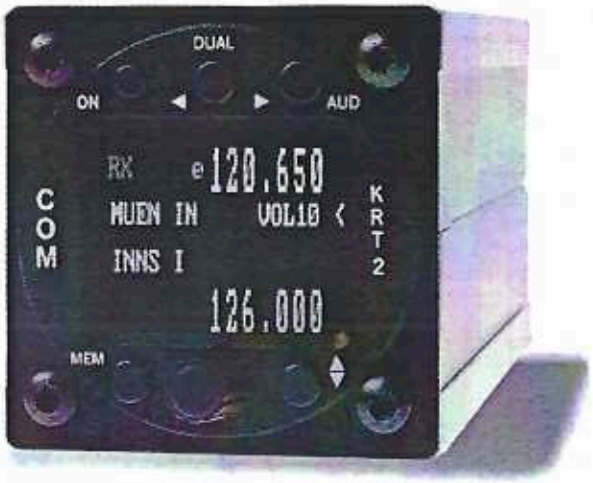
CESSNA AIRCRAFT COMPANY

WICHITA, KANSAS

KRT2

VHF Communication Transceiver

Standard, Landscape, Portrait



Part No 285942 = KRT2-S (Standard)
Part No 285945 = KRT2-L (Mini Landscape)
Part No 286048 = KRT2-P (Mini Portrait)
Part No 285946 = KRT2-RC (Remote Control)

Operation- and Installation Manual

Record of Revisions

Revision	Date	Chapter(s)	Topic
0100	24.03.2017	all	Initial Release
0101	16.01.2018	5.3, 7.1	FCC supplements, chap. 7.1 (periodic maintenance) amended

Table 1: Record of Revisions

Service Bulletins (SB)

Service Bulletins must be inserted in the manual and added to this table.

No SB	No Rev.	Release date	Date added	Name

Table 2: Service Bulletins

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1. GENERAL

This manual contains information about the physical, mechanical and electrical properties as well as a description for the operation and installation of the VHF airborne transceiver KRT2.

1.1 Symbols

WARNING
Non-compliance may cause personnel injury due to radiation or fire.

CAUTION
Non-compliance may cause damage or incorrect operation of the transceiver.

INFORMATION

1.2 Acronyms

Abbreviation	Description	Definition
BAT	Battery (Electrical)	Check DC source
DIM	Dimming	Display brightness setting
EXT	Exterior / External	External Audio input level setting
INT	Intercom level	Intercom volume level setting
PTT	Push-To-Talk	Transmitter activation
RC	Remote control	KRT2-RC remote control for KRT2
SQ	Squelch	Squelch setting
VOX	Voice operated intercom	Voice level setting for intercom activation

Table 3: Acronyms

1.3 Customer Service

In order to process returned units most expeditiously, please use the email support.krt@tq-avionics.com on the website www.tq-avionics.com

	Suggestions which will improve this manual are very much appreciated at: info@tq-avionics.com.

	Information concerning software updates is available under support.krt@tq-avionics.com.

1.4 KRT2 Transceiver properties

- VHF airborne transceiver
- Frequency range 117.975 to 137.000 MHz
- Channel spacing 8,33 / 25 kHz (2278 channels)
- Fast channel selection
- 2 separate microphone inputs (standard or dynamic)
- Audio-input for other audio devices
- Installation: Standard panel cut-out (57 mm)
- Integrated Intercom
- 100 user definable frequencies with up to 8 character/spaces identifiers

	Continuous transmissions will be turned off after 2 minutes. (Stuck microphone function)

2. Installation limitation

The conditions and tests required for (E)TSO approval of this article are minimum performance standards. It is the responsibility of those installing this article either on or within a specific type or class of aircraft to determine that the aircraft installation conditions are within the (E)TSO standards. (E)TSO articles must have separate approval for installation in aircraft.

2.1 Installation

For installation hints, data, electrical connections and mounting instructions please see section 6 "Installation".

2.2 Aircraft Radio

The KRT2 was designed as a closed unit for installation in a cockpit environment of the general aviation with the following limitations:

Installation must be in accordance with the applicable EASA or FAA requirements.
The classification of the software approval is suitable for aircraft type.

The failure classification identified in accordance with FAA AC 23.1309-1D is:

MINOR

The Software level is:

LEVEL D

2.3 Quantitative safety objective identification

In accordance with EASA regulations, the goal is a safety objective for the VHF COM radio in the KRT-2 VHF Communication Transceiver System of 1 x 10E-4 per flight hour for Class I airplanes and 1 x 10E-5 per flight hour for Class II Airplanes.

2.4 Deviations

None

3. CONTROL general

3.1 Control elements overview

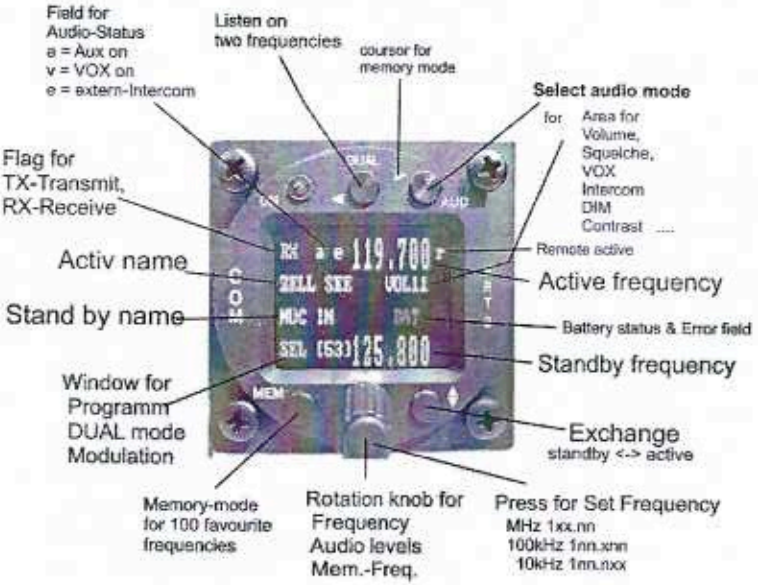


Figure 1: KRT2-S Front View

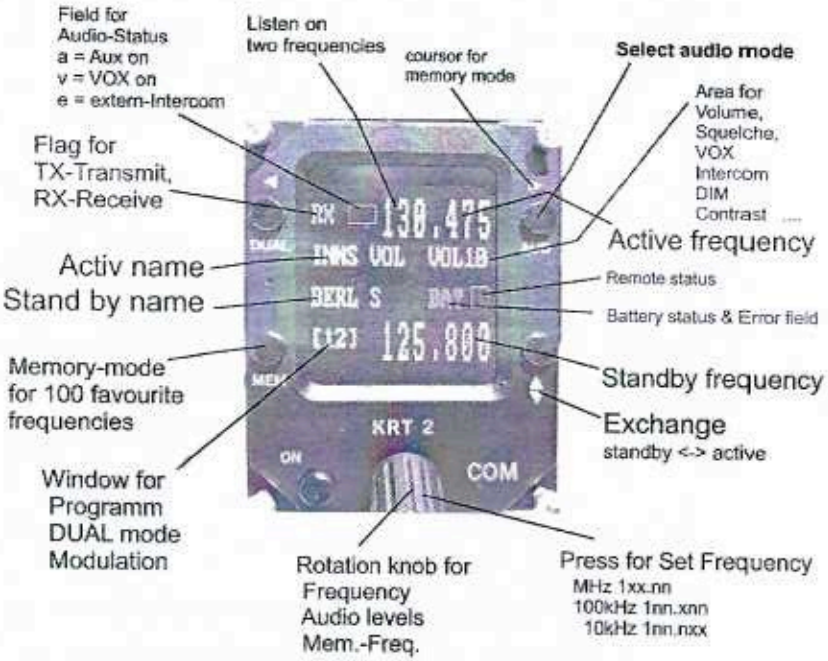


Figure 2: KRT2-P front view

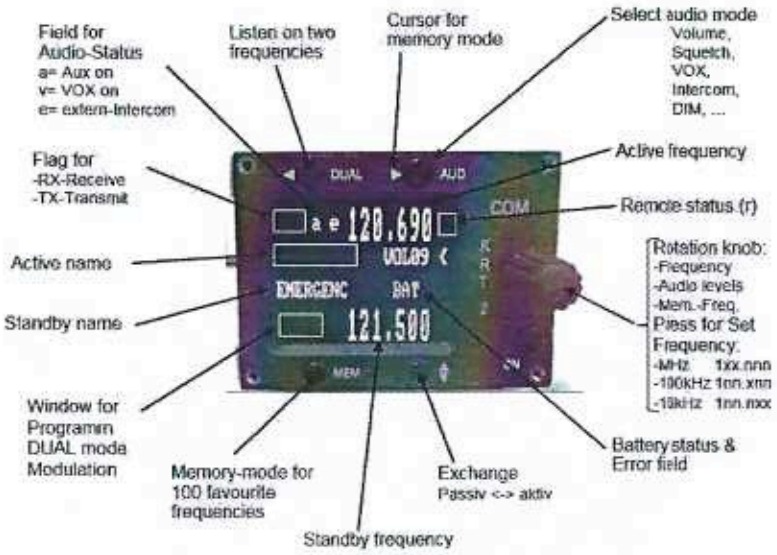


Figure 3: KRT2-L front view

All functions and performances of the normal size unit (57mm round) and the Portrait format (Mini) are identical.

The only differences are the text areas on the display Compare Figure 2: KRT2 Front view and Figure 3: KRT2 Mini Front view for more details.

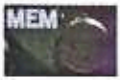
Button	Function	Usage
	ON / OFF	Self-locking switch
	DUAL WATCH	1. Scanning between the Active and Standby frequencies 2. Positioning cursor to the left when programming the station identifier
	AUDIO SELECT	1. Stepping through the audio menus 2. VOL SQ VOX TX INT EXT DIM CON SIT and MIC 3. Positioning cursor to the right when programming the station identifier
	FAVORITES	1. Frequency and identifier selection from the favourites list 2. Programming of favourites (frequency and identifier)
	EXCHANGE	Exchange of the Active and Standby frequencies
	TURNING KNOB	Pressing for Selection of the frequency range to: MHz, 100kHz, 10kHz Toggles between frequency and identifier when programming the favourites Sets all variable values in any menu • Volume setting of headsets and speakers • MHz/kHz selection of the standby • frequency in 3 different ranges • Favourite selection • Alpha character selection when programming favourites • Change of microphone settings

Table 4: KRT2 Controls

3.2 Display

Indication	Meaning	Remark
RX	Reception	RX is displayed during reception (squelch opened)
TX	Transmission	Transmitter operates normally
Te	Transmitter was turned off automatically after 2 min continuous operation	
119.700	Frequency	
ZELL SEE	Frequency station identifier	Displayed when frequency and identifier are stored in the favourite list
MUC IN	Standby frequency station identifier	Displayed when frequency and identifier are stored in the favourite list
VOL	Receiver volume level (default after a certain time delay)	When AUD is pressed the corresponding Audio Menu item and setting is displayed
DUAL	DUAL function is activated	DUAL function is deactivated by DUAL, FREQ or MEM
[03] (MEM)	Favourite list index (0-99)	When frequency and identifier are stored at this index ex:[03] they are displayed
125.100 upper	Active frequency	Displayed in large fonts.
125.800 lower	Standby/DUAL - frequency	Displayed in large fonts.
<	The pointer indicates what the turning knob will change VOL SQ VOX.....etc. Standby frequency	Arrow is positioned according to the button pressed (AUD or FREQ)
BAT	Supply voltage is low <10,5V	Low or defective battery / generator.
A-match	Antenna error	Bad antenna match
a v e	Status of certain Audio menu functions	a = AUX. Input active v = VOX active e = external Intercom switch active

Table 5: KRT2 Display

3.3 Menu levels

Displayed	Signification	Remark
VOL	Volume	Default
SQ	Squelch	
VOX	Voice operated	Voice operated intercom
DIM	Display brightness	
BAT(tst)	DC source check	
INT	Intercom - Volume	
EXT	Volume of external devices	
TX(m)**	PTT button selection	Left/Right/Both
SIT	Side tone	During transmitter operation
MIC	Setup-Menu for Microphones	Service-Menu without radio opera-tion.

Table 6: KRT2 Menu Levels

3.4 Self-test error reports

Display	Meaning	Remark
Er_PLL	Internal error, no transmission	Return the transceiver for maintenance
Er_ADC	Internal error, operation limited	Return the transceiver for maintenance
Er_FPA	Internal error; unit not usable	Return the transceiver for maintenance
Er_I2C	Internal error; unit not usable	Return the transceiver for maintenance
Er_D10	Internal error; reception corrupt	Return the transceiver for maintenance
Error_3V3	Internal error; unit not usable	Return the transceiver for maintenance
Key_Block	Internal error; unit not usable	Return the transceiver for maintenance

Table 7: KRT2 Built In Tests (BIT)

4. OPERATION

4.1 General

In the normal operating mode in which the turning knob always is connected to the volume (VOL). The normal operating mode can be left by pressing the AUD, FREQ or MEMORY button.
When not in the normal mode and there is no pilot action for more than 10 seconds the unit returns to the normal mode.

4.2 ON / OFF switching

ON / OFF switching is done by pushing the self-locking switch.

After power up the following display will be displayed:

Device-name
KRT2

Software Version e.g. V8.6

(Example)

The unit starts in the normal operating mode using and displaying the data last used.

4.3 Frequency selection

There are two different frequency selection methods:

- Direct Input
- Selection from the favourite list (index 0-99)

4.3.1 Direct frequency selection

The Standby-Frequency is set with the turning knob in 3 different ranges. The selected range is highlighted and can be changed with the FREQ button. Frequency ranges are:

1xx.nnn

1nn.xnn

1nn.nxx

Press the FREQ button once or several times until the desired frequency range is highlighted. The unselected digits are displayed as dotted digits.

When the pointer is not next to the Standby Frequency window, it will be repositioned with the first pressing of the FREQ button.

Exchanges the Active and Standby frequencies.

When the Exchange button is not pressed, the Standby frequency display will return to its normal appearance after 20 seconds.

4.3.2 Frequency selection from favourites list

By pressing

and operating the turning knob a specific favourite list position can be accessed [xx] (xx = index 0 ... 99). When frequency and station identifier have been defined, they will be displayed in the Standby and station identifier windows.

The favourite identifiers list can be sorted in alphabetic order (see 4.3.3 Storing and Editing Favourites).

To exit this menu, press the key.

The selection procedure can be terminated with either the AUD or FREQ buttons. Without pressing any of these buttons the unit will return to its normal operating mode after about 15 seconds.

4.3.3 Storing and editing favourites

Any displayed Standby Frequency can be associated with an identifier and both can be stored as favourites in the favourite list. The frequency and identifier of a favourite can be edited.

First press the

button and go to the desired favourite list position to be edited using the turning knob (index [00 ...99]).

Press the MEMORY button a second time and „–EDIT–„ will show up in the program window.

In the identifier window a blinking cursor will show up under the extreme left character.

The turning knob selects the desired character.

The AUD button positions the curser one character to the right. The DUAL button positions the cursor one character to the left and simultaneously erases this character.

The station identifier can consist of a maximum of 8 characters/spaces.

To change frequency just press the FREQ button and follow the normal direct input procedure to edit the frequency, see "4.3.1 Direct Frequency Selection".

To quit the frequency input press the MEMORY button again in order to go to the station identifier window for editing the identifier if required. Using the buttons FREQ and MEMORY it is possible at any time to toggle between identifier and frequency inputs.

Keep in mind the watch dog timer which will terminate the input mode after 15 sec.

Termination and Save

To save the entered identifier, press the key as the cursor is on the station name, "SAVE" will appear and the system will go back to the favourite selection.

A **sorting process** can be activated by pressing the "MEM" button again from the EDIT-mode.

"SORT?" will be displayed for 20 seconds and it should be activated with or skipped with "MEM".
When activated all 99 favourites will be sorted in alphabetical order and the process can take several minutes.

During the sorting procedure "RUN nn" is displayed in the program window, with nn being the running index.
After a 6 seconds time out or by ending the sort, the transceiver resumes its normal operating mode.

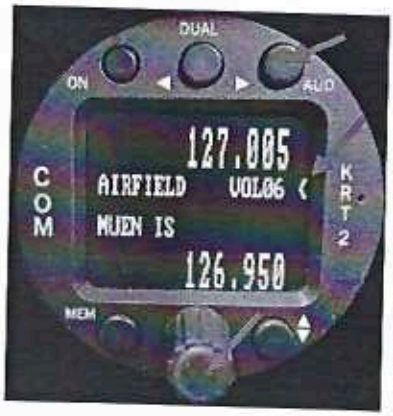
When the "MEM" button is pressed at the time "RUN nn" is displayed, the sorting procedure is terminated. The favourite list is then partially sorted and the transceiver resumes its normal operating mode.

Example:

1. Button MEM -> SEL [23] = Select location
2. Button MEM -> -EDIT- = Input of name
   Rotation knop to select the characters
   For cursor use (AUD) (DUAL)
   Frequency setting -> press rotating button
   Use button MEM to go back to -EDIT-
3. Button -> shortly SAVE -> back to 1.) (do nothing = cancelling)
4. Button MEM -> question for "SORT?" Yes = No = MEM or do nothing.

If the function is abandoned before completion, it will be exited after a 6 seconds time out with no activity. The data will not be saved.

4.4 AUD – Audio menu



Any action in the Audio Menu requires the pointer (<) to be next to the Audio menu window (see picture). When the pointer is next to the Standby frequency window, the pointer can be repositioned by pressing the AUD button once.

VOLnn is the Audio menu default display. No action on any control for more than 10 seconds will result in the VOLnn being displayed.

Audio Menu items can be accessed in the following order by repeatedly pressing the AUD button.

VOL (default), SQ, VOX, TXm**, INT, EXT, DIM, BATtst, SIT, MIC

4.4.1 VOL – Volume

Turning the turning knob changes the receiver volume.

VOLnn Range: 01 – 20

The VOL setting only concerns the receiver and not the intercom system. Intercom volume values are set in the INT audio menu.

4.4.2 SQ – Squelch

Pressing the AUD button once enables the turning knob to change the squelch level values.

SQnn Value range: 01 – 10

The Squelch setting is depending on several factors.

For engine driven airplanes an initial setting of 3-5 is recommended. For gliders a setting of 2 is recommended. The lower the Squelch level value the higher is the input sensitivity. A high sensitivity setting is susceptible to noise from other sources like ignition strobe-lights etc.

Standard SQ-level is 3...5. Higher setting will suppress weaker input signals. 01 = Squelch off, 02 = for long range. Squelch does not influence the intercom system.

4.4.3 VOX – Intercom voice trigger level setting

Pressing the AUD button twice enables the turning knob to change the voice level which triggers the intercom.

The intercom voice trigger level must be set to a value which prevents normal cockpit noise from being heard in the earphones. The intercom system should only be activated when talking at a normal voice level into the microphone.

The higher the trigger level the louder the voice must be in order to trigger the intercom system.
VOX on condition is indicated by flag "v".
VOXnn Range: 01 – 10

	The larger the value, the louder one must speak in order to activate the Intercom connection. VOX on condition is indicated by flag "v". For installation with loudspeakers set VOX to 10 to switch it off.

4.4.4 Manual Intercom

With cockpit loud background noise or with uncompensated microphones the Intercom can be controlled manually by using an external switch. Therefore the VOX system must be activated permanently by selecting VOX: 01.
To turn off the Intercom the talk switch (default closed) must be opened, which will be indicated by "e".

Deactivation of the intercom is done with an opened external microphone button (Ground connection to pin 12), this is indicated with an "e" on the display.

This mode only works with a deactivated external audio input (see "4.4.7 EXT – External Audio Input Volume").

In gliders the VOX has to be set to 10 in order to disable the speaker control.

4.4.5 TXm – PTT switch selection

Pressing the AUD button three times enables the turning knob to enable certain PTT switches.

On transmission the microphone which is related to the PTT-L/R will be enabled. The equivalent indication is TX (TXm**), TX1 (TXm*-) TX2 (TXm-*).
On transmission the PTT-L/R related microphone will be the only one activated.

TXm** *- Left / -* Right / ** Both

4.4.6 INT – Intercom volume

Pressing the AUD button four times enables the turning knob to set the intercom volume.

INTnn Range: 1 – 9

4.4.7 EXT – External audio input volume

Pressing the AUD button five times enables the turning knob to set the external audio input volume. External audio inputs can be audio alarms, voice alarms, Vario, etc. The required level is 200mVpp (6Vpp max).
Activation occurs for settings >00 and will be indicated by "a".

- 00 = turning off,
- 01 = lowest gain without threshold
- 09 = highest gain with interference suppression threshold.

EXTnn Range: 0 - 9

4.4.8 DIM – Display brightness

Pressing the AUD button six times enables the turning knob to set the display brightness.

DIMnn Range: 01 – 16

The display consumption at 12V is between 10mA und 70mA. In addition the battery voltage in Volts is indicated.

4.4.9 BAT – Battery test

Pressing the AUD button seven times enables the turning knob to display the battery voltage.

4.4.10 SIT – Side tone

Pressing the AUD button eight times enables the turning knob to set the side tone volume (for gliders has to be set to 01).

SITnn Range: 1 – 9

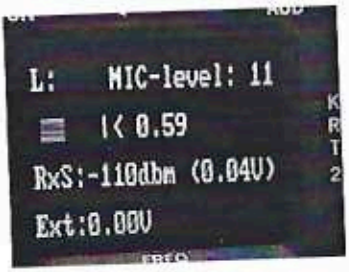
4.4.11 MIC – Setup

This mode is for microphone setup and test only without using the PTT. It is a service mode and it is not designed for normal operation.

Each of the two microphone input channels can be configured individually, which enables different microphone types to be used.

A maximum of two microphones of same type may be connected to each microphone input channel (see chapter 5.6.1 Microphone-Connection).

The MIC – Setup is the last item of the Audio menu and can be accessed by pressing the AUD button nine times.



By pressing the DUAL button repeatedly L, R and AUTO can be selected (Left "L" in the example).

L: Left = Left microphone.

R: Right = Right microphone.

AUT: Auto mode.

Menus L and R:

By means of the turning knob the displayed microphone input channel amplifier gain (MIC-level 01 = low gain, 09 =high gain) can be selected individually. The microphone signal level is dynamically displayed with a bar and a numeric value.

The initial MIC-level should be 0,5; the engine should be running, use a headset or earphone and speak at a normal voice level to fine-tune the MIC-level. Whenever a new MIC level is selected, the dynamic bar indicator should then be at about 50%.

Special hint:

During MIC-Setup activation, the speaker state will be taken into account (SQU on/off), if the state is active an acoustic feet-back may be heard.

The range of the MIC-level for standard microphones is 1 to 9.
MIC levels 10 and 11 are special settings for low microphone levels like dynamic micro-phones often used in gliders.

- Level 10 is used for non-amplified Electret microphones with an 8 volt supply voltage.
- Level 11 is for dynamic microphones only.

For high gain selection (> 9) the use of side tone can lead to a feedback at transmission. In that case the side tone should be set to SIT=01.

The new values will be stored upon exiting the microphone setup menu. AUTO must be selected in order to activate the AUTO menu, else use L or R to activate the Left and & Right microphone menu.

Menu Auto

In the AUTO mode only the left microphone impedance is measured. When a dynamic microphone is recognized, the left channel is switch to 11, the right channel is unaffected. When returning to the standard configuration, the original L and R values are restored.

A simultaneous operation on Electret and Dynamic microphones is not possible (there is just one DC source for both ports). For dynamic mode this source must be turned off automatically

The display of microphone type (lower right side) is done upon activation of the microphone menu. When an Electret microphone is recognized, the values used are the one that were saved upon leaving the menu.

The MIC submenu is terminated by pressing the AUD button.

Additional indications

Additional indications for test purposes:

- RxS : RF receiver input level (from Automatic Gain Control)
- Ext : External audio input voltage
- Mic : d (dynamic) or s (standard)

The microphone type is displayed when the microphone menu is entered.

This Symbol appears on right side of STBY Frequency in AUTO-mode if the dynamic microphone has been recognized and activated after transmission. In that case the intercom voice triggering level VOX is set to 10 (switched off).

To reactivate the intercom after a microphone change, a short press of the PTT is required.

4.4.12 Menu lock

For school operation the menus area TXm to MIC-setup can be locked.

To lock or unlock the button combination AUD & FREQ has to be pressed simultaneously for > 2 seconds.

In lock condition there will be displayed an "L" at the right end of the third line.
To store those condition the setting of SQnn should be changed before turning off.

VOL	SQ	VOX	TXm	INT	EXT	DIM	CON	SIT	MIC
Available			Locked and not available						

Table 8: KRT2 Menu Lock

4.5 DUAL watch

Because the communication transceiver KRT2 contains only one receiver, DUAL watch is achieved by alternating between the Active and Standby frequencies.

The DUAL button activates and deactivates the dual watch function. Deactivation also can take place by pressing either the FREQ or MEMORY buttons.
The frequencies to be watched should be selected prior to the DUAL watch selection.

The frequencies to be used must be selected before using dual watch and the squelch level SQ must be set to 02 or more in order to eliminate noise. Scanning Frequencies is only possible when differentiating between radio noise and radio transmissions.

There must be radio noise suppression in order to recognize reception. The squelch level SQ must be set to 02 or higher.

When DUAL watch is activated, "DUAL" is displayed on the lowest line. The pointer next to the DUAL display indicates the frequency on which there is reception.
The Active frequency always has priority and the receiver remains on the active frequency as long as there is reception on this frequency.
When there is no reception on both the Active frequency and the Standby frequency the receiver scans both frequencies 5 times per second.

When reception is detected on the Active frequency the reception on the standby frequency is interrupted until the end of the reception.

When a reception is detected on the Standby frequency the receiver stays on the Standby frequency, however it switches to the Active frequency every 2 seconds for a duration of 0.3 seconds. If a signal is detected on the active frequency the reception will be switch to that frequency.
The pointer next to the DUAL display indicates on which frequency there is reception.

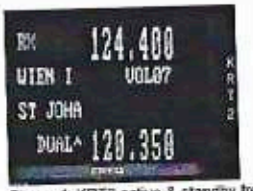


Figure 4: KRT2 active & standby frequencies

Standby and Active frequencies can be exchanged when in the DUAL mode. The transmitter operates on the Active frequency only.

Summary:

- Select the Standby frequency to be monitored in addition to the active frequency.
- With the AUD button and turning knob set SQnn to 2 or higher.
- With the DUAL button activate the DUAL watch function.
- When there is no reception on both the Active frequency and the Standby frequency the receiver scans both frequencies.
- When scanning the Active frequency always has priority.
- Deactivate the DUAL watch function with the DUAL, FREQ or MEMORY buttons.

4.6 Transmitter Operation

The unit transmits on the active frequency (upper line) as long as a PTT (press to talk) switch is pressed.

Transmission Reception

Figure 5: KRT2 TX & RX operations

"TX" indicates normal transmitter operation.
"RX" indicates a receiver operation.

In the lower left corner of the display the carrier modulation is dynamically displayed. It corresponds to the side tone which is not available on gliders when no earphones are in use.

In order to avoid the blocking off of the frequency by unintentional long transmissions (stuck microphone) the transmitter is switched off after two minutes and the display changes from „TX" to „Te". To resume transmission the PPT switch first must be released and then be pressed again.

While transmitting the external audio input will be turned off automatically and the microphone input, selected during the pre-setting in the TXm-menu, will be selected.

The differential speaker output will be turned off to prevent an audio feedback to the microphone. For the same reason; the speaker is turned off when the intercom (VOX) is active.

The output for the headset will carry the side tone.

4.6.1 Two PTT configuration

There are two different PTT assigned for each of the left and right side microphones. This enables the deactivation of the unused one preventing additional noise and unintentional talking on transmission.

In case there is just one PTT and multiple headsets in use both the PTT-L and PTT-R must be tied together, see chapter "4.4.5 TXm – PTT Switch Selection".

4.6.2 Self-Test monitor

Software diagnostics operate continuously in the background in order to test the system.

The field for battery status & error (see Control Elements Overview) is used to display warnings and in case of hardware failure, different error reports will be displayed here.

The warnings are:

BAT Low battery voltage (becomes active < 10,5V)

At transmission

A-match Bad antenna match or defective antenna.

Also while transmitting the TX-flag (left top) will change to:

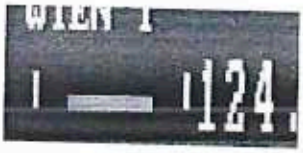
Te if transmission time has exceeded (> 2 minutes).

All other reports starting with Er_.... Indicating a major hardware failure and consequently the radio has to be returned to the factory.

4.6.3 Optical side tone

Especially when used in gliders, where headsets are generally not worn and thus no side tone is heard, it is very helpful to see if the microphone is operating properly.

At left lower side, there is a modulation indicator that depicts the voice level. When there is no modulation it becomes a small dot approximately in the centre.
Also if it lays far off centre, it indicates that there is bad antenna matching.



4.7 Resetting to factory settings

Returning to the factory settings can only be initiated during power-up.
To do this, during power-up the MEM and DUAL buttons must be pressed simultaneously and the display will show "SET DEFAULTS". When the buttons are released the resetting to the factory settings takes place. When resetting is completed "DONE" is displayed.

4.8 SET UP - Menu

During power-up the MEM button must be pressed.
The message "protect mode" is displayed, just wait until it changes.

There are two functions available within the Set-up menu:

- ERASE – Erasing of the favorites (frequency and identifier)
- Channel Spacing – 25kHz / 8,33kHz

Programming of the Set-up is done with the lower 3 buttons next to the symbols (Exit, S, E). Their function is described on the display.

To exit the SET-UP menu, use the MEM button. The unit remains powered and the normal operating mode is resumed.

4.8.1 ERASE – erasing of favourites list

When in the SET UP – Menu select the "ERASE" sub-menu with the buttons next to the symbols (Exit, Y).

Erasing the favourites (frequency and identifier) starts after the button has been pressed again. This procedure may last a few minutes during which time the message "ERASING" is displayed.

All INFO frequencies and identifiers that were stored on delivery are lost and all favourite index positions (01 to 99) are available to the user.

4.8.2 Channel spacing

When in the SET UP – Menu pressing the FREQ (S) button will change the KRT2 into the Channel Space submenu.

The desired channel spacing then can be selected and the (X) then indicates the actual channel spacing.

TRIG

TT31 Mode S Transponder
Operating Manual

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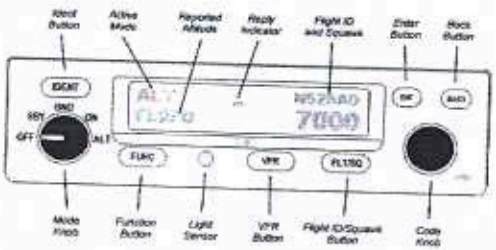
EN / DE / FR

Front Panel



Display

The display shows the operating mode of the transponder, the reported pressure altitude, and the current squawk code and Flight ID. The reply indicator is active when the transponder replies to interrogations.

The pressure altitude is displayed as a Flight Level, which is the pressure altitude in hundreds of feet. When non-standard atmospheric conditions apply, this may not match the altimeter indicated altitude, but will be correctly displayed by the ATC radar.

Mode Selector Knob

The left hand knob controls the power to the transponder and the operating mode.

OFF	Power is removed from the transponder.
SBY	The transponder is on, but will not reply to any interrogations.
GND	The transponder will respond to Mode S ground interrogations from surface movement radar.
ON	The transponder will respond to all interrogations, but altitude reporting is suppressed.
ALT	The transponder will respond to all interrogations.

When airborne, the transponder should always be set to ALT unless otherwise directed by Air Traffic Control. When you are taxiing on the ground, the transponder should be set to GND mode. If your installation includes a squat switch or is configured with an automatic air/ground system it will switch automatically and you do not need to manually select the GND position.

EN

Push Buttons

IDENT	Press the IDENT button when ATC instructs you to "Ident" or "Squawk Ident". This activates the SPI pulse in the transponder replies for 18 seconds. IDENT will appear in the display.
FUNC	Pressing the FUNC button provides access to the flight timer, stopwatch, ADS-B monitor (depending on installation) and altitude monitor function.
VFR	Pressing the VFR button sets the transponder to the pre-programmed conspicuity code. Pressing the button again restores the previous squawk code.
FLT/SQ	Pressing FLT/SQ alternates the primary display between squawk code and Flight ID.
ENT	The ENT button enters a digit in the code selector.
BACK	The BACK button goes back to the previous digit in the code selector.

Code Selector Knob

The right hand knob is used to set squawk codes and the Flight ID. The FLT/SQ button selects which will be updated. Turning the knob will highlight the first digit on the display, and the digit can be changed as required. Press the ENT button to advance to the next digit. When ENT is pressed on the last digit, the new squawk code or Flight ID will replace the previous value. If the code entry is not completed within 7 seconds, the changes are ignored and the previous code restored.

1200	VFR code in the USA
7000	VFR code commonly used in Europe
7500	Hijack code
7600	Loss of communications
7700	Emergency code

The Flight ID should correspond to the aircraft call sign entered on your flight plan. If no flight plan is active, the aircraft registration should be used as your Flight ID. Use only letters and digits. If the Flight ID is less than 8 characters long, entering a blank character will end it.

Flight Timer

The Flight Timer records the time for which the transponder has been powered on and operating in flight mode – either ON or ALT. Press the FUNC button to display the Flight Timer.

Stopwatch

The stopwatch can be used as a convenient timer. Press the FUNC button to display the stopwatch. Pressing ENT will reset and start the timer. Pressing ENT again will stop the timer.

Altitude Monitor

The Altitude Monitor activates an audio annunciator or annunciator light (depending on installation) when the aircraft pressure altitude differs from the selected altitude by more than 200 feet. Press the FUNC button to display the altitude monitor enable screen. Pressing ENT toggles the altitude monitor at the current altitude.

When altitude monitoring is in use, a small deviation pointer appears adjacent to the altitude display on the transponder.

ADS-B Monitor

The ADS-B Monitor is only available on installations that include an ADS-B position source. The ADS-B Monitor provides a display of the position information that is being transmitted in ADS-B position reports. This can provide confirmation that the correct information is being transmitted, particularly where the GPS source is remote from the transponder.

In the event that valid position information is NOT available from the GPS, the latitude and longitude display will be replaced by dashes; if no valid latitude and longitude is shown then ADS-B position information is NOT being transmitted.

Loss of ADS-B position information will also result in a WARNING message being displayed.

Warning Messages

If the transponder detects a problem, the screen will indicate WARNING and a brief statement of the problem. Depending on the nature of the problem, your transponder may not be replying to interrogations. Note the message on the screen and pass that information to your avionics maintenance organisation. Press ENT to clear the message; if the fault is still present the message will reappear.

Fault Annunciation

If the transponder detects an internal failure, the screen will indicate FAULT and a brief statement of the problem. No replies will be made to interrogations when a fault is detected.

Some FAULT indications can be recovered by switching the transponder off and back on again, although in all cases a FAULT code implies that there is a fault with the transponder or the installation. Note the FAULT message at the

bottom of the screen and pass that information to your avionics maintenance organisation.

Configuration Mode

The system is configured when it is first installed by your avionics supplier. Configuration items include the Mode S aircraft address, the interface to the other aircraft systems, the aircraft category, and the pre-programmed values for VFR squawk code. To view or change these settings you must use Configuration Mode.

Do not use Configuration Mode in flight. Check with your avionics installer before changing the configuration.

To enter configuration mode, hold down the FUNC button whilst switching on the transponder. Configuration items can be changed using the Code Knob and the ENT and BACK buttons. Pressing FUNC advances to the next configuration item.

When configuration is complete, switch the transponder off. When it is switched back on the transponder will use the new configuration.

Low Temperature Operation

The TT31 is certified to operate correctly down to -20C, but at low temperatures the display may be impaired. On a cold day you may need to wait for the cockpit to warm up to ensure normal operation.